

Emergence, molecular evolution and spatiotemporal dynamics of avian influenza viruses in Nigeria, 2006–2026: a systematic review

Abstract

Background: Avian influenza viruses (AIVs) have caused recurrent epizootics in Nigeria since 2006.

Objectives: To synthesise Nigeria-specific evidence on AIV emergence, distribution, host range and molecular dynamics, 2006–2026.

Methods: Seven bibliographic sources, including African Journals Online, were searched on 16–20 September 2026, with a Google Scholar update on 28 September and a dated update search on 9 October 2026 (rebuilt strategies with no record caps, and citation chasing on all included reports). Eligible reports gave Nigeria-specific laboratory-confirmed AIV data from birds. Two **AI** agents screened and extracted independently, with author verification. Risk of bias used JBI checklists and a molecular check; synthesis was without meta-analysis (SWiM), and certainty used GRADE-prognosis logic.

Results: We included 127 reports of 113 studies (2006–2026): 56 prevalence or surveillance studies, 31 outbreak or isolate series, 6 analytical cross-sectional and 20 secondary or modelling studies, covering all six geopolitical zones. Highly pathogenic H5N1 was confirmed in early 2006 (low certainty; 85 studies informed first detections), with major epizootics in 2006–2008 and 2015–2017 (low; 31 studies informed the occurrence series). Laboratory-confirmed detections spanned 26 states and the Federal Capital Territory in 2006–2010 (low; 81 studies). Commercial, backyard, live-bird-market and resident wild birds were infected (low; 90 studies). Clade 2.2 was followed by clades 2.3.2.1c and 2.3.4.4b, with intra-clade reassortment (low; 35 studies). Introduction numbers, non-H5N1 first-detection dates, migratory-bird infection and antibody exposure (35 studies) were of very low certainty.

Limitations: **AI**-agent screening; title-and-metadata screening of many records; 98 reports not retrieved; largely passive surveillance.

Conclusions: The evidence indicates repeated HPAI waves with clade turnover; introductions and inter-wave persistence remain uncertain, and active surveillance with genome sequencing is needed.

Registration: Not registered.

Funding: [Author to complete: funding source(s), or a statement that the review received no specific funding]

Introduction

Avian influenza viruses (AIVs) are important pathogens of domestic and wild birds and a continuing threat to poultry production, food security, wildlife conservation and public health. Influenza A viruses have segmented RNA genomes, which allows reassortment when different viruses co-infect a host; mutation, host switching and geographic dissemination can also produce genetically distinct viral populations [1,2].

The epidemiology of highly pathogenic avian influenza (HPAI) has changed substantially over the past two decades. The goose/Guangdong lineage of H5 viruses has diversified into multiple phylogenetic clades, and clade 2.3.4.4b has become prominent since 2020–2021 because of its geographic expansion and broadening host range [2–6]. Its expansion has involved reassortment, repeated geographic introductions and establishment in new wild-bird and poultry populations [2,4,7,8]. Outbreak counts

alone therefore cannot describe AIV dynamics; temporal patterns, geography and viral evolutionary relationships also matter.

Africa has experienced repeated AIV incursions in domestic and wild birds. A systematic review of birds sampled in sub-Saharan Africa during 2000–2019 identified 19 haemagglutinin/neuraminidase subtype combinations and substantial ecological variation in subtype occurrence [9]. More recent molecular surveillance in Southern Africa and Nigeria has documented circulation and diversification of HPAI H5 viruses in poultry and wild birds [10,11]. The potential contribution of migratory and other wild birds to long-distance virus movement remains relevant to the interpretation of surveillance data [8,10,12,13].

Nigeria is an important setting in which to study these processes. The country recorded HPAI H5N1 in 2006, followed by extensive poultry outbreaks and later epidemic waves [14,15]. A major re-emergence was documented in 2015, and further Nigerian outbreaks have been described up to 2025 [15,16].

Nigerian molecular studies indicate considerable evolutionary complexity. Adesola et al. described a temporal progression from clade 2.2 in 2006 to later 2.3.2/2.3.2.1f and 2.3.4.4b detections [14], and Shittu et al. reported widespread circulation of West African group 2 (WA2) clade 2.3.2.1c H5N1 in 2015 [16]. Laleye et al. reported independent introductions of H5Nx clade 2.3.4.4b viruses [17] (not retrieved; title and abstract only), and Meseko et al. described H5N1, H5N2 and H5N8 detections in poultry during 2021–2022, including evidence interpreted as multiple introductions and regional evolution [11]. H5 clade 2.3.4.4 viruses were detected in apparently healthy wild birds in the Hadejia-Nguru wetland in 2022 [12], AIV testing has been reported in captive wild birds and indigenous chickens [13], H9N2 was reported in commercial poultry in 2024 [18], and a 2026 investigation reported a new reassortant H5N1 clade 2.3.4.4b genotype [19]. These reports show why subtype, clade or sublineage, reassortant genotype and new introduction must be kept distinct: the labels are not interchangeable.

Two earlier reviews frame the gap. Kalonda et al. reviewed AIV detections in birds across sub-Saharan Africa from 2000 to 2019 (68 studies) [9], and Akanbi and Lakes gave a broad narrative review of avian influenza outbreaks in Nigeria from 2006 to 2025, based on PubMed and Scopus [15]. Neither review combined the spatial and temporal record with clade, reassortment and introduction evidence under explicit rules for counting overlapping datasets, appraising molecular reporting and rating certainty.

The objectives of this systematic review were to synthesise Nigeria-specific evidence from 2006 to 2026 on: (1) the timing of first detections and epidemic waves; (2) the geographic distribution of laboratory-confirmed AIV by state, zone and epoch; (3) the range of host systems and species with confirmed infection, and with antibody evidence of exposure; and (4) clade turnover, diversification, reassortment and introductions, each graded by the strength of the molecular evidence.

Methods

Protocol and registration

This review was not prospectively registered. Methods were set out in a dated internal protocol note, frozen on 16 September 2026 and re-frozen with amendments on 19 September 2026; subsequent amendments are dated and listed in Appendix B. The protocol is available from the corresponding author on request. The review is reported in accordance with PRISMA 2020 [20,21], with PRISMA-S for the searches [22] and the SWiM guideline for synthesis without meta-analysis [23].

Eligibility criteria

The question was framed with the CoCoPop (condition, context, population) form of PEO, rather than PICO, because no intervention was assessed. The condition was AIV infection or detection, including HPAI and low-pathogenicity avian influenza (LPAI), with subtype, clade or lineage, reassortment, mutation and diversity evidence. The context was Nigeria, from 1 January 2006 to the search dates in September 2026. The population comprised domestic poultry (commercial, backyard or village, and live-bird markets), wild, migratory and wetland birds, captive birds, and Nigeria-linked virus isolates.

A report was eligible if it gave Nigeria-specific data, or data from which the Nigerian component could be extracted, on laboratory-confirmed AIV in avian hosts, and reported at least one of: epidemiological evidence (surveillance, outbreak or prevalence data); molecular or phylogenetic evidence; or spatial or temporal data. Eligible reports were primary research, surveillance, outbreak investigation and molecular-epidemiology reports with extractable data, published in English or translatable (see Search strategy). Peer-reviewed articles and defined grey literature (theses, conference papers, preprints and agency reports) were eligible; grey literature was examined in a sensitivity analysis. Host category alone was not a route to inclusion. Reports were excluded if they lacked Nigeria-specific data, concerned non-avian influenza only, had no epidemiological, molecular or spatiotemporal data, contained no primary data (editorials, news items, narrative reviews and abstracts without data), were experimental only without field relevance, or duplicated a dataset already represented, in which case the most complete report was retained and the companions were linked to it. During verification the author set three further rules: secondary or global analyses that reuse Nigerian data were eligible if they produced Nigeria-specific analytic results, whereas narrative compilations, reprinted tables and model predictions were not; official outbreak records described as laboratory-confirmed were eligible even without a named assay, and were flagged for sensitivity analysis; and reports that only reprinted official outbreak tables were excluded.

Information sources

PubMed/MEDLINE (via the National Library of Medicine E-utilities), Scopus (Elsevier) and ScienceDirect (Elsevier) were searched on 16 September 2026, together with supplementary exports from OpenAlex and Crossref (1,000 records each, the export cap). The Web of Science Core Collection was searched through the Clarivate Starter API on 19 September 2026, and African Journals Online (AJOL) was searched on site on 20 September 2026 with nine exact queries. A broader Web of Science browser search on 20 September 2026 supplied counts only, because no complete record export was available; its records were therefore not imported. A dated Google Scholar update search (four queries, relevance-ranked, first 200 records per query) was run through SerpApi on 28 September 2026, after the protocol freeze, and is reported as identification by other methods. A dated update search was then run on 9 October 2026: every database strategy was rebuilt with the facet blocks on separate lines and the record caps removed (PubMed, OpenAlex with cursor paging, Crossref with cursor paging, the Web of Science Starter API and AJOL), and backward and forward citation chasing was performed on all included reports through the OpenAlex and Crossref reference and citation APIs. The rebuild corrected a PubMed strategy that had missed about half the eligible field (140 records with the original string versus 209 with the improved string), and every string as executed is documented in Appendix A. Records already in the register were removed by DOI, PMID and normalised title; the 52 new records were screened under the same rules as before. Google Scholar alone was re-run earlier because the 17 September 2026 Google Scholar supplement had no persistent export, so its records could never be screened; it was replaced by a dated re-run of the same four queries with the full output saved, and the re-run showed that eligible

Nigerian reports were missing from the register. Backward and forward citation chasing on the included reports was performed in the October 2026 update search; the two earlier reviews were not used as citation sources. No authors were contacted.

Search strategy

Each strategy combined an AIV concept block, a Nigeria concept block and a 2006–2026 date limit, with molecular, epidemiological/spatiotemporal and wild-bird/host terms run as separate facet lines rather than embedded in every block. The start date of 1 January 2006 was set by the review question as the first recognised HPAI H5N1 epidemic in Nigeria. Eligible reports were in English, or had an English abstract and a full text that could be translated with the translation method logged, as specified in the protocol. No included report required translation; one Indonesian-language report was assessed in full text and excluded for lacking Nigerian data (no translation method was logged), and two French-language reports were among those not retrieved (Appendix B). The PubMed base strategy was: (avian influenza[tiab] OR avian flu[tiab] OR bird flu[tiab] OR fowl plague[tiab] OR avian influenza virus[tiab] OR AIV[tiab] OR highly pathogenic avian influenza[tiab] OR HPAI[tiab] OR low pathogenic avian influenza[tiab] OR LPAI[tiab] OR H5N1[tiab] OR H5N2[tiab] OR H5N6[tiab] OR H5N8[tiab] OR H9N2[tiab] OR H7N9[tiab] OR H5Nx[tiab]) AND (Nigeria[Mesh] OR Nigeria[tiab] OR Nigerian[tiab]) AND ("2006/01/01"[dp] : "2026/09/30"[dp]). The full strategies as run for every source, with dates, filters, caps and record counts, are given in Appendix A. The original strategies were not peer reviewed with the PRESS checklist; the October 2026 rebuilt strings were self-assessed against PRESS and the assessment is reported in Appendix A, but no independent PRESS reviewer was available.

Selection process

Records were de-duplicated by DOI, PubMed identifier (PMID) or normalised title plus year; a later check with normalised DOIs found two further duplicate pairs. Records were screened in two stages, title/metadata and then full text, by two large language model agents working independently and blind to each other's decisions, using a piloted screening form. From the amendment of 19 September 2026 until it was replaced on 27 September 2026, Reviewer 2 was an independent Codex agent (named Epicurus in the working log), and the main Codex session acted as arbiter. [Author to confirm: identity of Reviewer 1 before 27 September 2026] From 27 September 2026, Reviewer 1 was an executor agent (Anthropic Claude Opus 5.5) and Reviewer 2 a separate reviewer agent (Claude Sonnet 5, and Claude Sonnet 5.5 from 29 September 2026); disagreements were adjudicated by a third AI agent (the coordinating session) against the protocol criteria. At the title stage a record retained by either reviewer proceeded to full text. Each full-text exclusion was assigned one reason from a fixed hierarchy, with page-level evidence and the source of the full text recorded.

Agreement between the two AI reviewers was 97.0% at the title/metadata stage ($\kappa = 0.88$; 1,877 records) and 86.6% at full text ($\kappa = 0.76$; 216 reports), or 95.3% ($\kappa = 0.90$) where both reviewers gave a definitive decision. The full-text figures were calculated on the 216 reports assessed when the full-text stage first closed; three further database reports assessed on 29 September 2026 (two sent to full text during author verification and one advanced through its duplicate record in the update search) are not included. For the Google Scholar update search, agreement was 94.3% ($\kappa = 0.77$; 228 records) at the title stage and 19 of 20 reports at full text; for the October 2026 update search, agreement was 92.3% ($\kappa = 0.85$; 52 records) at the title stage, and the 4 retained reports were abstract-only. These figures describe agreement between AI agents, not between humans.

Full texts were sought through identifier resolution and legal open-access routes (Crossref, OpenAlex, Europe PMC, PubMed, Semantic Scholar, Unpaywall, institutional repositories and Google Scholar version clusters). No CAPTCHA was bypassed, no login or shadow library was used, and institutional access was not available to the review team. Reports for which no verified full text was obtained were recorded as not retrieved and were not assessed for eligibility; this replaced the original rule of two documented attempts plus author contact (dated amendments, Appendix B).

AI screening decisions were then verified by the author (M.O.). The verification set comprised 429 decisions: all full-text inclusions and exclusions, all duplicate-record removals, a seeded random 10% sample of title-stage exclusions, and all title-stage adjudication overrides. Of these, 107 were designated priority decisions (reviewer disagreements, adjudication overrides, borderline cases, duplicate removals and the most recent screening batches). For each priority decision, the author reviewed a summary of the record, the **AI** decision and the extracted evidence, and then gave a decision. For accessibility reasons, these summaries were prepared by an **AI** assistant, and the author's decisions were given in writing and transcribed by the assistant. The author changed six of the 107 priority decisions: four full-text exclusions became inclusions, and two title-stage exclusions were sent to full-text assessment. The remaining 322 lower-priority decisions were confirmed in a single statement rather than individually. During report-to-study linkage, a further report included by the **AI** reviewers was found to be a multi-country sequence compilation whose only Nigerian datum was already reported in included studies; the author excluded it. During data extraction, two further included reports were found to contain no Nigeria-specific data (passing narrative mentions only); the author excluded both. The author later re-checked all 107 priority decisions a second time, entering each answer directly in a one-row-per-screen tool that showed the current decision and the reviewers' evidence; all 107 were reconfirmed without change. This second pass is reported as a reconfirmation, not as an independent re-review.

Data collection process and study linkage

Reports were linked into studies before extraction. A study was defined as a set of reports sharing samples, animals or outbreaks; separate studies that reused the same data were flagged as overlapping so that those data could be counted once in the synthesis. Links were proposed by one **AI** agent from the full texts, checked against the texts by a second, and adjudicated by the coordinating session. Data were extracted with a form piloted by two extractors working blind on the same 10 studies, which led to a revised version (adding, for example, per-report access level, a findings-row rule and categories for inherited reassortants and hypothesised introductions). Each study was then extracted independently and blind by two **AI** agents (Reviewer 1, Claude Opus 5.5; Reviewer 2, Claude Sonnet 5.5), with every value tied to a verbatim locator in the source text; disagreements were adjudicated by the coordinating session against the texts. Authors were not contacted for missing data.

Data items

The form captured study design and data source; setting (states, geopolitical zones, sites, host systems and species); study period; sampling strategy, unit and denominators; laboratory methods (serology, antigen detection, RT-PCR, virus isolation, pathotyping, sequencing platform and segments, and phylogenetic method, model and support); one findings row per evidence type, subtype, host system, state and year, with evidence type, subtype, pathotype and its basis, clade or lineage and the nomenclature system, date, number positive, number tested, number characterised and number of outbreaks; authors' "first" claims; accession availability, reassortment (claimed, demonstrated or inherited) and introduction

(phylogeographic, clustering only or hypothesised); risk factors and spatial results; overlaps with other reports; and internal inconsistencies. Host systems were coded as commercial poultry (C), backyard or village poultry (B), live-bird markets (L), wild migratory birds (Wm), wild resident birds (Wr), captive birds (Cap) and other (O). Values not reported were recorded as not reported, and nothing was imputed. The funding sources of the included studies were not extracted.

Risk of bias assessment

Each study was appraised with the JBI checklist matching its design [24,25]: the JBI Prevalence checklist (55 studies, including reviews of official records), the JBI Analytical Cross-sectional checklist (5 studies) and the JBI Case Series checklist (30 studies). The Case Series checklist was added by a dated minor amendment (3 October 2026), because outbreak investigations and small isolate series fitted neither the Prevalence nor the Cohort tool listed in the protocol. Twenty secondary or modelling studies received no JBI checklist but a structured note on the limitations of their data sources. Studies with molecular data (37) were also assessed with a seven-item molecular transparency check covering sampling (M1), assay transparency (M2), sequencing transparency (M3), accession availability (M4), phylogenetic method and support (M5), clade nomenclature (M6) and reassortment discipline (M7). A "No" on M4 or M5 capped the study's molecular claims at "suggestive". Each item was answered Yes, No, Unclear or N/A with a verbatim quote, and an overall judgement of Include, Exclude or Seek further information was given; scores were never summed. Two **AI** appraisers worked independently and blind, and the coordinating session adjudicated against the texts using written conventions. Item-level agreement before adjudication was 868 of 1,094 items (79.3%). Because authors were not contacted, "Seek further information" was treated as inclusion with concerns. By author decision, a study was classed as high risk if the author designated it, if it had a "No" on the item for a valid identification method, or if only its abstract could be appraised.

Synthesis methods

Meta-analysis was not attempted, because study designs, units (birds, flocks, outbreaks and sequences) and assays were too heterogeneous to combine. Synthesis followed SWiM [23], with five groupings matched to the review question: S1, temporal (first detections and epidemic waves); S2, geographic (the 36 states and the Federal Capital Territory [FCT], in six zones); S3, host range (host systems and species); S4, molecular (subtype, then clade or sublineage, then reassortment, then introduction); and S5, exposure (serology, and antigen or RNA surveys in apparently healthy birds), kept separate because antibody shows exposure rather than infection. Epochs were 2006–2010, 2011–2014, 2015–2020 and 2021–2026.

The standardised metrics were: the date (year and month) of first laboratory-confirmed detection by subtype, clade, host system and state; occurrence counts taken from one designated primary dataset per period; the number of states with laboratory-confirmed detection per epoch; the host systems and species with confirmed detection, and separately with antibody evidence only; clade assignments, with reassortment classed as shown (segment incongruence), inherited or claimed only, and introductions as phylogeographic, clustering only or hypothesised; and, for S5, the proportion positive (n/N) with an exact (Clopper–Pearson) 95% confidence interval calculated from the reported counts. Proportions were reported as ranges and never pooled, and no vote counting on statistical significance was used.

"Laboratory-confirmed" meant a sequence, isolate, RNA detection or laboratory-confirmed case with the method not reported. Official reports that stated no assay were shown separately and were not counted. Laboratory evidence was prioritised in the order sequence, isolate, RNA, laboratory-confirmed case and

official report, and molecular claims from studies not capped as "suggestive" were prioritised. Each data stream reused across reports was counted once, under one primary study; the other studies were cited as also reporting it (24 streams; Appendix F). Results were tabulated by epoch, then state or zone, then risk of bias, with low-concern studies listed first, and every table entry was traced to an extracted findings row. Tables and confidence intervals were produced with Python scripts that asserted each of these rules.

Heterogeneity and sensitivity analyses

Heterogeneity was explored narratively within each grouping, against five sources: surveillance intensity (passive versus active surveillance, and submission bias by distance from the national reference laboratory at Vom) as against true emergence; assay class and cut-off; host system; official records as against primary data; and multi-country studies, of which only the Nigerian component was used. Two sensitivity analyses were planned before synthesis: (i) exclusion of the 11 high-risk studies; and (ii) exclusion of grey literature (theses, preprints, a public lecture, a seminar paper and a conference paper), applied report by report so that a mixed study kept its peer-reviewed findings. A third, (iii), as (ii) with four reports of uncertain peer-review status also treated as grey, was added during synthesis, before certainty rating, and a fourth, (iv), restricting state counts to states with sequence or isolate evidence, was added post hoc after certainty rating. Each table was rebuilt on the filtered data and compared with the main analysis.

Reporting bias assessment

No statistical test for small-study effects was possible without a meta-analysis. The risk of bias due to missing results was judged narratively, as the publication-bias domain of the certainty assessment, with reference to the reports not retrieved, the representation of grey literature and of small surveys in Nigerian journals, and the fact that removal-only sensitivity analyses cannot test claims of absence.

Certainty assessment

Certainty was rated for each synthesis claim using GRADE logic for prognosis [26]. Two **AI** raters (Reviewer 1 and Reviewer 2) judged certainty independently, and the coordinating session resolved disagreements against the tables. By convention, observational evidence started at low certainty; a claim stayed at low certainty when no domain (risk of bias, inconsistency, indirectness, imprecision and publication bias) raised a serious concern, and was rated very low when at least one did. A claim resting on a single study was downgraded for imprecision, and nothing was rated up. Where the evidence was too sparse to rate, confidence was described as not gradable. Wording in the Results and Discussion follows these ratings: low certainty supports "probably" or "the evidence indicates", and very low certainty supports "may" or "is uncertain".

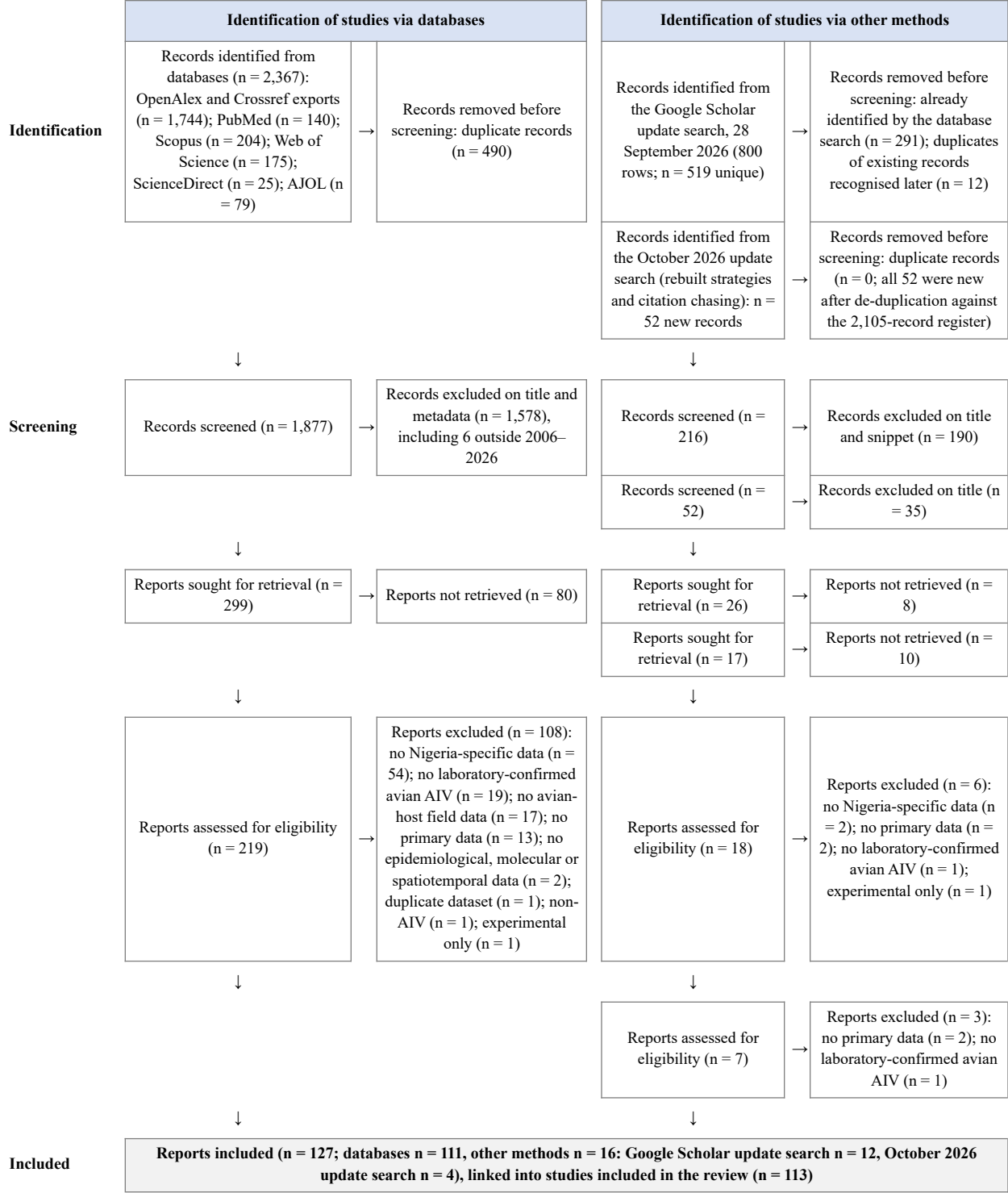
Results

Study selection

The database searches identified 2,367 records: 1,744 from the OpenAlex and Crossref exports, 140 from PubMed, 204 from Scopus, 175 from Web of Science, 25 from ScienceDirect and 79 unique AJOL article pages (from 144 query rows). After 490 duplicates were removed, 1,877 records were screened and 1,578 were excluded on title and metadata, including 6 outside the 2006–2026 window. Of 299 reports sought, 80 were not retrieved and 219 were assessed in full text; 108 were excluded, most often for lacking Nigeria-specific data (54) (Figure 1). The Google Scholar update search returned 800 rows (519 unique),

of which 291 were already in the register and 228 were new; 12 of these were later found to duplicate existing records, leaving 216 screened. After 190 title/snippet exclusions, 26 reports were sought, 8 were not retrieved, 18 were assessed and 6 were excluded. In total, 127 reports were included (111 from the database searches, 12 from the Google Scholar update search and 4 from the October 2026 update search), which were linked into 113 studies; eleven studies had more than one included report. The 117 full-text exclusions are listed with reasons in Appendix D, and the 98 reports not retrieved are characterised in Appendix B.

Figure 1. PRISMA 2020 flow diagram of the identification, screening and inclusion of reports and studies.



Study characteristics

The 110 included studies were reported between 2006 and 2026 (Appendix C). By the year of their first report, 28 studies were published in 2006–2010, 26 in 2011–2015, 30 in 2016–2020, 26 in 2021–2025 and 3 in 2026. Forty-one studies had a study period starting in 2006. By design, 56 studies were prevalence or surveillance studies (including reviews of official outbreak records), 31 were outbreak investigations or isolate series, 6 were analytical cross-sectional studies and 20 were secondary analyses of public sequences or official databases, or modelling studies. Thirty-seven studies reported molecular data, and 33 reported sequences. Forty-six studies named a single state as their whole setting and 25 named no state (national or country-level analyses, or state not reported); all six geopolitical zones were represented, with the north-west, north-central and south-west zones studied most often. Ninety-nine studies were peer-reviewed, five were grey literature only (a doctoral thesis, a conference paper, a public lecture, a preprint and a repository record), five combined peer-reviewed and grey reports, and four had a report of uncertain peer-review status. Seven studies could be read and appraised only as abstracts. Eighty-five studies contributed to the first-detection chronology, 81 to the state $\times$ epoch matrix, 90 to the host-range table, 35 to the clade table, 37 to the molecular methods audit, 35 to the exposure tables and 31 to the occurrence series. Four studies (Fasanmi et al., Cecchi et al., Si et al. and one repository record) contributed no synthesisable Nigerian data [27–30], and one contributed only an authors' claim [31].

Risk of bias in studies

All 113 studies were retained after appraisal (Appendix E). Overall judgements were Include for 104 studies, Include (high risk, by author decision) for 2 [27,32] and Seek further information for 7 that could be appraised only from their abstracts [16,30,33–37]. Fourteen studies met the high-risk rule [16,27,30,32–43]. Common concerns in the prevalence studies were convenience sampling, sample sizes without justification and cut-offs or laboratories that were not reported; in the case series, consecutive and complete inclusion of cases was rarely reported. Thirteen molecular studies were capped as "suggestive" because accession numbers or the phylogenetic method and support were not reported [44–57]; their molecular claims are reported below as suggestions only. By synthesis grouping, the contributing studies included the following numbers of high-risk, capped and abstract-only studies: S1, temporal (90 studies), 9, 10 and 4; S2, geographic (81), 7, 7 and 3; S3, host range (90), 10, 8 and 5; S4, molecular (T4 and T5; 38), 2, 13 and 2; and S5, exposure (35), 3, 1 and 2.

S1. Temporal: epidemic waves and first detections

First detections. The earliest dated H5N1 evidence was from January 2006: the Kaduna index isolate and genome [58] and farm cases in Kaduna, Plateau and Kano in the same month [59]. January 2006 is the date the outbreak was reported; laboratory confirmation was dated 6–7 February 2006 (by the reference laboratory in Padova) and official notification 8 February, according to background statements in two studies [38,39,60]. One capped study suggests that its Ibadan isolates pre-dated the official report, including a strain named 228-5/2005, but its row is dated January 2006 and no 2005 detection is documented [53]. Several authors claimed Nigeria's outbreak as the first in Africa, dating it to January or 7–8 February 2006 [47,61,62]. Other subtypes were first detected later (Table 1): H5N2 in wild birds in February 2007 [63], H5N8 in November 2016 [64], H5N6 in June 2019 and H9N2 in 2019, both from the 2019 national live-bird-market surveillance [65–67]. H9 antibody was present from 2013 [68]. Authors' "first" claims are listed in Appendix F with the earliest evidence alongside.

Table 1. Earliest dated evidence by subtype, clade or sublineage, and host system (condensed; the full chronology, including states, is in Appendix F).

Category	Key	Earliest dated evidence (outbreak-report or sampling date)	Evidence at earliest date	References (earliest rows)
Subtype	H5N1	2006-01	sequence	[58,59]
	H5 (N not determined)	2006 (earliest month-dated row 2006-12)	isolate	[49,69]
	H5N2	2007-02 (wild birds)	sequence	[63]
	H5N8	2016-11	sequence	[64]
	H5N6	2019-06	isolate	[65–67]
	H9N2	2019	sequence	[65–67]
	H9 (N not determined)	2024-10 to 2025-07 (window)	RNA	[70]
Clade or sublineage	2.2	2006 (earliest month-dated row 2006-03)	sequence	[71–73]
	"Qinghai lineage" (pre-WHO name)	2006-01	sequence	[58]
	2.2.2 (suggested; capped study)	2006-01	sequence	[53]
	EMA-1/EMA-2 (within 2.2)	2006	sequence	[73]
	Sublineage III/EMA-3 (within 2.2)	2008-05 to 2008-07	isolate	[74,75]
	2.3.2.1c	2015-01	sequence	[76,77]
	2.3.2.1c WA1; WA2	2015-01; 2015	sequence	[77–79]
	2.3.4.4 group B (as labelled)	2016-11	sequence	[64]
	2.3.4.4 (stated)	2019-02	isolate	[57]
	2.3.4.4b	2019-06 (H5N6)	isolate	[65–67]
	H9 G1 lineage	2019	sequence	[65–67]
	H5N1/H9N2 reassortant (H5N2)	2021	sequence	[11]
	2.3.4.4b genotype EA-2020-C	2023 (sampled 2023; introduced January 2021 per [19])	sequence	[19]
	2.3.4.4b genotype EA-2024-DV	2024	sequence	[19]
	Commercial (C)	2006-01	sequence	[58,59]
	Backyard/village (B)	2006 (earliest month-dated row 2006-02)	sequence	[80]
Host system	Live-bird market (L)	2006	sequence	[52,71,72]
	Wild resident (Wr)	2006-03 (species conflict in source)	sequence	[71,72]
	Captive (Cap)	2006-04	sequence	[71,72]
	Other (O)	2021 (earliest month-dated row 2021-02)	RNA	[81,82]
	Wild migratory (Wm)	2022-03 to 2022-04	sequence	[12]

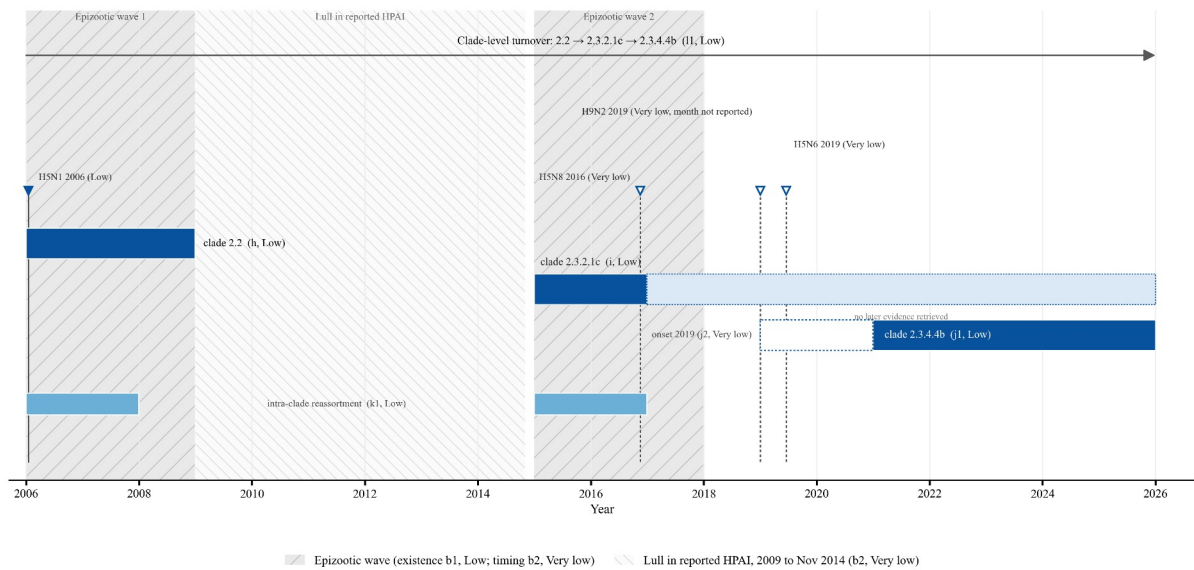
Dates are year-month of the earliest laboratory-confirmed row (sequence, isolate, RNA or laboratory-confirmed case); official reports without a stated assay are not counted. 2006-01 is the date the H5N1 outbreak was reported; laboratory confirmation was dated 6–7 February 2006 according to background statements [38,39,60]. Clade labels are as used by the studies; "2.3.4.4 group B" was mapped to 2.3.4.4b by this review.

Epidemic waves. Counted once from the designated primary dataset for each period, there were 145 positive diagnostic samples in 2006 and 154 in 2007 [52], 2 laboratory-confirmed cases in 2008 and none in 2009–2010 [60]. The re-emergence began with an index outbreak at the Sabon Gari live-bird market in Kano on 24 December 2014 [83], followed by 513, 248 and 50 confirmed cases in 2015, 2016 and 2017

[84]. No national dataset covered 2011–2013 or 2018–2020, for which only state-level evidence existed [57,65–67,85]. National records gave 454 H5N1, 12 H5N8 and 1 H5N2 outbreaks in 2021–2022 [11], and 28 outbreaks were reported in January–April 2024 [19]. The units changed from samples to cases and outbreaks, so the figures were not summed, and series for the same year disagreed: for 2015, other series gave 542 [44,45] and 265 [86]. The evidence indicates two major epizootics, in 2006–2008 and 2015–2017 (low certainty; Figure 3), which rests on order-of-magnitude contrasts between hundreds of confirmations per year and 0–2 in 2008–2010, because the exact counts were internally inconsistent in both principal series [52,84]. No HPAI outbreak was reported between 2009 and November 2014 in the retrieved sources, but laboratory evidence from these years exists: H5 RNA in live-bird-market ducks in Oyo in 2011–2012 (20/155), with three H5N2 isolates and an LPAI cleavage site in the two sequenced [87], and antibody-positive surveys (S5). Whether this was a true lull in virus circulation is uncertain (very low certainty), because the absence of reports is not the absence of virus and evidence for these years is sparse.

Heterogeneity. The early counts came from passive submissions to the National Veterinary Research Institute (NVRI) in 2006–2007 [52], with national live-bird-market surveillance in 2007–2008 [41,52] and market sampling in 2008 [74,75]; the zero years came from passive reports [60] or a global outbreak database [88]. Counts for 2015–2017 and 2021–2022 were official records, whereas those for 2006–2007 were laboratory samples, and many 2006 confirmations stated no assay. Three multi-country studies contributed their Nigerian strains only [64–67,77]. The first-detection dates of H5N8, H5N6 and H9N2 may partly reflect when testing for non-H5 subtypes began, which is documented in virus surveys only from 2019 [65–67]; type-A RNA surveys in 2012–2013 were negative (0/300 and 0/461) [68,89]. These dates are therefore uncertain (very low certainty).

Figure 3. Detection timeline and clade turnover of avian influenza virus in Nigeria, 2006–2026. Hatched bands show the two epizootic waves (2006–2008 and 2015–2017; existence low certainty) and the lull in reported HPAI from 2009 to November 2014 (very low certainty). Horizontal bars show successive clades: 2.2 (2006–2008), 2.3.2.1c from 2015, and 2.3.4.4b (2021–2026); the 2.3.2.1c bar is solid only over its evidenced interval (2015 to 2016) and continues as a lighter dashed band, because no later evidence was retrieved. Triangles mark the first detections of H5N1 (2006, low certainty), H5N8 (2016), H5N6 (2019) and H9N2 (2019, month not reported); the non-H5N1 first detections are of very low certainty.



S2. Geographic distribution

Laboratory-confirmed detections were recorded in 26 states and the FCT in 2006–2010, 2 states in 2011–2014 (Kano, from the December 2014 index event, and Oyo, from LPAI H5N2), 21 states and the FCT in 2015–2020, and 14 states and the FCT in 2021–2026 (Table 2 and Figure 2). Sequence or isolate evidence covered 17 states and the FCT, 2 states, 19 states and the FCT, and 10 states and the FCT in the same epochs. With any laboratory evidence, all six zones were covered in 2006–2010, 2015–2020 and 2021–2026. The evidence indicates that the virus spread widely in 2006–2010 (low certainty), consistent with the official national figure of 25 states and the FCT in 2006–2007 [52]. The later counts are floors that reflect where sequencing and reporting took place, not a contraction: official figures were 21, 27 and 26 states in 2015, 2016 and 2017 [84] and 31 of 37 administrative regions in 2021–2022 [11]. The extent of spread in the later epochs is therefore uncertain (very low certainty).

Table 2. States with avian influenza evidence by epoch (compact state × epoch matrix; the full matrix with subtypes, host systems and contributing references is in Appendix F).

Zone	State	2006–2010	2011–2014	2015–2020	2021–2026	Undated or multi-epoch
NC	Benue	S	–	–	–	–
	FCT	S	–	S	S	–
	Kogi	†	†	–	–	§ †

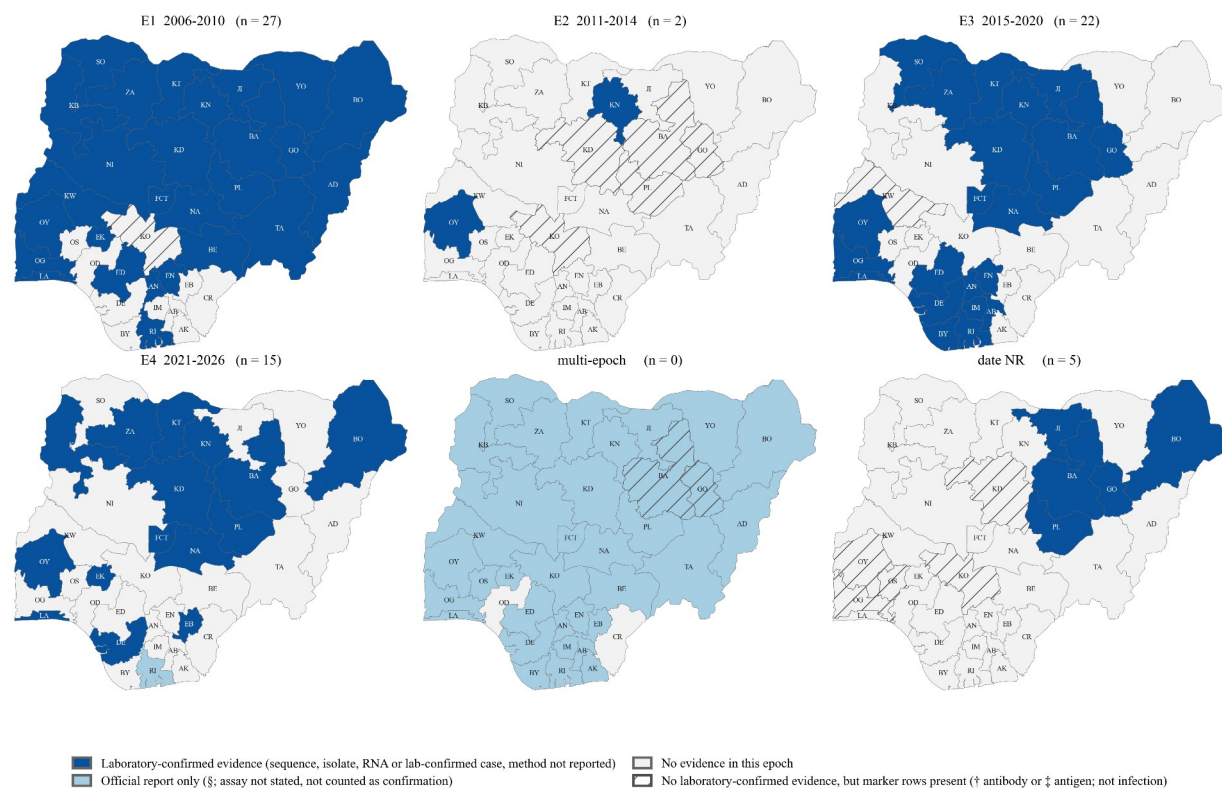
Zone	State	2006–2010	2011–2014	2015–2020	2021–2026	Undated or multi-epoch
	Kwara	L	–	†	–	–
	Nasarawa	L	–	S	S	–
	Niger	L	–	–	–	–
	Plateau	S	†	S	S	S
NE	Adamawa	L	–	–	–	–
	Bauchi	S	†	S	S	† L
	Borno	S	–	–	L	L
	Gombe	S	†	L	–	† L
	Taraba	S	–	–	–	–
	Yobe	L	–	–	–	–
NW	Jigawa	S	–	S	–	S
	Kaduna	S	†	S	S	†
	Kano	S	S	S	S	–
	Katsina	S	–	S	S	–
	Kebbi	S	–	–	S	–
	Sokoto	S	–	S	–	–
	Zamfara	L	–	S	S	–
SE	Abia	–	–	S	–	–
	Anambra	S	–	L	–	–
	Ebonyi	–	–	–	L	–
	Enugu	L	–	S	–	–
	Imo	–	–	S	–	–
SS	Akwa Ibom	–	–	–	–	§
	Bayelsa	–	–	S	–	–
	Cross River	–	–	–	–	–
	Delta	–	–	S	L	–
	Edo	L	–	S	–	–
	Rivers	L	–	S	§	–
SW	Ekiti	S	–	–	S	–
	Lagos	S	–	S	S	–
	Ogun	S	–	S	–	†
	Ondo	–	–	–	–	–
	Osun	–	–	–	–	§ †
	Oyo	S	S	S	L	†
All	Any laboratory evidence	26 states + FCT	2 states	21 states + FCT	14 states + FCT	
	Sequence or isolate	17 states + FCT	2 states	19 states + FCT	10 states + FCT	

S, sequence or isolate evidence; L, laboratory evidence by RNA or laboratory-confirmed case only; § official report only (no assay stated; not counted); † antibody only (exposure); ‡ antigen only; –, no evidence. The FCT is counted separately from the 36 states. All states except Cross River and Ondo appear in an undated WAHIS compilation for 2006–2023 [90]; in the last column § is shown only for states with no laboratory evidence. Positives assignable only to a group of states (Lagos, Ogun or Oyo; Jigawa or Yobe; Kaduna or Kano) are not placed [12,49,91]. Later counts are floors set by sequencing and reporting coverage.

The earliest state evidence was dated January 2006 for Kaduna [58], Kano and Plateau [59] and, from a capped study, Oyo [53]; several other states were dated only by 2006–2008 series windows [42,92–95].

Some positives could not be assigned to one state: they were in Lagos, Ogun or Oyo [49], Jigawa or Yobe [12], and Kaduna or Kano [91]. One study claimed Gombe's first H5N1 in 2015 [96,97], whereas the earliest Gombe evidence in this review is from May–July 2008 [74,75]. Only Cross River and Ondo had neither laboratory nor official evidence; Akwa Ibom, Kogi and Osun had official reports only [90]. Negative testing was sparse: 11 states sampled in 2008 [74,75], 0 of 110 turkeys antibody-positive in Ondo [98] and RNA-negative birds in Kogi [99]. Whether these states were truly free of infection is uncertain (very low certainty). The 2006–2008 data came largely from NVRI submissions [42,52,92–95,100], and live-bird-market surveillance added detections outside outbreak reporting, in Gombe in 2008 [74,75] and in Sokoto and Ogun in 2019 [65–67].

Figure 2. Laboratory-confirmed avian influenza virus evidence in Nigeria by state and epoch, 2006–2026. Four small multiples show E1 (2006–2010), E2 (2011–2014), E3 (2015–2020) and E4 (2021–2026), with two further panels for rows that span epochs ("multi-epoch") and rows with no stated date ("date NR"). Dark blue denotes laboratory-confirmed evidence (sequence, isolate, RNA, or a lab-confirmed case with the method not reported); light blue denotes an official report only (assay not stated), which is not counted as confirmation; pale grey denotes no evidence in that epoch; hatching marks cells with antibody (†) or antigen (‡) marker rows but no laboratory-confirmed evidence. Two-letter codes identify states; the Federal Capital Territory is shown separately. Counts are named-state floors that record where sequencing and reports exist rather than the true extent of infection.



S3. Host range

Laboratory-confirmed detection was documented in all three domestic systems from 2006 (Table 3). In commercial poultry the earliest detection was in January 2006 [58,59], with H5N8 from February 2019 [57], H5N2 in 2021 [11] and H9N2 in a 2020–2021 window [91]. In backyard and village poultry the earliest month-dated detection was February 2006 [80], in chickens, ducks, geese, guinea fowl, pigeons and turkeys. In live-bird markets detection dated from 2006 [52,71,72], with LPAI H5N2 in 2012 [87] and H5N6, H5N8 and H9N2 in 2019 [65–67]. The evidence indicates confirmed infection in commercial, backyard, live-bird-market and wild resident birds (low certainty).

Table 3. Host systems and species with confirmed detection and with antibody evidence only.

Host system	Status	Subtypes with confirmed detection (earliest)	Species stated positive	Antibody only (exposure)	References
Commercial poultry (C)	Confirmed	H5N1 (2006-01); H5N8 (2019-02); H9N2 (2020–2021 window); H5N2 (2021); H9, N not determined (2024–2025)	chicken, layer, broiler	type A, H5, H7, H9 antibody	[11,57–59,70,91]
Backyard/village (B)	Confirmed	H5N1 (2006; earliest month-dated 2006-02); H5N8 (2019-02)	chicken, duck, goose, guinea fowl, pigeon, turkey	type A, H5 antibody	[57,80]
Live-bird market (L)	Confirmed	H5N1 (2006); LPAI H5N2 (2012); H5N6 (2019-06); H5N8 (2019-07); H9N2 (2019)	chicken, duck, guinea fowl, layer, pigeon	type A, H5, H7, H9 antibody	[52,65–67,71,72,87]
Wild resident (Wr)	Confirmed	H5N1 (2006-03, species conflict); H5N2 (2007-02 HPAI genotype; 2008 LPAI); H5N1 (2022)	vulture (conflict), spur-winged goose, white-faced whistling duck, African jacana, square-tailed nightjar	dove, egret, parrot, pigeon	[12,63,71,72,101]
Wild migratory (Wm)	Confirmed (single study)	H5N1 clade 2.3.4.4b (2022-03 to 2022-04)	ruff, white stork	type A antibody	[12,99]
Wild, migrant status unresolved	Official report only (§)	none	–	dove, egret, pigeon	[89,90,96,97]
Captive (Cap)	Confirmed (grey literature)	H5N1 (2006-04, Jos Wildlife Park; 2021-08, duck, Delta)	duck	crane, duck, goose, ostrich, parrot, peafowl, stork	[13,71,72,81,82,102]
Other (O)	Confirmed (grey literature)	H5N8 RNA in an imported pheasant (2021-02); H5N1 (2021–2022)	pheasant	dove, quail, white stork	[11,43,81,82,103]

Confirmed detection means sequence, isolate, RNA or laboratory-confirmed case on a row stating which species were positive. Positivity was not established for crane, Muscovy duck, ostrich and peafowl/peacock (named only on mixed-species rows); black stork, dove, egret, mallard, quail and weaver had antibody only. Ostriches in one study were not necropsied and are excluded [92–95].

In wild resident birds, the first dated detection was a vulture isolate in March 2006, although the same strain is named as chicken in the thesis table, so this date carries a species conflict [71,72]. Spur-winged geese and white-faced whistling ducks carried H5N2 in February 2007 (an HPAI genotype) [63] and in 2008 (LPAI) [101], and H5N1-positive birds in 2022 included an African jacana and a square-tailed nightjar [12]. Infection of migratory birds was confirmed only in stated Palearctic migrants, ruff and white stork, with clade 2.3.4.4b H5N1 in March–April 2022, in a single study [12]; this finding is of very low certainty. The jacana was coded as resident in one part of that report and as an intra-African migrant in another, so it was not counted as a migrant. Captive birds were positive at Jos Wildlife Park in April 2006 [71,72] and a captive duck in Delta in August 2021 [81,82], and H5N8 RNA was found in an imported pheasant in February 2021 [81,82]. Captive and other hosts may therefore be affected (very low certainty), because each rests on single rows from grey literature. Positivity was not established for crane,

Muscovy duck, ostrich and peafowl, and black stork, dove, egret, mallard, quail and weaver showed antibody only. Poultry detections came mainly from passive submissions of sick flocks [42,80,92–95], whereas wild-bird and market detections came from active surveillance at few sites [12,63,65–67,101]. Species-level host range beyond the domestic poultry species could not be graded.

S4. Molecular: subtype, clade or sublineage, reassortment and introduction

2006–2008. H5N1 viruses of the first wave belonged to clade 2.2, shown by the studies' own sequences with phylogenetic support in six studies not capped as "suggestive" [71–75,104–106] (Table 4); the evidence indicates clade 2.2 in this wave (low certainty). The index genome was labelled "Qinghai lineage", a name that pre-dates the unified H5 nomenclature [58]. Clades 2.2.1 and 2.2.2, and the EMA-1, EMA-2 and EMA-3 and I/II-Nigeria sublineages, all lie within clade 2.2. Sublineage III (EMA-3) was first dated May–July 2008 [74,75]; a capped study claimed it was novel to Africa [52], and another capped study suggests clade 2.2.2 [53]. Reassortment was shown by segment incongruence in four studies: an EMA-1/EMA-2 reassortant with a 4:4 constellation [73], a 2:6 reassortant in 10 of 12 viruses [105], reassortants between sublineages [106] and two further events, R2 and R4 [104]. Only one study gave a phylogeographic link for introductions in this wave, estimating four (three in the winter of 2005–2006 and one in 2008) [64]. Elsewhere, clustering suggests two introductions [53,73] or three [49,107], and one study argued for continuous local circulation after 2006 [106]. The number of independent introductions within a clade is therefore uncertain (very low certainty).

Wild-bird H5N2, 2007–2008. These viruses had no WHO clade. One study reported an HPAI-genotype lineage [63], and a single study showed an LPAI inter-lineage (gene-pool) reassortant combining an African NS gene with Eurasian wild-bird segments [101].

Table 4. Clade, reassortment and introduction verdicts by period, applying the review's language rules.

Period	Subtypes	Clade or sublineage	Clade verdict	Reassortment	Introduction	References
2006–2008	H5N1	2.2 (with 2.2.1, 2.2.2, EMA-1/2/3, I/II-Nigeria, sublineage III)	Shown: own sequences with phylogenetic support in six uncapped studies; 2.2.2 and EMA-3 novelty are suggested by capped studies	Shown (segment incongruence): EMA-1/EMA-2 4:4; 2:6 in 10/12 viruses; inter-sublineage reassortants; R2 and R4	Phylogeographic link in one study (four introductions); elsewhere clustering suggests two or three; one study argues for local circulation after 2006	[49,52,53,64,71–75,104–107]
2007–2008 (wild birds)	H5N2	No WHO clade (HPAI-genotype lineage; LPAI Eurasian gene pool)	Shown (study lineages)	LPAI inter-lineage (gene-pool) reassortant shown in one study	Not shown	[63,101]
2011–2012	H5N2 (LPAI)	No WHO clade (wild-duck-associated LPAI cluster)	Shown (two sequences)	None claimed	Not shown	[87]
2014/15–2017	H5N1; H5N8	2.3.2.1c (WA1, WA2); "2.3.4.4 group B" (H5N8, 2016)	Shown in four uncapped studies; one re-analysis relabels as "2.3.2.1f"; 2.3.4.4b mapping of the 2016 H5N8 is this review's	WA1×WA2 intra-clade reassortment shown in one study; H9N2-derived PB2 inherited	Phylogeographic link in one study: clade introduced into West Africa (tMRCA May–November 2014), Nigeria the main point of introduction; WA1/WA2 clustering suggests at least two introductions	[14,16,64,76–79]

Period	Subtypes	Clade or sublineage	Clade verdict	Reassortment	Introduction	References
2018–2020	H5N6; H5N8; H9N2	2.3.4.4b (H5N6, H5N8); H9 G1 lineage; 2.3.4.4 stated (2019 farms)	Shown (2019 national market surveillance)	Inherited genotypes (shown elsewhere)	Not shown in Nigeria	[48,57,65–67]
2021–2022	H5N1; H5N8; H5N2	2.3.4.4b	Shown in uncapped studies	H5N1/H9N2 reassortant H5N2 from one farm (supplement unavailable)	Clustering suggests separate introductions; one continuous lineage not shown	[11,12,56]
2023–2025	H5N1	2.3.4.4b genotypes EA-2020-C and EA-2024-DV	Shown	One study reported EA-2024-DV as a reassortant genotype; where it arose is not shown (not counted as Nigerian reassortment)	EA-2020-C considered introduced in January 2021; the same study proposes a single EA-2024- DV introduction by an uncertain route	[19]

A reassortant requires segment incongruence shown in the study; an introduction requires a phylogeographic link; otherwise "clustering suggests". Studies capped as "suggestive" (no accessions, or no phylogenetic method and support) can only suggest. Subtype, clade or sublineage, reassortant and introduction are not interchangeable.

2014/15–2017. Clade 2.3.2.1c was first sequenced in January 2015 [76,77], with 2015–2016 genomes in a further study [78,79]; the evidence indicates clade 2.3.2.1c from 2015 (low certainty), and the high-risk abstract-only study is not needed for this conclusion [16]. Intra-clade reassortment between the West African groups WA1 and WA2 was shown in one study [78,79], and the H9N2-derived PB2 segment was inherited rather than arising in Nigeria [76,77]. A re-analysis of public sequences labelled the same strains "2.3.2.1f" [14]. The same phylogeographic study dated the introduction of clade 2.3.2.1c into West Africa to May–November 2014 (time to the most recent common ancestor), with Nigeria as the main point of introduction [64], while the co-circulation of WA1 and WA2 suggests at least two introductions [77]. H5N8 detected in November 2016 carried the study's label "2.3.4.4 group B" [64]; mapping it to 2.3.4.4b is this review's interpretation, so the onset of clade 2.3.4.4 in Nigeria, in 2016 or 2019, is uncertain (very low certainty).

2019–2025. Clade 2.3.4.4b was first dated June 2019 (H5N6), followed by H5N8 in July 2019, both inherited genotypes, and H9N2 of the G1 lineage was found in 2019 [65–67]. In 2021–2022, H5N1, H5N8 and an H5N2 virus described as an H5N1/H9N2 reassortant were reported from national surveillance, the H5N2 from one farm with the supporting supplement unavailable [11], and wild-bird H5N1 was found in 2022 [12]. Genotype EA-2020-C was sampled in 2023 and was considered introduced in January 2021 [19]. One study reported genotype EA-2024-DV from 2024 as a reassortant whose HPAI-derived segments matched European 2.3.4.4b viruses and whose internal segments (PB2, PB1 and NP) matched Siberian LPAI H3N8/H4N6 viruses, and proposed a single introduction by an uncertain route [19]. Where this reassortment arose was not shown, so it is not counted as a Nigerian reassortment, and the genotype rests on a single study (very low certainty). The evidence indicates that clade 2.3.4.4b was present in 2021–2026 (low certainty) [11,12,19]. Clustering suggests separate introductions of clade 2.3.4.4 viruses, but one continuous lineage was not shown [11].

Taken together, the evidence indicates clade-level turnover from 2.2 to 2.3.2.1c to 2.3.4.4b (low certainty), and intra-clade reassortment in 2006–2007 and 2015–2016 (low certainty) [73,78,79,104–106]. The WA1×WA2 genotype [78,79] and the inter-subtype H5N2 reassortant [11] each rest on a single study (very low certainty). Methods ranged from complete genomes with maximum-likelihood or Bayesian support to haemagglutinin-only, cleavage-site or stated clades (Appendix F); re-analyses of public

sequences were counted once, under their primary studies [14,50,51,55,64,107–110]; and sequencing gaps occurred in 2009–2014 and 2018.

S5. Exposure in apparently healthy birds

The exposure data comprised 110 survey rows, of which 72 were independent estimates (Table 5). Antibody to the nucleoprotein (type A) was reported in 30 rows, ranging from 0/195 in wild resident birds [111] to 1/1 in a single dove [103]. Within systems, the range was 0/110 [98] to 119/377 [112] in commercial poultry, 0/168 to 87/184 in live-bird markets [13] and 90/450 [62] to 85/200 [113] in backyard birds; captive birds gave 5/48 [13]. In wild birds, the values were 0/195 [111], 2/44 [99], 4/90 [89] and 5/146 (wild resident with market birds) [114]. H5 antibody by haemagglutination inhibition (HI) ranged from 0/79 [43] to 48/396 [115] across 9 rows, and by ELISA from 9/136 [116] to 6/6 [102] across 4 rows, with 19/127 in wild resident birds [102]. H5 antibody in poultry can reflect vaccination, and authors raised the possibility of unapproved vaccine use [112,117,118]. All H7 and H9 results were follow-up tests on screen-positive sera, so none was independent [13,68,119–121]. H9 exposure is partly corroborated by H9N2 virus detections in 2019–2025 [18,65–67,91], whereas no H7 virus was detected in any study. The authors' claims of first H7 (2018) and H9 (2020) serology are reported as claimed [119,120], but H7 and H9 reactors were present from 2013 [68].

Table 5. Exposure and virus detection in apparently healthy birds: ranges across independent survey rows, with n/N and exact 95% confidence intervals (never pooled).

Assay class	Host system	Indep. rows	Lowest n/N (95% CI)	Highest n/N (95% CI)
NP/type-A antibody (ELISA/AGID)	Backyard (B)	3	90/450 (0.164–0.240) [62]	85/200 (0.356–0.497) [113]
	Commercial (C)	10	0/110 (0.000–0.033) [98]	119/377 (0.269–0.365) [112]
	Captive (Cap)	1	5/48 (0.035–0.227) [13]	–
	Live-bird market (L)	10	0/168 (0.000–0.022) [13]	87/184 (0.399–0.548) [13]
	Market and wild resident	1	5/146 (0.011–0.078) [114]	–
	Other (O) (small N)	2	4/20 (0.057–0.437) [89]	1/1 (0.025–1.000) [103]
	Wild, status unresolved	1	4/90 (0.012–0.110) [89]	–
	Wild migratory and resident	1	2/44 (0.006–0.155) [99]	–
	Wild resident (Wr)	1	0/195 (0.000–0.019) [111]	–
H5 antibody (HI)	Backyard (B)	2	14/480 (0.016–0.049) [122,123]	23/547 (0.027–0.062) [124]
	Commercial and backyard	1	22/450 (0.031–0.073) [96,97]	–
	Commercial, backyard and market	1	6/200 (0.011–0.064) [125]	–
	Commercial, backyard and wild resident	1	16/500 (0.018–0.051) [96,97]	–
	Live-bird market (L)	2	10/232 (0.021–0.078) [124]	48/396 (0.091–0.158) [115]
	Other (O)	2	0/79 (0.000–0.046) [43]	3/203 (0.003–0.043) [43]
H5 antibody (ELISA)	Commercial (C) (small N)	1	6/6 (0.541–1.000) [102]	–
	Market and captive	1	8/72 (0.049–0.207) [102]	–
	Market and wild resident	1	9/136 (0.031–0.122) [116]	–
	Wild resident (Wr)	1	19/127 (0.092–0.224) [102]	–
antibody, AGID H5 + HI H5/H7/H9 panel (reactivity NR)	Live-bird market (L)	1	6/4,228 (0.0005–0.003) [52]	–
	Other (O)	1	0/3,100 (0.000–0.001) [52]	–

Assay class	Host system	Indep. rows	Lowest n/N (95% CI)	Highest n/N (95% CI)
antigen (ELISA/rapid test)	Commercial, backyard and market	1	0/400 (0.000–0.009) [125]	–
	Wild resident (Wr)	1	4/204 (0.005–0.049) [111]	–
RNA (RT-PCR)	Backyard (B)	4	0/373 (0.000–0.010) [101]	17/28 (0.406–0.785) [126]
	Backyard and market	1	0/710 (0.000–0.005) [99]	–
	Live-bird market (L)	4	3/8,753 (0.0001–0.001) [52]	7/30 (0.099–0.423) [127]
	Market, wild and other	1	0/300 (0.000–0.012) [89]	–
	Market and wild resident	1	1/26 (0.001–0.196) [116]	–
	Other (O)	1	0/7,861 (0.000–0.0005) [52]	–
	Wild migratory and resident	4	0/154 (0.000–0.024) [101]	25/226 (0.073–0.159) [12]
	Wild resident (Wr)	1	3/651 (0.001–0.013) [101]	–
virus isolation	Live-bird market (L)	4	0/400 (0.000–0.009) [35]	2/43 (0.006–0.158) [74,75]

Proportions are n/N with exact (Clopper–Pearson) 95% confidence intervals computed from the reported counts; ranges span independent rows and are never pooled. Antibody shows exposure only, and H5 antibody in poultry may reflect vaccination. H7 and H9 antibody results were follow-up tests on screen-positive sera and are not independent, so they are not shown (Appendix F). NP, nucleoprotein; AGID, agar gel immunodiffusion; HI, haemagglutination inhibition.

RNA in apparently healthy birds was reported in 17 rows, from 0/300 [89] to 17/28 type-A RNA in backyard ducks in Borno [126]; in live-bird markets the range was 3/8,753 [52] to 7/30 [127], and in wild birds up to 25/226 for H5N1 [12]. Virus isolation ranged from 0/400 [35] to 2/43 [74,75]. The assays (nucleoprotein ELISA or agar gel immunodiffusion, HI and H5 ELISA kits) differed in target and cut-off, samples were mostly convenience samples, and some denominators were very small. All antibody-based estimates of exposure are therefore uncertain (very low certainty), and H7 exposure is less certain than H9 exposure.

Sensitivity analyses

No synthesis conclusion changed in any sensitivity analysis (details in Appendix F). Removing the 14 high-risk studies (i) removed Niger from the 2006–2010 state list, because it rested on one study [42]; reduced the no-stated-date group from five states to four (Plateau lost); and removed two host-by-assay groups from the exposure tables [37,43]; no first-detection date and no clade verdict changed. Restricting the analysis to peer-reviewed reports (ii) removed Delta from 2021–2026 and the 2006 vulture, so the earliest wild-resident date moved from March 2006 to February 2007, and the earliest captive-bird date moved to a 2015–2017 window; the Delta, captive, "other" and pheasant changes depended on excluding the preprint of one study, whose published version, read as an abstract only, states the same events [81,82]. Treating the four reports of uncertain status as grey as well (iii) moved the earliest dates for Kano to December 2006 and for Plateau to 2006, but left the national January 2006 date unchanged. The post hoc restriction to states with sequence or isolate evidence (iv) gave 17 states and the FCT, 2 states, 19 states and the FCT, and 10 states and the FCT across the four epochs; the south-south zone was lost in 2006–2010, leaving Anambra as the only south-eastern state, and both southern zones lost representation in 2021–2026, but the conclusion of wide spread in 2006–2010 held. Removal-only analyses could not test claims of absence, such as the 2009–2014 lull or the states without evidence.

Reporting biases

Small serological surveys in Nigerian journals were well represented (about 13 of the 35 studies contributing independent exposure estimates), and grey literature was included. However, 98 reports were not retrieved, of which about 40 appeared from their titles to be Nigerian avian laboratory or

epidemiological reports (Appendix B), and one of these was a molecular study of clade 2.3.4.4b introductions [17]. Missing results were therefore a concern for the claims of absence (the inter-wave lull and the states without evidence) and could not be excluded for the exposure estimates.

Certainty of evidence

Nine claims were rated low certainty and fourteen very low certainty; species-level host range was not gradable (Table 6). Claims of low certainty concerned the timing of the 2006 emergence and the two major epizootics, the wide spread in 2006–2010, infection of the four main host systems, the three successive clades and intra-clade reassortment. Claims of very low certainty concerned wave boundaries and sizes, the first-detection dates of the non-H5N1 subtypes, the later state counts, states without evidence, captive and other hosts, migratory birds, the onset of clade 2.3.4.4, single-study reassortant genotypes, the number of introductions and all antibody-based exposure.

Table 6. Summary of findings and certainty of evidence (GRADE-prognosis logic; observational evidence starts at low certainty and is rated very low when any domain raises a serious concern).

Claim	Certainty	Main reasons	Publication bias	Contributing studies
S1 Temporal				
(a) HPAI H5N1 was confirmed in Nigeria in early 2006 (outbreak reported January; laboratory confirmation 6–7 February 2006)	Low	Not downgraded: anchor study not high risk or capped; report dates, confirmation dates and claims agree; several independent studies; unchanged in sensitivity runs	Not downgraded: unretrieved reports could only make the date earlier, and none is known to	[38,39,47,53,58–62]
(b1) Two major epizootics, 2006–2008 and 2015–2017	Low	Not downgraded: order-of-magnitude contrasts robust to internally inconsistent counts; all series show the same peaks	Not downgraded: peaks appear in laboratory and official series	[52,60,84]
(b2) Wave boundaries and the 2009 to November 2014 lull in reported HPAI	Very low	Indirectness (absence of reports is not absence of virus; H5 RNA and antibody found); publication bias (sparse sources; 88 reports not retrieved)	Downgraded: sparse sources; 88 reports not retrieved	[60,87,88]
(b3) Wave sizes (occurrence counts)	Very low	Inconsistency (513, 542 and 265 for 2015); risk of bias/indirectness (no assay stated; units change)	Not assessed separately; rating already Very low	[11,44,45,52,84,86]
(c) H5N8, H5N6 and H9N2 first detected in 2016-11, 2019-06 and 2019	Very low	Imprecision (single detections); indirectness (non-H5 testing documented only from 2019)	Concern, not scored separately: missing reports could hold earlier detections	[64–68,89]
S2 Geographic				
(d1) Wide spread in 2006–2010 (26 states + FCT; 17 + FCT with sequence or isolate)	Low	Not downgraded: mostly non-high-risk sequence or isolate evidence; consistent with official 25 states + FCT; robust in runs (i)–(iv)	Not downgraded: the official figures agree (25 states + FCT in 2006–07)	[42,52,71,72,92–95,107]
(d2) Extent in later epochs (21 + FCT; 14 + FCT) as a measure of spread	Very low	Indirectness (floors set by sequencing and reporting); inconsistency (below official figures)	Concern, not scored separately: later counts are floors; missing reports could add states	[11,84]
(e) Only Cross River and Ondo lack both laboratory and official evidence	Very low	Indirectness (absence of reports is not absence of infection); imprecision (few negative surveys)	Concern, not scored separately: 88 reports not retrieved (about 33 appeared to be Nigerian avian reports; Appendix B); removal-only runs cannot test absence	[74,75,90,98,99]
S3 Host range				
(f1) Confirmed detection in commercial, backyard, live-bird-market and wild resident birds	Low	Not downgraded: several non-high-risk studies per system; wild resident status does not depend on the 2006 culture or the jacana	Not downgraded	[12,52,58,59,63,71,72,80,101]

Claim	Certainty	Main reasons	Publication bias	Contributing studies
(f2) Confirmed detection in captive and "other" birds	Very low	Imprecision and risk of bias (single rows from grey literature)	Not assessed separately; rating already Very low	[71,72,81,82]
(g) H5N1 (2.3.4.4b) in Palearctic migrants (ruff, white stork), 2022	Very low	Imprecision (single study); jacana migrant status inconsistent within the study	Not assessed separately; rating already Very low	[12]
Species-level host range beyond domestic poultry	Not gradable	Narrative confidence only	—	—
S4 Molecular				
(h) Clade 2.2 and its sublineages, 2006–2008	Low	Not downgraded: six uncapped, non-high-risk sequence studies agree once sublineages are nested in 2.2	Not downgraded: many independent sequence datasets	[71–75,104–106]
(i) Clade 2.3.2.1c from 2015	Low	Not downgraded: four uncapped sequence studies; only discordant label from a re-analysis	Not downgraded	[64,76–79]
(j1) Clade 2.3.4.4b present in 2021–2026	Low	Not downgraded: several uncapped studies	Not downgraded	[11,12,19]
(j2) Onset of 2.3.4.4 in 2016 or 2019	Very low	Imprecision (single detections); indirectness (2016 label mapped by the review)	Not assessed separately; rating already Very low	[64–67]
(k1) Intra-clade reassortment shown by segment incongruence in several studies (2006–2007 and 2015–2016)	Low	Not downgraded. Excludes the LPAI H5N2 gene-pool reassortant (single study) and EA-2024-DV (place of reassortment not shown)	Not downgraded	[73,78,79,104–106]
(k2) The WA1×WA2 genotype	Very low	Imprecision (single study)	Not assessed separately; rating already Very low	[78,79]
(k3) The inter-subtype H5N1/H9N2 reassortant H5N2 (one farm)	Very low	Imprecision (single study; supplement unavailable)	Not assessed separately; rating already Very low	[11]
(k4) EA-2024-DV genotype (single study)	Very low	Imprecision; place of reassortment not shown	Not assessed separately; rating already Very low	[19]
(l1) Clade-level turnover: 2.2, then 2.3.2.1c, then 2.3.4.4b	Low	Not downgraded: each clade supported by several independent sequence studies	Not downgraded	[11,19,73,76–79,104]
(l2) Multiple independent introductions within a clade	Very low	Risk of bias/indirectness (clustering is not a phylogeographic link); inconsistency (two, three or four by source)	Not assessed separately; rating already Very low	[19,49,53,64,73,106,107]
S5 Exposure				
(m) Antibody exposure, by host system and assay class	Very low	Risk of bias (convenience samples); inconsistency (0/168 to 87/184 in markets); imprecision (small N); indirectness (exposure only; vaccination)	Not downgraded separately (small serosurveys well represented, about 12 of 34 studies), but cannot be excluded with 88 reports not retrieved	[13,35,62,98,112,113]
(n) H9 and H7 exposure	Very low	Indirectness (HI cross-reactivity; vaccination); risk of bias (screen-positive subsets); imprecision (no independent proportion). H7 weaker than H9	Not assessed separately; rating already Very low	[18,65–68,91,119–121]

Certainty rated by two independent **AI** raters with adjudication; observational evidence starts at low; a serious concern in any domain gives very low; single-study claims are downgraded for imprecision; nothing is rated up. Low certainty: confidence is limited and the true picture may differ substantially. Very low certainty: the finding is very uncertain.

Discussion

Principal findings

This review brought together 113 Nigerian studies from 2006 to 2026 and synthesised them with explicit rules for counting overlapping datasets, appraising molecular reporting and matching wording to certainty. Four findings were supported at low certainty, which means confidence in them is limited; the

true picture may differ substantially. First, HPAI H5N1 was probably confirmed in Nigeria in early 2006: the outbreak was reported in January and laboratory confirmation followed on 6–7 February. Second, the evidence indicates two major epizootics, in 2006–2008 and 2015–2017, separated by years with few or no reported outbreaks; national records also document 467 H5Nx outbreaks in 2021–2022. Third, the virus probably spread widely in 2006–2010, to at least 26 states and the FCT, and infected commercial, backyard, live-bird-market and wild resident birds. Fourth, the molecular record indicates clade-level turnover, from clade 2.2 in 2006–2008 to clade 2.3.2.1c from 2015 and clade 2.3.4.4b in 2021–2026, with intra-clade reassortment shown by segment incongruence in both of the first two waves.

Much of the remaining picture is uncertain. Whether virus circulation truly paused between 2009 and late 2014 is uncertain, because H5 RNA and antibody were detected in these years and sources for them are sparse. The first-detection dates of H5N8 (2016), H5N6 (2019) and H9N2 (2019) may partly reflect when testing for non-H5 subtypes began rather than when the viruses arrived. The later state counts are floors set by where sequencing and reporting took place, so they cannot show whether the geographic extent contracted. Infection of Palearctic migratory birds rests on a single 2022 study, and captive and other hosts rest on single rows from grey literature. Most importantly for interpreting the molecular dynamics, the number of independent introductions within each clade is uncertain: only two studies gave a phylogeographic link, and the counts from clustering disagreed (two, three or four), with one study arguing for continuous local circulation after 2006. All antibody-based estimates of exposure, including those for H7 and H9, were of very low certainty.

Comparison with earlier reviews

The narrative review by Akanbi and Lakes covered Nigerian avian influenza from 2006 to 2025 using PubMed and Scopus [15], and the systematic review by Kalonda et al. covered AIV detections in birds across sub-Saharan Africa from 2000 to 2019, with 68 studies [9]. The present review differs from both in scope and in method. It searched seven sources, including AJOL, OpenAlex and Crossref, added a Google Scholar update search, and included 110 Nigerian studies. More importantly, it integrated the spatiotemporal record with clade, reassortment and introduction evidence in a single framework: each dataset was counted once, under one primary study, across 24 reused data streams; official records without a stated assay were separated from laboratory-confirmed evidence; molecular claims were graded by whether accession numbers and phylogenetic support were reported; and subtype, clade or sublineage, reassortant and introduction were kept distinct under fixed rules. For example, the review counted a reassortant only where segment incongruence was shown, so the EA-2024-DV genotype was not counted as a Nigerian reassortment because its place of origin was not shown; it counted an introduction only where a phylogeographic link was given; and it showed that the smaller numbers of affected states after 2010 are floors set by sequencing and reporting coverage, so they cannot be read as a contraction. The review also extends the period covered to 2026, including the 2023–2025 genotypes.

Implications for surveillance and research

The findings point to three practical needs. First, surveillance in Nigeria has depended heavily on passive submissions to the national laboratory at Vom, and the evidence for inter-wave periods, for several states and for non-H5 subtypes is thin. Sustained active surveillance in live-bird markets, which detected virus outside outbreak reporting in 2008 and 2019, and in wetlands used by wild and migratory birds, would make claims of absence and of first detection testable. Second, whole-genome sequencing of a representative sample of outbreak viruses, with accession numbers deposited and phylogeographic

methods reported, is needed to distinguish new introductions from local persistence and to locate where reassortment occurs; 13 of the molecular studies could support only suggestive claims because these elements were missing. Third, national outbreak records should state the laboratory method and use consistent units, because conflicting series (for example, 513, 542 and 265 records for 2015) limited every estimate of wave size. Serological surveys would be more informative if they reported vaccination status, assay cut-offs and sampling frames.

Strengths and limitations of the evidence

The evidence base was dominated by observational studies with convenience sampling and passive reporting. The 2006–2008 evidence rests largely on passive submissions to NVRI, so detection probably varied with distance from the laboratory, reporting behaviour and access to diagnosis. Many confirmations did not state the laboratory method, the units of occurrence changed after 2007, and serology could not separate infection from vaccination. Most reports re-used a few national datasets; several links between reports were only probable or possible because record-level identifiers were missing, and state counts are floors based on named states. Four studies could be appraised only from their abstracts [16,33–35], and for three further studies a companion report was available only as an abstract [81,82,96,97,122,123]. Some conclusions rested on grey literature alone, namely the 2006 wild-resident date, the vulture detection and the captive and "other" host dates; four studies were wholly grey and four partly grey.

Strengths and limitations of the review process

The review followed a dated protocol with graded amendments, used dual independent decisions with recorded evidence for screening, extraction, appraisal and certainty rating (report-to-study links were proposed by one agent and checked by another), traced every synthesis entry to an extracted findings row and tested its conclusions in four sensitivity analyses. Its process also had important limitations. Screening, extraction, appraisal and certainty rating were conducted by AI agents with author confirmation, rather than by two independent human reviewers. The human check covered 107 priority decisions individually, but it relied on AI-prepared summaries rather than an independent re-reading of each full text, and the remaining 322 verified decisions were confirmed in a single statement; the author's second pass over the 107 decisions was a reconfirmation, not an independent re-review. Most title-stage exclusions were made on title and metadata alone, because many database records lacked abstracts. These factors increase the risk that relevant reports were missed or misclassified.

The review was not registered in PROSPERO or elsewhere, and some sources were added around the protocol freeze, including a post-freeze Google Scholar update search and a dated October 2026 update search (rebuilt strategies with the record caps removed, and citation chasing on the included reports); all such changes are dated in Appendix B. Ninety-eight reports were not retrieved, because institutional access was unavailable and authors were not contacted; about 40 of them appeared to be relevant Nigerian reports, including a molecular study of clade 2.3.4.4b introductions [17]. Citation chasing was limited to the included reports (forward and backward, OpenAlex and Crossref only); the original search strategies were not peer reviewed, the Web of Science browser results could not be exported, and the original OpenAlex and Crossref exports were capped at 1,000 records each (the October 2026 rebuild removed the caps). Eligibility was limited to English-language or translatable reports; no included report required translation, and two French-language reports were not retrieved. The post hoc sensitivity analysis (iv) was

added after the certainty ratings had been made. Finally, removal-only sensitivity analyses cannot test claims of absence, which are among the least certain findings.

Conclusions

The evidence indicates that HPAI H5N1 was first confirmed in Nigeria in early 2006, that two major epizootics followed in 2006–2008 and 2015–2017, and that the H5 viruses turned over from clade 2.2 to clade 2.3.2.1c and then to clade 2.3.4.4b, with intra-clade reassortment in the first two waves. Infection was confirmed in commercial, backyard, live-bird-market and resident wild birds, and was widespread in 2006–2010. How often viruses were introduced, whether they persisted between waves, when non-H5N1 subtypes arrived and what role migratory birds played all remain uncertain. Answering these questions will require sustained active surveillance across states and host systems, genome sequencing with public deposition of sequences and reporting of phylogeographic methods, and national outbreak records that state the laboratory method.

Declarations

Funding. [Author to complete: funding source(s) and the role of any funder, or a statement that the review received no specific funding]

Competing interests. [Author to complete: competing interests of each author, or "The authors declare no competing interests"]

Author contributions. [Author to complete: contributions of each author (for example, using the CRediT taxonomy)]

Acknowledgements. [Author to complete: acknowledgements]

Data availability. [Author to complete: data availability statement, including where the search files, screening register, extraction and risk-of-bias data are held]

References

1. Xie R, Edwards KM, Wille M, Wei X, Wong SS, Zanin M, et al. The episodic resurgence of highly pathogenic avian influenza H5 virus. *Nature*. 2023;622(7984):810–817. doi:10.1038/s41586-023-06631-2.
2. Fusaro A, Zecchin B, Giussani E, Palumbo E, Agüero-García M, Bachofen C, et al. High pathogenic avian influenza A(H5) viruses of clade 2.3.4.4b in Europe—why trends of virus evolution are more difficult to predict. *Virus Evol*. 2024;10(1):veae027. doi:10.1093/ve/veae027.
3. Webby RJ, Uyeki TM. An update on highly pathogenic avian influenza A(H5N1) virus, clade 2.3.4.4b. *J Infect Dis*. 2024;230(3):533–542. doi:10.1093/infdis/jiae379.
4. Chrzastek K, Lieber CM, Plemper RK. H5N1 clade 2.3.4.4b: evolution, global spread, and host range expansion. *Pathogens*. 2025;14(9):929. doi:10.3390/pathogens14090929.
5. Capelastegui F, Goldhill DH. H5N1 2.3.4.4b: a review of mammalian adaptations and risk of pandemic emergence. *J Gen Virol*. 2025;106(6):002109. doi:10.1099/jgv.0.002109.
6. Xie Z, Yang J, Jiao W, Li X, Iqbal M, Liao M, et al. Clade 2.3.4.4b highly pathogenic avian influenza H5N1 viruses: knowns, unknowns, and challenges. *J Virol*. 2025;99(6):e00424-25. doi:10.1128/jvi.00424-25.
7. Koopmans MPG, Barton Behravesh C, Cunningham AA, Adisasmito W, Almuhairi S, Bilivogui P, et al. The panzootic spread of highly pathogenic avian influenza H5N1 sublineage 2.3.4.4b: a critical appraisal of One Health preparedness and prevention. *Lancet Infect Dis*. 2024;24(12):e774–e781. doi:10.1016/S1473-3099(24)00438-9.
8. Peacock TP, Moncla L, Dudas G, VanInsberghe D, Sukhova K, Lloyd-Smith JO, et al. The global H5N1 influenza panzootic in mammals. *Nature*. 2025;637(8045):304–313. doi:10.1038/s41586-024-08054-z.
9. Kalonda A, Saasa N, Nkhoma P, Kajihara M, Sawa H, Takada A, et al. Avian influenza viruses detected in birds in sub-Saharan Africa: a systematic review. *Viruses*. 2020;12(9):993. doi:10.3390/v12090993.
10. Abolnik C, Phiri T, Peyrot B, de Beer R, Snyman A, Roberts D, et al. The molecular epidemiology of clade 2.3.4.4B H5N1 high pathogenicity avian influenza in Southern Africa, 2021–2022. *Viruses*. 2023;15(6):1383. doi:10.3390/v15061383.
11. Meseko C, Milani A, Inuwa B, Chinyere C, Shittu I, Ahmed J, et al. The evolution of highly pathogenic avian influenza A (H5) in poultry in Nigeria, 2021–2022. *Viruses*. 2023;15(6):1387. doi:10.3390/v15061387.
12. Olawuyi K, Orole O, Meseko C, Monne I, Shittu I, Zecchin B, et al. Detection of clade 2.3.4.4 highly pathogenic avian influenza H5 viruses in healthy wild birds in the Hadejia-Nguru wetland, Nigeria 2022. *Influenza Other Respir Viruses*. 2024;18(2):e13254. doi:10.1111/irv.13254.
13. Daodu OB, Jegede HO, Aiyedun JO, Oludairo OO, Olorunshola ID, Daodu OC, et al. Surveillance for avian influenza virus in captive wild birds and indigenous chickens in Nigeria. *Trop Anim Health Prod*. 2020;52(5):2387–2393. doi:10.1007/s11250-020-02265-y.
14. Adesola RO, Onoja BA, Adamu AM, Agbaje ST, Abdulazeez MD, Akinsulie OC, et al. Molecular epidemiology and genetic evolution of avian influenza H5N1 subtype in Nigeria, 2006 to 2021. *Virus Genes*. 2024;60(5):501–509. doi:10.1007/s11262-024-02080-9.
15. Akanbi OB, Lakes T. Review of avian influenza outbreaks in Nigeria 2006–2025. *Open Vet J*. 2025;15(12):6146–6165. doi:10.5455/OVJ.2025.v15.i12.2.
16. Shittu I, Meseko CA, Sailleau C, Barbe V, Mullers E, Quenel AS, et al. Highly pathogenic avian influenza (H5N1) in Nigeria in 2015: evidence of widespread circulation of WA2 clade 2.3.2.1c. *Arch Virol*. 2017;162(3):841–847. doi:10.1007/s00705-016-3149-4.
17. Laleye AT, Bianco A, Shittu I, et al. Genetic characterization of highly pathogenic avian influenza H5Nx clade 2.3.4.4b reveals independent introductions in Nigeria. *Transbound Emerg Dis*. 2022;69(2):423–433. doi:10.1111/tbed.14000.

18. Akanbi OB, Alaka OO, Olaifa OS, Meseko CA, Inuwa B, Ohore OG, et al. Pathology and molecular detection of influenza A subtype H9N2 virus in commercial poultry in Nigeria, 2024. *Open Vet J.* 2024;14(9):2381–2391. doi:10.5455/OVJ.2024.v14.i9.26.
19. Meseko C, Zecchin B, Go-Maró EW, Dianati M, Mkpuma N, Inuwa B, et al. Emergence and rapid spread of a new reassortant high pathogenicity H5N1 clade 2.3.4.4b avian influenza virus in Nigeria. *Influenza Other Respir Viruses.* 2026;20(5):e70260. doi:10.1111/irv.70260.
20. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ.* 2021;372:n71. doi:10.1136/bmj.n71.
21. Page MJ, Moher D, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars. *BMJ.* 2021;372:n160. doi:10.1136/bmj.n160.
22. Rethlefsen ML, Kirtley S, Waffenschmidt S, Ayala AP, Moher D, Page MJ, et al. PRISMA-S: an extension to the PRISMA Statement for Reporting Literature Searches in Systematic Reviews. *Syst Rev.* 2021;10(1):39. doi:10.1186/s13643-020-01542-z.
23. Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, et al. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *BMJ.* 2020;368:l6890. doi:10.1136/bmj.l6890.
24. Aromataris E, Munn Z, editors. *JBIM Manual for Evidence Synthesis.* JBI; 2020. doi:10.46658/JBIMES-20-01.
25. Munn Z, Moola S, Lisy K, Riitano D, Tufanaru C. Methodological guidance for systematic reviews of observational epidemiological studies reporting prevalence and cumulative incidence data. *Int J Evid Based Healthc.* 2015;13(3):147–153. doi:10.1097/XEB.0000000000000054.
26. Iorio A, Spencer FA, Falavigna M, Alba C, Lang E, Burnand B, et al. Use of GRADE for assessment of evidence about prognosis: rating confidence in estimates of event rates in broad categories of patients. *BMJ.* 2015;350:h870. doi:10.1136/bmj.h870.
27. Fasanmi OG, Ahmed SSU, Oladele-Bukola MO, El-Tahawy AS, Elbestawy AR, Fasina FO. An evaluation of biosecurity compliance levels and assessment of associated risk factors for highly pathogenic avian influenza H5N1 infection of live-bird-markets, Nigeria and Egypt. *Acta Tropica.* 2016;164:321–328. doi:10.1016/j.actatropica.2016.08.030.
28. Cecchi G, Ilemobade A, Le Brun Y. Agro-ecological features of the introduction and spread of the highly pathogenic avian influenza (HPAI) H5N1 in northern Nigeria. *Geospatial Health.* 2008;3(1):7–16. doi:10.4081/gh.2008.227.
29. Si Y, Skidmore AK, Wang T, De Boer WF, Debba P, Toxopeus AG, et al. Spatio-temporal dynamics of global H5N1 outbreaks match bird migration patterns. *Geospatial Health.* 2009;4(1):65–78. doi:10.4081/gh.2009.211.
30. Fasina FO, et al. *Epidemiologic and Economic Analyses on Highly Pathogenic Avian Influenza H5N1 in Nigeria and Egypt.* Utrecht University Repository. 2015.
31. Hogerwerf L, Wallace R, Ottaviani D, Slingenbergh J, Prosser DJ, Bergmann L, et al. Persistence of Highly Pathogenic Avian Influenza H5N1 Virus Defined by Agro-Ecological Niche. *EcoHealth.* 2010;7(2):213–225. doi:10.1007/s10393-010-0324-z.
32. Gwamzhi D, Dasat SG. Prevalence of avian influenza virus (AIV) (subtype A/H5N1) in commercial poultry farms in Jos South Local Government Area of Plateau State, Nigeria for sustainable development. *FUDMA Journal of Sciences.* 2019;3(3):198–201.
33. Ekong PS, Ducheyne E, Carpenter TE. Spatio-temporal epidemiology of highly pathogenic avian influenza (H5N1) outbreaks in Nigeria, 2006-2008. *Preventive Veterinary Medicine.* 2012;103(2-3):170–177. doi:10.1016/j.prevetmed.2011.10.001.

34. Cattoli G, Fusaro A, Monne I. Evidence for differing evolutionary dynamics of A/H5N1 viruses among countries applying or not applying avian influenza vaccination in poultry. *Vaccine*. 2011;29(50):9368–9375. doi:10.1016/j.vaccine.2011.09.127.
35. Aiki-Raji CO, Adebiyi AI, Agbajelola VI, Adetunji SA, Lameed Q, Adesina M, et al. Surveillance for low pathogenic avian influenza viruses in live-bird markets in Oyo and Ogun States, Nigeria. *Asian Pacific Journal of Tropical Disease*. 2015;5(5):369–373. doi:10.1016/S2222-1808(14)60799-4.
36. Glaji YA, et al. Retrospective survey of spatio-temporal spread of avian influenza in and around Maiduguri, Borno state northeastern Nigeria. *Journal of Veterinary Advances*. 2013;2(7). doi:10.14196/sjva.v2i7.862.
37. NR, et al. [Title unavailable - IJAAR 2017 5(3); abstract-only record: H5N2 antibody seroprevalence in birds, Maiduguri]. *International Journal of Advance Agricultural Research*. 2017;5(3). doi:10.33500/ijaar.2017.05.003.
38. Wakawa AM, Sa'idu L, Abdu PA, Joannis TM. Review of highly pathogenic avian influenza outbreaks in poultry in Zaria, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2008;7(1):1–4.
39. Wakawa A, Sa'idu L, Kazeem H, Fatihu M, Adamu J, Mamman P, et al. Highly pathogenic avian influenza I water fowls in Zaria, Nigeria. *Nigerian Veterinary Journal*. 2008;29(2):55–58. doi:10.4314/nvj.v29i2.3594.
40. Adegboye OA, Kotze D. Epidemiological analysis of spatially misaligned data: a case of highly pathogenic avian influenza virus outbreak in Nigeria. *Epidemiology and Infection*. 2014;142:940–949. doi:10.1017/S0950268813002136.
41. Waziri NE, Nguku P, Olayinka A. Evaluating a surveillance system: live-bird market surveillance for highly pathogenic avian influenza, a case study. *Pan African Medical Journal*. 2014;18(Suppl 1):11. doi:10.11694/pamj.supp.2014.18.1.4188.
42. Akanbi OB, Ekong PS, Odita CI, Taiwo VO. The effect of highly pathogenic avian influenza (HPAI) H5N1 outbreaks on mixed species poultry farms in Nigeria. *Journal of World's Poultry Research*. 2016;6(3):147–152.
43. Osanyintuyi SG, Adebayo IA, Arogbodo JO. Sero-epidemiological Survey of Avian Influenza and Newcastle Disease Antibodies in Japanese Quails (*Coturnix Coturnix Japonica*). *Animal and Veterinary Sciences*. 2025;13(6):175–178. doi:10.11648/j.av.s.20251306.11.
44. Akanbi OB, Meseko CA, Odita CI, Shittu I, Rimfa AG, Ugbe D, et al. Epidemiology and clinicopathological manifestation of resurgent highly Pathogenic Avian Influenza (H5N1) virus in Nigeria, 2015. *Nigerian Veterinary Journal*. 2016;37(3):175–186.
45. Ifende VI, Meseko CA, Odita CI, Akanbi BO, Moses GD, James AS, et al. Highly pathogenic avian influenza in Nigeria: Overview of 2015 outbreaks. In: *NVRI Seminar Series 2015*. 2015. p. 147–152.
46. Akanbi BO, Fereidouni S, Taiwo VO, Starick E, Okewole PA, Binder A, et al. Formalin-fixed and paraffin-embedded tissues of chickens are useful for retrospective studies on pathology of H5N1 Highly Pathogenic Avian Influenza Virus (HPAI) outbreaks in Nigeria. *Nigerian Veterinary Journal*. 2017;38(3):183–192.
47. Ducatez MF, Tarnagda Z, Tahita MC, Sow A, de Landtsheer S, Londt BZ, et al. Genetic characterization of HPAI (H5N1) viruses from poultry and wild vultures, Burkina Faso. *Emerging Infectious Diseases*. 2007;13(4):611–613. doi:10.3201/eid1304.061356.
48. Sanogo IN, Guinat C, Dellicour S, Diakité MA, Niang M, Koita O, et al. Genetic insights of H9N2 avian influenza viruses circulating in Mali and phylogeographic patterns in Northern and Western Africa. *Virus Evolution*. 2024;10(1):veae011. doi:10.1093/ve/veae011.
49. Ducatez MF, Olinger CM, Owoade AA. Molecular and antigenic evolution and geographical spread of H5N1 highly pathogenic avian influenza viruses in western Africa. *Journal of General Virology*. 2007;88:2297–2306. doi:10.1099/vir.0.82939-0.

50. Li R, Jiang Z, Xu B. Global spatiotemporal and genetic footprint of the H5N1 avian influenza virus. *International Journal of Health Geographics*. 2014;13:14. doi:10.1186/1476-072X-13-14.
51. Eng C, Tong JC, Tan TW. Distinct Host Tropism Protein Signatures to Identify Possible Zoonotic Influenza A Viruses. *PLoS ONE*. 2016;11(2):e0150173. doi:10.1371/journal.pone.0150173.
52. Joannis TM, Meseko CA, Oladokun AT. Serologic and virologic surveillance of avian influenza in Nigeria, 2006-7. *Eurosurveillance*. 2008;13(42):pii=19007. doi:10.2807/ese.13.42.19007-en.
53. Aiki-Raji CO, Aguilar PV, Kwon YK. Phylogenetics and pathogenesis of early avian influenza viruses (H5N1), Nigeria. *Emerging Infectious Diseases*. 2008;14(11):1753–1755. doi:10.3201/eid1411.080557.
54. Ameji NO, Oladele OO, Adanu AW, Jambalang AR, Inuwa B, Haruna AM, et al. Qualitative Assessment of the Clinico-Pathological Features of Highly Pathogenic Avian Influenza H5N1 Outbreaks in Commercial Poultry and Peri-Domestic Birds in Northern Nigeria. *Journal of Biosciences and Medicines*. 2022;10(9):273–288. doi:10.4236/jbm.2022.109019.
55. Meseko CA, Ehizibolo DO, Vakuru C. Migratory waterfowls from Europe as potential source of highly pathogenic avian influenza infection to Nigeria poultry. *Nigerian Veterinary Journal*. 2018;39(1):1–15. doi:10.4314/nvj.v39i1.1.
56. Odita CI, Hambolu MA, Olasoju TI, Makanjuola OA, Sito O, Meseko CA. Cluster poultry farming, the bane of highly pathogenic avian influenza spread in Lagos State Nigeria. *Nigerian Veterinary Journal*. 2022;43(4):1–9. doi:10.4314/nvj.v43i4.1.
57. Ameji NO, Oladele OO, Meseko CA, Mshelia GD, Lombin LH. Outbreak of highly pathogenic avian influenza subtype H5N8 in two multi-age chicken farms in Jos, Plateau State, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2019;17(3):60–65. doi:10.4314/sokjvs.v17i3.11.
58. De Benedictis P, Joannis TM, Lombin LH. Field and laboratory findings of the first incursion of the Asian H5N1 highly pathogenic avian influenza virus in Africa. *Avian Pathology*. 2007;36(2):115–117. doi:10.1080/03079450601161406.
59. Kumbish PR, Joannis TM, Jambalang AR, Damina MS, Hussaini BA, Akanbi BO, et al. Clinico-pathological features of highly pathogenic avian influenza (HPAI-H5N1) outbreaks in Commercial chickens in Nigeria. *Vom Journal of Veterinary Science*. 2006;(Special Edition):13–22.
60. Oladokun AT, Meseko CA, Ighodalo E, John B, Ekong PS. Effect of intervention on the control of Highly Pathogenic Avian Influenza in Nigeria. *Pan African Medical Journal*. 2012;13:14.
61. Wakawa AM, Abdu PA, Umoh JU, Lawal S, Miko RB. Serological evidence of mixed infections with avian influenza and Newcastle disease in village chickens in Jigawa State, Nigeria. *Veterinarski Arhiv*. 2009;79(2):151–155.
62. Ameji ON, Abdu PA, Sa'idu L. Sero-Prevalence of Avian Influenza, Newcastle and Gumboro Disease in Chickens in Kogi State, Nigeria. *Bulletin of Animal Health and Production in Africa*. 2011;59(4):411–418.
63. Gaidet N, Cattoli G, Hammoumi S. Evidence of infection by H5N2 highly pathogenic avian influenza viruses in healthy wild waterfowl. *PLoS Pathogens*. 2008;4(8):e1000127. doi:10.1371/journal.ppat.1000127.
64. Fusaro A, Zecchin B, Vrancken B. Disentangling the role of Africa in the global spread of H5 highly pathogenic avian influenza. *Nature Communications*. 2019;10:5310. doi:10.1038/s41467-019-13287-y.
65. Sulaiman L, Shittu I, Fusaro A. Live Bird Markets in Nigeria: A Potential Reservoir for H9N2 Avian Influenza Viruses. *Viruses*. 2021;13(8):1445. doi:10.3390/v13081445.
66. Shittu I, Bianco A, Gado D. First detection of highly pathogenic H5N6 avian influenza virus on the African continent. *Emerging Microbes & Infections*. 2020;9:886–888. doi:10.1080/22221751.2020.1757999.
67. Świątoń E, Fusaro A, Shittu I, Niemczuk K, Zecchin B, Joannis T, et al. Sub-Saharan Africa and Eurasia Ancestry of Reassortant Highly Pathogenic Avian Influenza A(H5N8) Virus, Europe, December 2019. *Emerging Infectious Diseases*. 2020;26(7):1557–1561. doi:10.3201/eid2607.200165.

68. Oluwayelu DO, Omolanwa A, Adebisi AI. Flock-based surveillance for low pathogenic avian influenza virus in commercial breeders and layers, southwest Nigeria. *African Journal of Infectious Diseases*. 2017;11(1):44–49. doi:10.21010/ajid.v11i1.5.
69. Meseko CA, Oladokun AT, Shehu B. An Outbreak Of Highly Pathogenic Avian Influenza (Hpa) In A Mixed Farm By The Introduction Of A Water Fowl. *Nigerian Veterinary Journal*. 2007;28(3):67–69. doi:10.4314/nvj.v28i3.3571.
70. Akintule AO, Oyediji LM, Tige DB, Iliyasu NM, Ehizibolo D. Prevalence of Avian Influenza Virus and Associated Risk Perception in Commercial Poultry Farms in Kaduna State and Metropolis [preprint]. Research Square. 2026. doi:10.21203/rs.3.rs-8223783/v1.
71. Fasina FO. Molecular and spatial-temporal epidemiology of highly pathogenic notifiable avian influenza (HPNAI) H5N1 in Nigeria [MSc dissertation]. University of Pretoria; 2008.
72. Fasina FO, Bisschop SP, Joannis TM. Molecular characterization and epidemiology of the highly pathogenic avian influenza H5N1 in Nigeria. *Epidemiology and Infection*. 2009;137(4):456–463. doi:10.1017/S0950268808000988.
73. Salzberg SL, Kingsford C, Cattoli G, Spiro D, Janies D, Aly MM, et al. Genome Analysis Linking Recent European and African Influenza (H5N1) Viruses. *Emerging Infectious Diseases*. 2007;13(5):713–718. doi:10.3201/eid1305.070013.
74. Meseko C, Oladokun A, Solomon P, Yakubu B. Detection of highly pathogenic avian influenza (H5N1) in apparently healthy ducks (*Anas sparsa sparsa*) in live bird markets, Nigeria. *Nigerian Veterinary Journal*. 2010;31(2):164–169. doi:10.4314/nvj.v31i2.68949.
75. Fusaro A, Joannis T, Monne I. Introduction into Nigeria of a distinct genotype of avian influenza virus (H5N1). *Emerging Infectious Diseases*. 2009;15(3):445–447. doi:10.3201/eid1503.081161.
76. Monne I, Meseko C, Joannis T. Highly Pathogenic Avian Influenza A(H5N1) Virus in Poultry, Nigeria, 2015. *Emerging Infectious Diseases*. 2015;21(7):1275–1277. doi:10.3201/eid2107.150421.
77. Tassoni L, Fusaro A, Milani A, Lemey P, Awuni JA, Sedor VB, et al. Genetically Different Highly Pathogenic Avian Influenza A(H5N1) Viruses in West Africa, 2015. *Emerging Infectious Diseases*. 2016;22(12):2132–2136. doi:10.3201/eid2212.160578.
78. Laleye A, Joannis T, Shittu I, Meseko C, Zamperin G, Milani A, et al. A two-year monitoring period of the genetic properties of clade 2.3.2.1c H5N1 viruses in Nigeria reveals the emergence and co-circulation of distinct genotypes. *Infection, Genetics and Evolution*. 2018;57:98–105. doi:10.1016/j.meegid.2017.10.027.
79. Laleye A, et al. Phylogenetic analysis of 2015-2016 Nigerian highly pathogenic avian influenza (HPAI) H5N1 viruses. *UpSpace Institutional Repository*. 2017.
80. Bello M, Lukshi BM, Sanusi M. Outbreaks of Highly Pathogenic Avian Influenza (H5N1) in Bauchi State, Nigeria. *International Journal of Poultry Science*. 2008;7(5):450–456. doi:10.3923/ijps.2008.450.456.
81. Atuman YJ, Inuwa B, Bakam J, Orioko E, Ikemefuna N, Emmanuel HS, et al. Exotic and aquatic zoological birds harbour highly pathogenic Avian Influenza Virus, 2021: continuous threat to poultry production, food security and public health in Nigeria. *Emerging Animal Species*. 2025;12:100055. doi:10.1016/j.eas.2025.100055.
82. Yakubu A, Inuwa B, Bakam J, Orioko E, Emmanuel HS, Meseko C. Exotic and zoological birds resident and imported into Nigeria harbour highly pathogenic avian influenza virus: Threat to poultry production, food security and public health [preprint]. *bioRxiv*. 2025. doi:10.1101/2025.01.22.634354.
83. Oluwayelu DO, Meseko CA, Ayinmode AB. Re-emergence of Highly Pathogenic Avian Influenza H5N1 in Nigeria, 2014-2016: Role of Social Network and Value Chain Forces in Interstate Transmission. *Virologica Sinica*. 2020;35(4):494–498. doi:10.1007/s12250-020-00201-w.
84. Okoli SC. Descriptive epidemiology of the outbreak of avian influenza in Nigeria: a retrospective review, 2015-2017. *PAMJ-One Health*. 2021;6:11. doi:10.11604/pamj-oh.2021.6.11.31265.

85. Gosomji I, Fagbamila I, Meseko C, Haruna A, Oladele O, Dakwal S, et al. Descriptive epidemiology and economic impact of highly pathogenic avian influenza in Plateau State, Nigeria: a retrospect from 2015 to 2022. *Proceedings of the Nigerian Academy of Science*. 2026;19(1):95–104. doi:10.57046/cczv5032.
86. Fasanmi OG, Kehinde OO, Laleye AT, Ekong B, Ahmed SSU, Fasina FO. National surveillance and control costs for highly pathogenic avian influenza H5N1 in poultry: A benefit-cost assessment for a developing economy, Nigeria. *Research in Veterinary Science*. 2018;119:127–133. doi:10.1016/j.rvsc.2018.06.006.
87. Coker T, Meseko C, Odaibo G. Circulation of the low pathogenic avian influenza subtype H5N2 virus in ducks at a live bird market in Ibadan, Nigeria. *Infectious Diseases of Poverty*. 2014;3:38. doi:10.1186/2049-9957-3-38.
88. Wu T, Perrings C. The live poultry trade and the spread of highly pathogenic avian influenza: Regional differences between Europe, West Africa, and Southeast Asia. *PLoS ONE*. 2018;13(12):e0208197. doi:10.1371/journal.pone.0208197.
89. Assam A, Abdu PA, Ademola AO, Augustine E, Lawal S. Avian influenza, Newcastle and Gumboro disease antibodies and antigens in apparently healthy wild birds in Kaduna State, Nigeria. *Bulletin of Animal Health and Production in Africa*. 2014;62(2):183–196.
90. Al-Mustapha AI, Adetunji V, Ogundijo OA. Animal Disease Burden in Nigeria, 2006-2023. *Transboundary and Emerging Diseases*. 2025;2025:1694850. doi:10.1155/tbed/1694850.
91. Mkpuma N, Oyedeji LM, Akintule AO, Chinyere CN, Agbato AO, Bitrus I, et al. Serological and Molecular Detection of Low Pathogenic Avian Influenza Subtype H9N2 in Commercial Poultry Flock in Kaduna and Kano State Nigeria. *Nigerian Veterinary Journal*. 2024;45(2):26–38. doi:10.4314/nvj.v45i2.3.
92. Akanbi OB. Characterization and comparative susceptibility of highly pathogenic avian influenza H5N1 virus infections in chickens and ducks [PhD thesis]. University of Ibadan; 2014.
93. Akanbi OB, Taiwo VO. Mortality and Pathology Associated with Highly Pathogenic Avian Influenza H5N1 Outbreaks in Commercial Poultry Production Systems in Nigeria. *International Scholarly Research Notices*. 2014;2014:415418. doi:10.1155/2014/415418.
94. Akanbi OB, Taiwo VO. Backyard Poultry Mortality Associated with Highly Pathogenic Avian Influenza (HPAI) H5N1 outbreaks in Nigeria. *IOSR Journal of Agriculture and Veterinary Science*. 2014;7(9):23–27. doi:10.9790/2380-07922327.
95. Akanbi OB, Taiwo VO. Husbandry Practices and Outbreak Features of Natural Highly Pathogenic Avian Influenza H5N1 in Turkey Flocks in Nigeria 2006-2008. *Journal of World's Poultry Research*. 2015;5(4):109–114.
96. Waziri MI, Abdu PA, Sa'idu L. Seroepidemiology and assessment of risk factors for the spread of avian influenza in birds in two Nigerian states. *Veterinary Medicine and Science*. 2017;3(4):227–238. doi:10.1002/vms3.73.
97. Musa WI, Sa'idu L, Bello M. Co-infections of domestic and wild birds with avian influenza and Newcastle disease viruses: implications for control and genetic mutations. *Veterinary Research Communications*. 2020;44(3-4):159–166. doi:10.1007/s11259-020-09783-y.
98. Oluwayelu DO, Aiki-Raji CO, Adigun OT. Serological Survey for Avian Influenza in Turkeys in Three States of Southwest Nigeria. *Influenza Research and Treatment*. 2015;2015:787890. doi:10.1155/2015/787890.
99. Ameji N, Sa'idu L, Abdu P. Detection of avian influenza antibodies and antigens in poultry and some wild birds in Kogi state, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2017;15(3):62–68. doi:10.4314/sokjvs.v15i3.9.

100. Ekong PS, Cardona CJ, Bryssinckx W. Spatial clustering of pathology submissions during the initial introduction and spread of avian influenza H5N1 in poultry in Nigeria in 2006–2007. *Veterinaria Italiana*. 2018;54(1):13–20. doi:10.12834/VetIt.870.4301.3.
101. Snoeck CJ, Adeyanju AT, De Landtsheer S. Reassortant low-pathogenic avian influenza H5N2 viruses in African wild birds. *Journal of General Virology*. 2011;92:1172–1183. doi:10.1099/vir.0.029728-0.
102. Musa WI, Abdullahi BY, Lawal S, Bello M, Abdu PA. Detection of antibodies to avian influenza, infectious bronchitis and Newcastle disease viruses in wild birds in three states of Nigeria. *Sokoto Journal of Veterinary Sciences*. 2017;15(Special Issue):28–35. doi:10.4314/sokjvs.v15i4.5.
103. Adebisi AI, Gamra O, Jubril A, Oluwayelu DO. Serological investigation of low pathogenic avian influenza and Newcastle disease virus antibodies in Japanese quails, 30 village weavers and one laughing dove in two states of Nigeria. *Bulletin of the National Research Centre*. 2022;46:25. doi:10.1186/s42269-022-00710-6.
104. Fusaro A, Nelson MI, Joannis T. Evolutionary dynamics of multiple sublineages of H5N1 influenza viruses in Nigeria from 2006 to 2008. *Journal of Virology*. 2010;84(7):3239–3247. doi:10.1128/JVI.02385-09.
105. Monne I, Joannis TM, Fusaro A. Reassortant avian influenza virus (H5N1) in poultry, Nigeria, 2007. *Emerging Infectious Diseases*. 2008;14(4):637–640. doi:10.3201/eid1404.071178.
106. Owoade AA, Gerloff NA, Ducatez MF. Replacement of sublineages of avian influenza (H5N1) by reassortments, sub-Saharan Africa. *Emerging Infectious Diseases*. 2008;14(11):1731–1735. doi:10.3201/eid1411.080555.
107. Cattoli G, Monne I, Fusaro A. Highly pathogenic avian influenza virus subtype H5N1 in Africa: a comprehensive phylogenetic analysis and molecular characterization of isolates. *PLoS ONE*. 2009;4(3):e4842. doi:10.1371/journal.pone.0004842.
108. Wallace RG, Fitch WM. Influenza A H5N1 Immigration Is Filtered Out at Some International Borders. *PLoS ONE*. 2008;3(2):e1697. doi:10.1371/journal.pone.0001697.
109. Liang L, Xu B, Chen Y, Liu Y, Cao W, Fang L, et al. Combining Spatial-Temporal and Phylogenetic Analysis Approaches for Improved Understanding on Global H5N1 Transmission. *PLoS ONE*. 2010;5(10):e13575. doi:10.1371/journal.pone.0013575.
110. Dhingra M, Artois J, Dellicour S, Lemey P, Dauphin G, Von Dobschuetz S, et al. Geographical and Historical Patterns in the Emergences of Novel Highly Pathogenic Avian Influenza (HPAI) H5 and H7 Viruses in Poultry. *Frontiers in Veterinary Science*. 2018;5:84. doi:10.3389/fvets.2018.00084.
111. Adamu HU, Balami AG, Abdu PA. Avian influenza, Gumboro and Newcastle disease antibodies and antigens in apparently healthy wild birds in Kano Metropolis, Nigeria. *Nigerian Veterinary Journal*. 2017;38(1):69–77.
112. Balami AG. Evidence of Antibodies to H5 Subtype Avian Influenza Virus in Commercial Layer Farms in Jos, Nigeria. *Advances in Animal and Veterinary Sciences*. 2014;2(9):524–529. doi:10.14737/journal.aavs/2014/2.9.524.529.
113. Bakre A, Adelakun O, Dauda U, Adesola R, Oladele O. Seroprevalence of avian influenza in free-range domestic ducks in some selected households in Oyo State, southwestern Nigeria. *Sokoto Journal of Veterinary Sciences*. 2022;20(4):268–271. doi:10.4314/sokjvs.v20i4.7.
114. Wungak YS, Orakpoghenor O, Bitrus I, Olawuyi KA, Osemeke OH, Ularanu HG, et al. Detection of antibodies to H5 and H9 subtypes of influenza viruses in wild birds in Zaria, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2021;19(4):160–165. doi:10.4314/sokjvs.v19i4.2.
115. Miko R, Abdu PA, Assam A, Sa'idu L. Avian Influenza H5-Subtype Antibodies in Apparently Healthy Local Poultry in Live Bird Markets in Jigawa State, Nigeria. *Bulletin of Animal Health and Production in Africa*. 2013;61(2):121–126.

116. Alonge PI, Oladele SB, Hassan FB, Orakpoghenor O, Samuel J. Serological and Molecular Detection of Avian Influenza Virus Hemagglutinin (H5) in Wild Birds in and around Zaria, Nigeria. *Sahel Journal of Veterinary Sciences*. 2021;18(1):27–31. doi:10.54058/saheljvs.v18i1.226.
117. Meseko CA, Olorunsola B, Chinyere CA, Olawuyi K. Re-current Epizootics of Highly Pathogenic Avian Influenza in Nigeria: Status of Vaccination as Alternate control. *Nigerian Veterinary Journal*. 2020;41(1):7–17. doi:10.4314/nvj.v41i1.2.
118. Obi T. The Battle Against Transboundary Animal Diseases In Nigeria And Some West African Countries [public lecture]. *Proceedings of the Nigerian Academy of Science*. 2015;8(1):59–78. doi:10.57046/lcjd2673.
119. Chinyere CN, Okwor EC, Meseko CA, Ezema WS, Choji ND, Amos DI, et al. Sero-Detection of Avian Influenza A/H7 in Nigerian Live-Bird Markets in Plateau State. *Nigerian Veterinary Journal*. 2020;41(2):161–174. doi:10.4314/nvj.v41i2.7.
120. Abiayi DC, Otolorin GR, Dzikwi-Emennaa AA, Meseko CA. Serological evidence of influenza A/H9 in indigenous birds and level of awareness at live bird markets, Plateau State. *Sokoto Journal of Veterinary Sciences*. 2021;19(4):175–182. doi:10.4314/sokjvs.v19i4.4.
121. Bakam JD, Inuwa B, Yiltawe W, Shittu I, Meseko C, Kia G. Serosurvey for H5, H7 and H9 avian influenza viruses in local chickens in live bird markets within Kaduna metropolis, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2023;21(3):119–125. doi:10.4314/sokjvs.v21i3.3.
122. Gugong VT, Ajogi I, Junaidu K, Okolocha EC, Ngbede EO, Abraham MN, et al. Avian influenza in village chickens, its awareness and presence of potential risk practices among rural dwellers. *Asian Pacific Journal of Tropical Disease*. 2012;2(4):282–285. doi:10.1016/s2222-1808(12)60061-9.
123. Gugong V, Ajogi I, Juniadu K, Okolocha E, Ngbede E, Hambolu S, et al. Avian influenza H5 subtype antibodies in village chickens in four local government areas of Kaduna state, Nigeria. *Veterinary World*. 2012;5(12):713–717. doi:10.5455/vetworld.2012.713-717.
124. Ameji ON, Sa'idu L, Abdu PA. Sero-Prevalence of Avian Influenza in Poultry in Kogi State, Nigeria. *Current Research in Poultry Science*. 2016;6(1):1–6. doi:10.3923/crpsaj.2016.1.6.
125. Musa IW, Abdu PA, Sackey AK. Survey for highly pathogenic avian influenza from poultry in two northeastern States, Nigeria. *Veterinary Medicine International*. 2013;2013:531491. doi:10.1155/2013/531491.
126. Umar HI, Inuwa B, Alamini I, Adamu A, Ismaila S, Meseko C, et al. Avian influenza A virus surveillance in reservoir domestic ducks (*Anas platyrhynchos domesticus*) in Maiduguri metropolitan community of north-eastern Nigeria: a need for one-health approach. *FUDMA Journal of Sciences*. 2023;7(6):339–342. doi:10.33003/fjs-2023-0706-1800.
127. Balami SB, Inuwa B, Gambo MS, Iyanda SO, Ndahi JJ, Adamu SG. Influenza A virus in pigeons (*Columba livia domestica*): prevalence and associated risk factors in Maiduguri Metropolis, Borno State, Nigeria. *Journal of Sustainable Veterinary & Allied Sciences*. 2025;7(2):180–185.
128. Ekong PS, Olugasa BO, Oyetunde IL, Waziri NE, Joannis TM. Spatial distribution of primary outbreaks of highly pathogenic avian influenza in Nigeria. In: *Proceedings of the XIII International Congress of the International Society for Animal Hygiene (ISAH)*; 2007; Tartu, Estonia. Vol. II. 2007. p. 740–743.
129. Nwankwo IO, Faleke OO, Garba J. Avian influenza virus infection in apparently healthy domestic birds in Sokoto, Nigeria. *Veterinaria Italiana*. 2012;48(3):309–312.
130. Pelletier ST, Rorres C, Macko PC. Models of highly pathogenic avian influenza epidemics in commercial poultry flocks in Nigeria and Ghana. *Tropical Animal Health and Production*. 2012;44:1681–1687. doi:10.1007/s11250-012-0124-2.

131. Fasina FO, Rivas AL, Bisschop SP. Identification of risk factors associated with highly pathogenic avian influenza H5N1 virus infection in poultry farms, in Nigeria during the epidemic of 2006-2007. *Preventive Veterinary Medicine*. 2011;98:204–208. doi:10.1016/j.prevetmed.2010.11.007.
132. Rivas AL, Chowell G, Schwager SJ. Lessons from Nigeria: the role of roads in the geo-temporal progression of avian influenza (H5N1) virus. *Epidemiology and Infection*. 2010;138:192–198. doi:10.1017/S0950268809990495.
133. Okoli SC, Ameh C, Adebawale AS. Evaluation of the avian influenza surveillance system in Enugu State Nigeria, 2015-2017. *PAMJ-One Health*. 2021;4:6. doi:10.11604/pamj-oh.2021.4.6.27496.
134. Okoli SC. Assessment of the risk of outbreak of avian influenza in poultry farms in Ebonyi State, Nigeria: a cross-sectional study, 18th to 25th January, 2022. *PAMJ-One Health*. 2022;8:1. doi:10.11604/pamj-oh.2022.8.1.34535.
135. Ogolo FO, Nwanga E, Orakpoghenor O, Markus TP, Bello AL. Retrospective Study of Highly Pathogenic Avian Influenza Outbreaks from 2015 to 2022 in Rivers State, Nigeria. *Research in Veterinary Science and Medicine*. 2023;3:2. doi:10.25259/RVSM_6_2023.
136. Abubakar A. Molecular Characterization of Avian Influenza H9N2 and H5N1 from Outbreak in a Backyard Layer Farm in Kaduna State, Nigeria. *Journal of Veterinary and Biomedical Sciences*. 2021;3(2):70–76. doi:10.36108/jvbs/1202.30.0270.
137. Ameji NO, Oladele OO, Jambalang AR, Adanu AW, Chinyere CN, Meseko C, et al. Multiple Outbreaks and Clinico-pathological Features of Highly Pathogenic Avian Influenza H5N1 and H5N8 in Poultry Farms in Jos Metropolis, Plateau State, Nigeria. *Journal of World's Poultry Research*. 2021;11(3):376–386. doi:10.36380/jwpr.2021.45.
138. Okoli SC, Ameh C, Balogun MS, Durosinlorun A, Vakuru C, Olaniran A. Retrospective Study of the Burden of Avian influenza in Kano State, Nigeria: A secondary Data Analysis 2015-2017. *East African Journal of Agriculture and Biotechnology*. 2019;1(1):12–23. doi:10.37284/eajab.1.1.40.
139. Saidu L, Wakawa AM, Abdu PA, Adene DF, Kazeem HM, Ladan KC, et al. Impact of Avian Influenza in Some States of Nigeria. *International Journal of Poultry Science*. 2008;7(9):913–916. doi:10.3923/ijps.2008.913.916.
140. Ja'afar-Furo MR, Balla HG, Tahir AS, Haskainu C. Incidence of Avian Influenza in Adamawa State, Nigeria: The Epidemiology, Economic Losses and the Possible Role of Wild Birds in the Transmission of the Disease. *Journal of Applied Sciences*. 2008;8(2):205–217. doi:10.3923/jas.2008.205.217.
141. Adidu-Omoniwa AM, Dzikwi-Emennaa AA, Shuaibu I, Philip NK, Duwong RK, Omoniwa DO, et al. Retrospective study of Highly Pathogenic Avian Influenza outbreaks from 2017 to 2021 in Plateau State, Nigeria. *Nigerian Veterinary Journal*. 2024;45(4):37–50. doi:10.4314/nvj.v45i4.4.
142. Wakawa A, Abdu P, Oladele S, Sa'idu L, Mohammed S. Risk factors for the occurrence and spread of Highly Pathogenic Avian Influenza H5N1 in commercial poultry farms in Kano, Nigeria. *Sokoto Journal of Veterinary Sciences*. 2012;10(2):40–51. doi:10.4314/sokjvs.v10i2.8.
143. Fasanmi OG. Epidemiological modelling for the control of highly pathogenic avian influenza H5N1 (HPAI H5N1) in Egypt and Nigeria [PhD thesis]. University of Pretoria; 2017.
144. Idachaba SE, Nafarnda WD, Egwu G, Ayuba H, Barde IJ. Retrospective Study of Avian Influenza Outbreak from 2015-2017; Resurgences in Plateau State Nigeria. *Acta Scientific Microbiology*. 2020;3(2):70–72.
145. Kumbish PR, Bello MK, Jambalang AR, Hussaini BA, Solomon P, Ekong PS, et al. Bacterial and Parasitic agents associated with outbreaks of Highly pathogenic Avian influenza (H5N1) in Nigeria. *Vom Journal of Veterinary Science*. 2006;(Special Edition):23–31.

Appendix A. Search strategies and PRISMA-S audit

A1. Sources, dates, record counts and reproducibility

Every strategy followed the architecture (AIV concept) AND (Nigeria concept) AND (date limit), with molecular/evolutionary, epidemiology/spatiotemporal and wild-bird/host concepts as separate facet lines; Nigeria was not embedded within the AIV or facet blocks. Table A1 distinguishes complete documented executions from abbreviated or count-only records. Counts must not be added across overlapping facets, or across imported and count-only layers; the records entering the PRISMA flow are those marked as imported.

Table A1. Search sources, dates, documented strategies, record counts and reproducibility status.

Source, platform and date	Documented search record	Records	Status in the PRISMA flow; reproducibility notes
PubMed/MEDLINE via NLM E-utilities; 16 Sep 2026	Full v3 base and molecular facet (A2); date limit 2006/01/01–2026/09/30	140 base; 53 molecular facet	Imported (140). Executed strings retained; an English filter is recorded in the run log but not encoded in the v3 script record.
Scopus via Elsevier; 16 Sep 2026	TITLE-ABS-KEY("avian influenza" OR AIV OR HPAI OR LPAI OR H5N1 OR H5N2 OR H5N6 OR H5N8 OR H9N2) AND TITLE-ABS-KEY(Nigeria OR Nigerian) AND PUBYEAR > 2005 AND PUBYEAR < 2027	204	Imported (204; manual RIS export). Base syntax documented; facet lines, interface settings and export configuration incomplete.
ScienceDirect via Elsevier; 16 Sep 2026	Abbreviated record only: single-line title/abstract/keyword variant of the Scopus string, 2006–2026	25	Imported (25; manual RIS export). Exact as-run string, filters and cap not documented.
OpenAlex (api.openalex.org); 16 Sep 2026	search "avian influenza Nigeria"; publication dates 2006/01/01–2026/09/30	6,455 total; 1,000 exported	Imported, with Crossref (1,744 rows after internal de-duplication of the two 1,000-record exports). Export capped at 1,000.
Crossref (api.crossref.org); 16 Sep 2026	query "avian influenza Nigeria"; 2006–2026; journal articles	299,088 total; 1,000 exported	As above. Relevance-ranked export capped at 1,000.
Web of Science Core Collection via Clarivate Starter API; 19 Sep 2026	TS=("avian influenza" OR AIV OR HPAI OR LPAI OR H5N1 OR H9N2) AND TS=(Nigeria OR Nigerian) AND PY=(2006-2026); db=WOS; four pages (50/50/50/25)	175 unique UIDs	Imported (175). No English filter recorded.
Web of Science Core Collection, browser (Free View); 20 Sep 2026	Expanded base plus molecular, epidemiology and wild-bird/host facet lines (A2); publication date 2006-01-01–2026-09-20; editions all	206 base; 77 molecular; 154 epidemiology; 70 wild-bird/host	Count only, not imported: no reliable record-level export. Not added to the 175 UIDs.
African Journals Online (on site); 20 Sep 2026	Nine exact mandatory-term queries (A2); no site date filter, year checked after capture	144 query rows; 79 unique article pages (73 dated 2006–2026)	Imported (79).
Google Scholar via SerpApi; 17 Sep 2026	Q1–Q4 (A2); relevance; cap 200 per query	140 retrieved; 99 unique	Superseded supplement: no persistent export, never screened; replaced by the 28 Sep re-run.
Google Scholar via SerpApi; 28 Sep 2026 (post-freeze update)	Q1–Q4 verbatim; as_ylo=2006, as_yhi=2026, no patents, no citations, hl=en; relevance; 20 per page; cap 200 per query	800 rows; 519 unique; 291 already in register; 228 new	Imported as identification via other methods (228; 12 later found to be duplicates). Full raw output retained.

A2. Search strings as run

PubMed v3 base (16 September 2026). (avian influenza[tiab] OR avian flu[tiab] OR bird flu[tiab] OR fowl plague[tiab] OR avian influenza virus[tiab] OR AIV[tiab] OR highly pathogenic avian influenza[tiab] OR HPAI[tiab] OR low pathogenic avian influenza[tiab] OR LPAI[tiab] OR H5N1[tiab] OR H5N2[tiab] OR H5N6[tiab] OR H5N8[tiab] OR H9N2[tiab] OR H7N9[tiab] OR H5Nx[tiab]) AND (Nigeria[Mesh] OR Nigeria[tiab] OR Nigerian[tiab]) AND ("2006/01/01"[dp] : "2026/09/30"[dp]). Result: 140.

PubMed v3 molecular facet. base AND (genome[tiab] OR molecular[tiab] OR phylogen*[tiab] OR clade[tiab] OR lineage[tiab] OR reassort*[tiab] OR mutation[tiab] OR evolution[tiab]). Result: 53 (a subset of the base set).

Scopus (16 September 2026). TITLE-ABS-KEY("avian influenza" OR AIV OR HPAI OR LPAI OR H5N1 OR H5N2 OR H5N6 OR H5N8 OR H9N2) AND TITLE-ABS-KEY(Nigeria OR Nigerian) AND PUBYEAR > 2005 AND PUBYEAR < 2027. Result: 204.

Web of Science Starter API (19 September 2026). TS=("avian influenza" OR AIV OR HPAI OR LPAI OR H5N1 OR H9N2) AND TS=(Nigeria OR Nigerian) AND PY=(2006-2026). Result: 175.

Web of Science browser (20 September 2026; count only). Base: ("avian influenza" OR "avian flu" OR "influenza A virus" OR "influenza A" OR H5N1 OR H5N2 OR H5N6 OR H5N8 OR H5Nx OR H7N1 OR H7N3 OR H7N4 OR H7N7 OR H9N2) AND (Nigeria OR Nigerian), 206. Molecular: base AND (molecular OR phylogen* OR sequence* OR genome* OR reassort* OR clade OR sublineage OR genotype), 77. Epidemiology: base AND (epidemiol* OR outbreak* OR surveillance OR prevalence OR incidence OR "risk factor*" OR spati* OR temporal OR spread), 154. Wild-bird/host: base AND ("wild bird*" OR migratory OR waterfowl OR duck* OR goose* OR "live bird market*"), 70. The interface suggested H7N9 for H7N4; the suggestion was not accepted.

AJOL (20 September 2026). +Nigeria +"avian influenza" (66); +Nigeria +"influenza A virus" (13); +Nigeria +"H5N1" (23); +Nigeria +"H5N8" (2); +Nigeria +"H9N2" (6); +Nigeria +"H7N7" (4); +Nigeria +"avian influenza" +molecular (8); +Nigeria +"avian influenza" +epidemiology (10); +Nigeria +"avian influenza" +"wild bird" (12). Displayed hits in parentheses; 144 query rows de-duplicated to 79 article pages.

Google Scholar (Q1–Q4; 17 and 28 September 2026). Q1 "avian influenza" Nigeria; Q2 H5N1 Nigeria; Q3 H9N2 Nigeria; Q4 poultry Nigeria wild birds.

A3. October 2026 update search (build 9 October 2026)

Every source was re-run with rebuilt strings in which the AIV, Nigeria and date facets sit on separate lines and the record caps are removed; the strings as executed are retained verbatim in the review's working files (W1 package, 9 October 2026), with the raw responses and a step-by-step PRESS self-assessment. PubMed (National Library of Medicine E-utilities) returned 209 records with the improved string, against 140 with the original string, the gain coming from bird-specific index terms and affiliation terms that the original strategy omitted. OpenAlex was paged with cursors to exhaustion (10,614 records, no 1,000-record cap); Crossref was paged with cursors, of which 6,000 of 16,292 matching records were downloaded before the download stopped for time; the Web of Science Starter API returned 194 records (its endpoint applies no publication-year filter, so the 2006–2026 window was applied locally, and it

returns no abstracts); AJOL returned 66 records of which only the first 20 could be captured because its pagination repeats page one for programmatic access, so the remaining titles would need a browser. Scopus and ScienceDirect could not be re-run from the available access and remain documented only as in A2. Google Scholar was not re-run because the SerpApi quota was spent. Backward and forward citation chasing on all 123 included reports used the OpenAlex and Crossref reference and citation APIs; 77 of 123 reports resolved to an indexed record (43 were blocked by the OpenAlex daily request budget and none by content), giving 2,195 candidate works, 1,745 of them new to the register and 11 relevance-positive. Every retrieved record was matched against the 2,105-record register by DOI, PMID and normalised title before screening. The screening packet and the blind dual review are described in B1; the four retained records were abstract-only, and one further record (a 2017 repository record of the 2015–2016 100-genome series) was linked as a probable companion of an included study rather than counted separately.

A4. PRISMA-S audit

Table A2. PRISMA-S reporting audit (16 items, numbered as in the published checklist [22]).

Item	PRISMA-S item	Status	Where reported; comment
1	Database name (with platform)	Yes	Methods (Information sources); Table A1: PubMed/MEDLINE via NLM E-utilities, Scopus and ScienceDirect via Elsevier, Web of Science Core Collection via Clarivate (Starter API and browser), OpenAlex and Crossref APIs.
2	Multi-database searching	Yes	Each database was searched separately (Table A1); none was searched through a multi-database platform in one query.
3	Study registries	Not applicable	No study registries were searched; registries are not a usual source for animal surveillance and molecular studies.
4	Online resources and browsing	Partial	AJOL searched on site and Google Scholar used as an update source (Table A1). Agency websites (for example, WOA) were not searched separately.
5	Citation searching	Yes	Backward and forward citation chasing on all 123 included reports through the OpenAlex and Crossref reference and citation APIs (9 October 2026 update search; A3).
6	Contacts	Yes (none)	No authors or experts were contacted for additional studies or data (Methods).
7	Other methods	Yes	Dated Google Scholar update search (28 September 2026) and dated October 2026 update search (rebuilt strategies and citation chasing), reported as identification via other methods (Figure 1).
8	Full search strategies	Partial	Original exact strings for PubMed, Web of Science (API and browser), AJOL, Google Scholar, OpenAlex and Crossref (A2); every October 2026 rebuilt string as executed, facet by facet (A3). Scopus facet lines and the exact ScienceDirect string were not documented and could not be re-run.
9	Limits and restrictions	Partial	Date limits documented for every source; English-language reports only at eligibility. An English filter is recorded inconsistently at search level (PubMed run log versus script; none for Web of Science); ScienceDirect filters not documented. The October 2026 rebuild removed the OpenAlex and Crossref record caps (A3).
10	Search filters	Not applicable	No published search filters were used.
11	Prior work	Partial	The architecture was rebuilt from the review's own draft term blocks; strategies of the earlier reviews were not reused as such.
12	Updates	Yes	Dated Google Scholar re-run on 28 September 2026 and a dated October 2026 update search (rebuilt databases, caps removed, citation chasing) after the protocol freeze.
13	Dates of searches	Yes	Table A1 (16, 19, 20 and 28 September 2026; 9 October 2026 update search in A3).
14	Peer review	No	No independent PRESS or second-person review of the strategies (stated as a limitation). The October 2026 strings were self-assessed against PRESS (A3).
15	Total records	Yes	Records per source in Table A1 and Figure 1; count-only layers identified.

Item	PRISMA-S item	Status	Where reported; comment
16	Deduplication	Yes	DOI, PMID or normalised title plus year, with manual checks; 490 duplicates removed before screening; a later normalised-DOI check found 2 further pairs, and 10 further duplicates were recognised at full text in the Google Scholar update search; the October 2026 update search matched every retrieved record against the 2,105-record register before screening (A3; Appendix B). Search files (raw exports, API responses and logs) are retained in the review's dated working files; post-freeze changes are dated in Table B2.

Appendix B. Screening and selection audit

B1. Flow arithmetic

Databases and registers: 2,367 records identified (OpenAlex and Crossref 1,744; PubMed 140; Scopus 204; Web of Science 175; ScienceDirect 25; AJOL 79) – 490 duplicates = 1,877 screened; 1,877 – 1,578 excluded on title/metadata (including 6 outside 2006–2026) = 299 reports sought; 299 – 80 not retrieved = 219 assessed; 219 – 108 excluded = 111 reports included. Other methods (Google Scholar update): 800 rows → 519 unique – 291 already identified = 228 new – 12 later found to be duplicates = 216 screened; 216 – 190 excluded on title/snippet = 26 sought; 26 – 8 not retrieved = 18 assessed; 18 – 6 excluded = 12 included. Update search (9 October 2026; rebuilt strategies and citation chasing): 17,037 records retrieved, 2,270 matched to the register, 48 candidates after corrected de-duplication (44 after probable-duplicate exclusion); citation chasing added 11 relevance-positive records; after merging the two sets (3 records in common) the screening packet was 52 records: 52 – 35 excluded on title = 17 sought; 17 – 10 not retrieved = 7 assessed; 7 – 3 excluded = 4 included. Total: 111 + 12 + 4 = 127 reports, linked into 113 studies (two provisional study identifiers were retired when their only reports were excluded at extraction, and identifiers were not renumbered). Full-text exclusion reasons (databases; other methods; update search): no Nigeria-specific data (54; 2; 0); no laboratory-confirmed avian AIV (19; 1; 1); no avian-host field data (17; 0; 0); no primary data (13; 2; 2); no epidemiological, molecular or spatiotemporal data (2; 0; 0); experimental only, without field relevance (1; 1; 0); duplicate dataset (1; 0; 0); non-AIV (1; 0; 0). Eleven studies had more than one included report; the linkage is shown in Appendix C.

B2. Agreement between AI reviewers

Table B1. Agreement between the two independent AI reviewers before adjudication (AI–AI agreement, not human–AI agreement).

Stage	Units	Agreement	Cohen's κ
Title/metadata screening (database records)	1,877 records	1,821/1,877 (97.0%)	0.88
Full text, three categories (retain/exclude/unsure)	216 reports	187/216 (86.6%)	0.76
Full text, both reviewers definitive	190 reports	181/190 (95.3%)	0.90
Full text, final reviewer pairing (batches 5–6)	46 reports	42/46 (91.3%)	0.83
Google Scholar update, title/snippet	228 records	215/228 (94.3%)	0.77
Google Scholar update, full text	20 reports	19/20	–
October 2026 update search, title stage	52 records	48/52 (92.3%)	0.85
Risk of bias, item level (all batches)	1,094 items	868/1,094 (79.3%)	–

Of the 56 title-stage disagreements, 31 were Reviewer 1 exclude/Reviewer 2 retain and 25 the reverse; in two further records both reviewers retained and adjudication excluded. Among concordant full-text exclusions, the same fixed reason was given in 32 of 43.

B3. Human verification

The author verified 429 AI decisions: all 122 full-text retains and 113 full-text exclusions then on the register, all 12 duplicate removals, and 182 title-stage exclusions (a seeded random 10% sample of 1,770, seed 20260929, plus 5 adjudication overrides). The 107 priority decisions were reviewed row by row from AI-prepared summaries, with the author's decisions given in writing and transcribed by the assistant at the author's request for accessibility reasons; the other 322 were confirmed in a single statement. The

author changed 6 of the 107 decisions: four full-text exclusions became inclusions (two under the rule on secondary analyses and two under the rule on official records), and two title-stage exclusions went to full text, where both were then excluded by the blind **AI** reviewers for lacking Nigeria-specific data. Three later exclusions were author decisions: one multi-country sequence compilation at report-to-study linkage (30 September 2026) and two reports without Nigeria-specific data at extraction (2 October 2026). The author's second pass over all 107 priority decisions, entered directly in a one-row-per-screen tool, reconfirmed all 107 without change; it took about 3.5 minutes and is reported as a reconfirmation, not an independent re-review. The 100% agreement of the blanket confirmation is not reported as an agreement statistic.

B4. Reports not retrieved

Eighty-eight reports were not retrieved from the original searches: 80 from the database searches and 8 from the Google Scholar update search; the October 2026 update search added 10 more, giving 98 in total. Of the 80, 27 were dated 2006–2010, 23 2011–2015, 15 2016–2020 and 15 2021–2026; 66 had a DOI, 19 a PMID and 12 neither; and 40 were known only from the OpenAlex and Crossref exports. On a provisional title-level judgement, about 40 of the 98 appeared to be Nigerian avian laboratory or epidemiological reports; three of these had a counterpart already assessed: a conference abstract of an included Kaduna wild-bird survey, an earlier version of an included 2021–2022 molecular study, and a second record of a report that was assessed in full text and excluded for lacking Nigeria-specific data. About 47 appeared to concern humans, mammals or countries other than Nigeria, risk, biosecurity or economics without laboratory data, or to be reviews. Two records with the same title (2019 and 2020) may be one report recorded twice. The 8 Google Scholar update-search reports comprised a review of H9N2 in Africa, a study of the live wild-bird trade in Kaduna, a phylogeographic study of H5N1 in Nigeria from 2006 to 2023, an economic study of the 2015 resurgence, a preprint of an included wild-bird study, two French-language reports and an impact assessment in Benue State. The 10 October-search reports were mostly older Nigerian reports not held in open repositories: molecular and phylogenetic accounts of the 2006–2008 viruses, spatial or retrospective outbreak reviews (Maiduguri, Lagos), awareness and preventive-practice surveys, and one probable news item. Not-retrieved reports were not assessed for eligibility and are not counted as exclusions on content.

B5. Protocol amendments

Table B2. Dated protocol amendments and author decisions (major and minor as graded in the review's amendment log).

Date (2026)	Grade	Amendment and reason
16 Sep	Major	Search library locked to PubMed (v3), OpenAlex/Crossref, Scopus and ScienceDirect; Web of Science, Google Scholar and AJOL moved to update searches because exports or automated access were not then available.
17 Sep	Minor	Google Scholar supplement run (140 records, 99 unique); it had no persistent export and was later superseded (28 Sep).
19 Sep	Major	Web of Science Starter API base query added (175 records) as a dated source update; Reviewer 2 assigned to an independent Codex agent (named Epicurus in the working log), with the main Codex session as arbiter. [Author to confirm: identity of Reviewer 1 before 27 September 2026]
20 Sep	Major	AJOL on-site search added (79 unique pages); Web of Science browser search run for counts only; operational search date fixed at 20 September 2026.
27 Sep	Major	Rule for unretrievable reports changed for deadline reasons: legal open-access routes plus one institutional-access attempt, with no author contact; reports still without full text recorded as not retrieved.

Date (2026)	Grade	Amendment and reason
27 Sep	Minor	Full-text reviewers changed: Reviewer 1 an executor agent (Claude Opus 5.5), Reviewer 2 a separate blind reviewer agent (Claude Sonnet 5); earlier adjudications unchanged.
28 Sep	Major	Institutional-access step dropped (not available to the team); legal open-access sweep extended; 80 database reports recorded as not retrieved.
28 Sep	Major	Post-freeze Google Scholar update search (re-run of Q1–Q4 with full raw output) replaced the unscreenable 17 September supplement; reported as identification via other methods.
29 Sep	Minor	Reviewer 2 model changed to Claude Sonnet 5.5 from the update-search full-text screen onwards.
29 Sep	Decision	Three eligibility rules set by the author during verification (secondary analyses; official outbreak records; reprinted tables); six priority decisions changed.
30 Sep	Decision	One multi-country sequence compilation excluded by the author (no primary Nigerian data).
2 Oct	Decision	Two reports excluded by the author at extraction (no Nigeria-specific data); their provisional study identifiers retired.
3 Oct	Minor	JBIC Case Series checklist added for outbreak investigations and isolate series; molecular cap rule fixed as a "No" on accessions (M4) or phylogenetic method and support (M5).
4 Oct	Decision	High-risk rule set by the author (author designation, a "No" on the valid-identification item, or abstract-only appraisal).
5 Oct	During synthesis	Sensitivity analysis (iii), treating four reports of uncertain peer-review status as grey literature, added during the synthesis build, before certainty rating.
7 Oct	Post hoc	Sensitivity analysis (iv), restricting state counts to sequence or isolate evidence, added after the independent certainty ratings.
8 Oct	Decision	Retrieval re-opened by the author through legal paid or loan routes (interlibrary loan, British Library On Demand, institutional access); author contact still not used. The document's search files, register, extraction and risk-of-bias data are held in the review's dated working files.
9 Oct	Major	Update search before submission: every database strategy rebuilt with the facet blocks on separate lines and the record caps removed (PubMed, OpenAlex, Crossref, Web of Science Starter API, AJOL), with a PRESS self-assessment; backward and forward citation chasing on all included reports (OpenAlex and Crossref); 52 new records screened, 4 retained as abstract-only studies; the four retained records were extracted, appraised (abstract-only) and added to the synthesis; one further record linked as a probable companion of an included study.

Appendix C. Characteristics of the included studies

Each row is one study; companion reports are cited with the primary report. Design is given as extracted; period is the data or sampling period. States are listed as named in the study (NR, not reported). Host systems: C, commercial poultry; B, backyard or village poultry; L, live-bird market; Wm, wild migratory; Wr, wild resident; Cap, captive; O, other. Evidence: Seq, sequence; Iso, isolate; RNA; Lab, laboratory-confirmed case (method not reported); Off, official report; Ab, antibody; Ag, antigen; Sec, secondary compilation. RoB tools: Prev, JBI Prevalence; CS, JBI Case Series; ACS, JBI Analytical Cross-sectional; none, secondary or modelling study (structured note only). Overall: Inc, Include; Inc-HR, Include (high risk, author decision); Seek, Seek further information (abstract only). Cap: "sugg." where molecular claims are capped as suggestive. HR: meets the high-risk rule. The last column maps each study to its identifier in the review's working files.

Table C1. Characteristics, risk-of-bias tool and judgement of the 113 included studies.

Reference	Design	Period	States	Hosts	Evidence	RoB tool	Overall	Cap	HR	Study ID (reports)
[111]	Cross-sectional survey	2014-11–2014-12	Kano	Wr	Ab, Ag	Prev	Inc	–	–	STU-001 (CAN-0006)
[115]	Cross-sectional survey	2008-08–2009-04	Jigawa	L	Ab	Prev	Inc	–	–	STU-002 (CAN-0007)
[89]	Cross-sectional survey	2012-03–2012-06	Kaduna	Wr, Wm, L, O	Ab	Prev	Inc	–	–	STU-003 (CAN-0008)
[44,45]	Outbreak investigation	2014-12–2015-12	21 states	C, B, L	Iso, Seq	CS	Inc	sugg.	–	STU-004 (CAN-0025, CAN-1902)
[46]	Molecular characterisation	2006–2007	NR	C	Ag, Seq	CS	Inc	sugg.	–	STU-005 (CAN-0031)
[47]	Phylogenetic/ phylogeographic analysis	2006	NR	C	Seq, Sec	none	Inc	sugg.	–	STU-006 (CAN-0033)
[32]	Cross-sectional survey	NR	Plateau	C	Iso	Prev	Inc-HR	–	yes	STU-007 (CAN-0053)
[38,39]	Case series	2006-12–2007-04	Kaduna	C, B	Ag	CS	Inc	–	yes	STU-008 (CAN-0060, CAN-1711)
[61]	Cross-sectional survey	NR	Jigawa	B	Ab, Iso	Prev	Inc	–	–	STU-009 (CAN-0065)
[62]	Cross-sectional survey	2009-10–2010-01	Kogi	B, L	Ab	Prev	Inc	–	–	STU-010 (CAN-0066)
[128]	Spatial-temporal analysis	2006-01–2007-01	Kaduna, Plateau, Kano, (partial)	C, B	Lab	Prev	Inc	–	–	STU-011 (CAN-0070)
[129]	Cross-sectional survey	2008-10–2009-03	Sokoto	L	NR	Prev	Inc	–	–	STU-012 (CAN-0086)
[60]	Other	2006-01–2010	27 states	C, B, L, Wm	Lab, Iso	Prev	Inc	–	–	STU-013 (CAN-0087)
[96,97]	Risk-factor study	2013	Bauchi, Gombe	C, B, L, Wr, Cap	Ab	ACS	Inc	–	–	STU-014 (CAN-0114, CAN-0147)
[16]	Molecular characterisation	2015	NR	NR	Seq	CS	Seek	–	yes	STU-015 (CAN-0129)
[31]	Modelling	2003–2008-12	NR	NR	Sec	none	Inc	–	–	STU-016 (CAN-0132)
[130]	Modelling	2006-01–2007	all 36 states and FCT (between-state model), Ogun (between-premises model)	C	Off	none	Inc	–	–	STU-017 (CAN-0142)
[13]	Cross-sectional survey	2016-03–2018-02	Kwara	L, Cap	Ab	Prev	Inc	–	–	STU-018 (CAN-0146)
[14]	Phylogenetic/ phylogeographic analysis	2006–2021	Kano, Kaduna, Plateau, (partial)	NR	Off, Seq, Sec	none	Inc	–	–	STU-019 (CAN-0155)
[83]	Spatial-temporal analysis	2014-12–2016-11	12 states	C, B, L	Off	Prev	Inc	–	–	STU-020 (CAN-0163)
[27]	Risk-factor study	2015-02–2015-05	NR (six South-west Nigerian states, not named)	L	Lab	ACS	Inc-HR	–	yes	STU-021 (CAN-0174)
[81,82]	Case series	2021-02–2021-08	Bauchi, Lagos, Sokoto, Delta	O, B, Cap	NR	CS	Inc	–	–	STU-022 (CAN-0191, CAN-1882)
[78,79]	Molecular characterisation	2015-01–2016-06	17 states	C, L	Seq	CS	Inc	–	–	STU-023 (CAN-0226, CAN-2139)
[131]	Risk-factor study	2006-02–2007-10	7 states	NR	Lab, Sec, Ab	ACS	Inc	–	–	STU-024 (CAN-0243)
[33]	Spatial-temporal analysis	2006–2008	NR	NR	Lab	Prev	Seek	–	yes	STU-025 (CAN-0244)

Reference	Design	Period	States	Hosts	Evidence	RoB tool	Overall	Cap	HR	Study ID (reports)
[86]	Economic	2006-01–2017-05	NR (national)	NR	Off	none	Inc	–	–	STU-026 (CAN-0266)
[34]	Phylogenetic/ phylogeographic analysis	NR	NR	NR	Seq	none	Seek	–	yes	STU-027 (CAN-0282)
[122,123]	Cross-sectional survey	NR	Kaduna	B	Ab	Prev	Inc	–	–	STU-028 (CAN-0329, CAN-1785)
[35]	Cross-sectional survey	NR	Oyo, Ogun	L	Ab, Iso	Prev	Seek	–	yes	STU-029 (CAN-0330)
[132]	Spatial-temporal analysis	2006-01–2006-06	NR (infected farms nationwide; states not listed)	NR	Off	Prev	Inc	–	–	STU-030 (CAN-0339)
[40]	Spatial-temporal analysis	2006-01–2006-12	21 states	C, B	Off	Prev	Inc	–	yes	STU-031 (CAN-0344)
[64]	Phylogenetic/ phylogeographic analysis	2006–2017-02	Kano, Bauchi, Plateau, Kaduna, Nasarawa, (partial)	NR	Seq, Sec, Iso	none	Inc	–	–	STU-033 (CAN-0379)
[58]	Outbreak investigation	2006-01	Kaduna	C	Iso, Seq	CS	Inc	–	–	STU-034 (CAN-0423)
[48]	Phylogenetic/ phylogeographic analysis	2019-04–2019-11	NR (Nigerian GISAID sequences; strain-name prefixes RV, KD, OG, AB, OY, IM, SK, EN, LA not decoded in the report)	NR	Seq, Sec	none	Inc	sugg.	–	STU-035 (CAN-0527)
[101]	Active surveillance	2007-12–2008-04	Plateau, NR (Hadejia-Nguru wetlands site; state not named)	Wm, Wr, B	Seq	Prev	Inc	–	–	STU-036 (CAN-0539)
[49]	Molecular characterisation	2006-01–2006-08	Oyo, Ogun, Lagos	C	Iso, Seq, Sec	Prev	Inc	sugg.	–	STU-037 (CAN-0541)
[12]	Cross-sectional survey	2022-03–2022-04	Jigawa, Yobe	Wr, Wm	Iso, Seq	Prev	Inc	–	–	STU-038 (CAN-0565)
[19]	Molecular characterisation	2023-01–2025-04	10 states	NR	Seq, Off	CS	Inc	–	–	STU-039 (CAN-0568)
[104]	Phylogenetic/ phylogeographic analysis	2006-01–2008-07	28 states	C, B, L	Seq	CS	Inc	–	–	STU-040 (CAN-0654)
[125]	Cross-sectional survey	2009–2012	Bauchi, Gombe	C, B, L	Ab, Ag	Prev	Inc	–	–	STU-041 (CAN-0712)
[98]	Cross-sectional survey	NR	Oyo, Osun, Ondo	C, B	Ab	Prev	Inc	–	–	STU-042 (CAN-0719)
[90]	Secondary database analysis	2006-01–2023-06	all 36 states + FCT reported animal disease events; HPAI reported in 36 states of the Federation except Cross River and Ondo; named for HPAI: Plateau, Kano, Lagos, (partial)	C, B, Wm, Wr	Off	Prev	Inc	–	–	STU-043 (CAN-0736)
[133]	Other	2015-01–2017-12	Enugu	C, L, Cap	Lab	Prev	Inc	–	–	STU-044 (CAN-0744)
[84]	Spatial-temporal analysis	2015-01–2017-12	10 states	C, L, Cap	Off	Prev	Inc	–	–	STU-045 (CAN-0745)
[134]	Risk-factor study	2022-01	Ebonyi	C, B, L	Lab, Off	none	Inc	–	–	STU-046 (CAN-0746)
[41]	Other	2006-01–2009-03	25 states + FCT with cases (not individually named); LBM surveillance covered all 36 states + FCT, (partial)	L, C, B	Lab, Iso	Prev	Inc	–	yes	STU-047 (CAN-0751)
[50]	Phylogenetic/ phylogeographic analysis	2005-01–2011-09	NR (Nigeria analysed as a single country unit in a global study; no Nigerian states given)	NR	Seq, Sec	none	Inc	sugg.	–	STU-048 (CAN-0768)
[87]	Active surveillance	2011-07–2012-07	Oyo	L	Iso, Seq	Prev	Inc	–	–	STU-049 (CAN-0782)
[103]	Cross-sectional survey	NR	Osun, Oyo	C, L, O	Ab	Prev	Inc	–	–	STU-050 (CAN-0825)
[100]	Spatial-temporal analysis	2006-01–2007-12	14 states	C, B	Lab, Off	Prev	Inc	–	–	STU-051 (CAN-0835)
[108]	Phylogenetic/ phylogeographic analysis	2006	NR (Nigeria analysed as a single locality; strain names carry SO, BA and numeric codes only)	NR	Seq, Sec	none	Inc	–	–	STU-052 (CAN-0859)
[107]	Molecular characterisation	2006-01–2007-02	12 states	NR	Seq	CS	Inc	–	–	STU-053 (CAN-0863)
[109]	Spatial-temporal analysis	2003-11–2006-12	NR (Nigeria analysed at country level; no state for the Nigerian sequences)	NR	Seq, Sec	none	Inc	–	–	STU-054 (CAN-0869)
[51]	Other	2006-02–2007-11	NR (country-level strain metadata only)	NR	Sec	none	Inc	sugg.	–	STU-055 (CAN-0902)
[88]	Secondary database analysis	2004–2016	NR (country-level panel)	NR	Off, Sec	none	Inc	–	–	STU-056 (CAN-0911)

Reference	Design	Period	States	Hosts	Evidence	RoB tool	Overall	Cap	HR	Study ID (reports)
[63]	Active surveillance	2007-02	Jigawa	Wr, Wm	Seq	Prev	Inc	–	–	STU-057 (CAN-0924)
[112]	Cross-sectional survey	2011-08–2011-09	Plateau	C	Ab, Lab	Prev	Inc	–	–	STU-058 (CAN-0947)
[68]	Active surveillance	2013-03–2013-10	Oyo	C	Ab	Prev	Inc	–	–	STU-059 (CAN-1293)
[135]	Secondary database analysis	2015–2022	Rivers	C	Off	Prev	Inc	–	–	STU-060 (CAN-1347)
[52]	Active surveillance	2006-01–2008-07	NR (national; 25 states + FCT reported infection; all 36 states + FCT sampled in active surveillance)	C, B, L	Iso, Off, Ab, Lab, Seq	Prev	Inc	sugg.	–	STU-061 (CAN-1357)
[73]	Molecular characterisation	2006	NR (strain names only; no Nigerian locations given)	NR	Seq	CS	Inc	–	–	STU-062 (CAN-1444)
[105]	Molecular characterisation	2007-01–2007-02	8 states	NR	Seq	CS	Inc	–	–	STU-063 (CAN-1446)
[106]	Molecular characterisation	2007-06–2007-11	Ogun, Lagos, Oyo, Ekiti	C	Seq	CS	Inc	–	–	STU-064 (CAN-1451)
[53]	Molecular characterisation	2006-01	Oyo	C	Seq	CS	Inc	sugg.	–	STU-065 (CAN-1452)
[76]	Molecular characterisation	2015-01	Kano, Lagos	B, L	Seq, Off	CS	Inc	–	–	STU-066 (CAN-1471)
[77]	Molecular characterisation	2015-01	Lagos	NR	Seq	CS	Inc	–	–	STU-067 (CAN-1473)
[126]	Active surveillance	2021-11–2021-12	Borno	B	NR	Prev	Inc	–	–	STU-068 (CAN-1501)
[110]	Secondary database analysis	2006–2015	Taraba, Kaduna, Ogun, Lagos	C, L, B, O	Sec	none	Inc	–	–	STU-069 (CAN-1535)
[65–67]	Active surveillance	2019-04–2019-12	10 states	L	Seq, Iso	Prev	Inc	–	–	STU-071 (CAN-1585, CAN-0491, CAN-1484)
[11]	Molecular characterisation	2021-01–2022-12	Bauchi, Kano, (partial)	C, B, O	Seq	CS	Inc	–	–	STU-072 (CAN-1602)
[136]	Outbreak investigation	2021-01	Kaduna	B	Ab, Off	CS	Inc	–	–	STU-073 (CAN-1630)
[137]	Case series	2021-01–2021-02	Plateau	C	Iso	CS	Inc	–	–	STU-074 (CAN-1637)
[138]	Secondary database analysis	2015-01–2017-12	Kano	NR, L	Off	Prev	Inc	–	–	STU-075 (CAN-1650)
[124]	Cross-sectional survey	2012-09–2013-02	Kogi	B, L	Ab	Prev	Inc	–	–	STU-076 (CAN-1662)
[80]	Outbreak investigation	2006-02–2006-05	Bauchi	C, B	Lab	CS	Inc	–	–	STU-077 (CAN-1666)
[139]	Secondary database analysis	2006-01–2006-03	Kano, Kaduna, Katsina, Jigawa	C, B	Off	Prev	Inc	–	–	STU-078 (CAN-1667)
[140]	Outbreak investigation	2007-01–2007-04	Adamawa	B	Lab	Prev	Inc	–	–	STU-079 (CAN-1672)
[28]	Spatial-temporal analysis	2006-01–2007-12	Kaduna, Kano, Jigawa, Yobe, Federal Capital Territory, (partial)	C, B, Wr	Sec	none	Inc	–	–	STU-080 (CAN-1678)
[29]	Spatial-temporal analysis	2003-12–2006-12	NR (Nigeria analysed as one of many countries; no state named)	C, B, Wm	Sec	none	Inc	–	–	STU-081 (CAN-1679)
[54]	Case series	2021-12–2022-03	Bauchi, Kaduna, Nasarawa, Plateau	C, B	Iso	CS	Inc	sugg.	–	STU-082 (CAN-1693)
[69]	Outbreak investigation	2006-12	Kano	B	Ag, Iso	CS	Inc	–	–	STU-083 (CAN-1710)
[74,75]	Active surveillance	2008-05–2008-07	11 states	L	Iso, Seq, Ab	Prev	Inc	–	–	STU-084 (CAN-1712, CAN-1453)
[55]	Other	2006–2016	NR (national review; Kebbi, Taraba, Bayelsa, Sokoto, Benue, Gombe and Hadejia-Nguru wetland states mentioned narratively)	C, B, L, Wm	Seq, Sec	none	Inc	sugg.	–	STU-085 (CAN-1714)
[117]	Cross-sectional survey	2015–2016	Lagos, Ogun, Plateau, Bauchi	C	Ab	Prev	Inc	–	–	STU-086 (CAN-1715, CAN-0058)
[119]	Cross-sectional survey	2018-05–2018-07	Plateau	L	Ab	Prev	Inc	–	–	STU-087 (CAN-1716)
[56]	Outbreak investigation	2021-05–2022-10	Lagos	C, B, L	NR	CS	Inc	sugg.	–	STU-088 (CAN-1717)
[141]	Secondary database analysis	2017–2021	Plateau	C	Off	Prev	Inc	–	–	STU-089 (CAN-1719)
[142]	Risk-factor study	2006–2009	Kano	C	Off	ACS	Inc	–	–	STU-090 (CAN-1724)
[99]	Cross-sectional survey	NR	Kogi	B, L, Wm, Wr	Ab	Prev	Inc	–	–	STU-091 (CAN-1729)
[102]	Cross-sectional survey	2015–2016	Bauchi, Gombe, Kaduna	Wr, Cap, L, C	Ab	Prev	Inc	–	–	STU-092 (CAN-1730)
[57]	Case series	2019-02	Plateau	C, B	Iso	CS	Inc	sugg.	–	STU-093 (CAN-1731)

Reference	Design	Period	States	Hosts	Evidence	RoB tool	Overall	Cap	HR	Study ID (reports)
[114]	Cross-sectional survey	NR	Kaduna	Wr, L	Ab	Prev	Inc	–	–	STU-094 (CAN-1734)
[120]	Cross-sectional survey	2020-03–2020-07	Plateau	L	Ab	Prev	Inc	–	–	STU-095 (CAN-1735)
[113]	Cross-sectional survey	2021-09–2022-08	Oyo	B	Ab	Prev	Inc	–	–	STU-096 (CAN-1737)
[121]	Cross-sectional survey	NR	Kaduna	L	Ab	Prev	Inc	–	–	STU-097 (CAN-1738)
[116]	Cross-sectional survey	2018-04–2018-12	Kaduna	Wr, L	Ab	Prev	Inc	–	–	STU-098 (CAN-1778)
[18]	Case series	2024-02	Oyo	C	NR	CS	Inc	–	–	STU-099 (CAN-1782)
[85]	Secondary database analysis	2015-01–2022-12	Plateau	C, B	Off	Prev	Inc	–	–	STU-100 (CAN-1796)
[91]	Case series	2020-01–2021-04	Kaduna, Kano	C	Ab, Iso	CS	Inc	–	–	STU-101 (CAN-1868)
[118]	Other	2006-02–2008-03	national (states not named)	B, L, C	Sec, Off	none	Inc	–	–	STU-102 (CAN-1875)
[70]	Cross-sectional survey	2024-10–2025-07	Kaduna	C	Ag	Prev	Inc	–	–	STU-103 (CAN-1884)
[42]	Secondary database analysis	2006–2008	21 states	B, C	Lab	ACS	Inc	–	yes	STU-104 (CAN-1907)
[143]	Other	2006-01–2016	7 states	NR	Sec	none	Inc	–	–	STU-105 (CAN-1914)
[127]	Cross-sectional survey	NR	Borno	L, B	NR	Prev	Inc	–	–	STU-106 (CAN-1919)
[43]	Cross-sectional survey	NR	NR	O	Ab	Prev	Inc	–	yes	STU-107 (CAN-1995)
[71,72]	Phylogenetic/ phylogeographic analysis	2006-02–2007-01	Plateau, Kaduna, Lagos, Oyo, Sokoto	C, L, B, Wr, Cap	Iso, Seq	CS	Inc	–	–	STU-108 (CAN-2005, CAN-0337)
[144]	Secondary database analysis	2015-01–2017	Plateau	C, B	Off	Prev	Inc	–	–	STU-109 (CAN-2009)
[92–95]	Passive surveillance/lab submissions	2006-01–2008-12	23 states	C, B, L, O	Lab	CS	Inc	–	–	STU-110 (CAN-2033, CAN-0714, CAN-1845, CAN-0039)
[59]	Case series	2006-01–2006-07	Kaduna, Plateau, Kano	C	Lab, Off	CS	Inc	–	–	STU-111 (CAN-2038)
[145]	Passive surveillance/lab submissions	2006-01–2006	NR	C, B	Lab	CS	Inc	–	–	STU-112 (CAN-2056)
[36]	Retrospective outbreak-record survey	2006-11–2007-01	Borno	NR	Off	CS	Seek	–	yes	STU-113 (CAN-2131)
[30]	Mixed secondary analyses	2006–2007	NR (Nigeria premises and live bird markets; states not named in the abstract)	C, B, L, O	NR	ACS	Seek	–	yes	STU-114 (CAN-2134)
[37]	Cross-sectional seroprevalence survey	NR	Borno	B, C, O	Ab	Prev	Seek	–	yes	STU-115 (CAN-2137)

Appendix D. Full-text reports excluded, with reasons

All 114 reports excluded after full-text assessment (108 from the database searches and 6 from the update search), with the single fixed reason assigned. Titles are given as recorded in the screening register; register identifiers allow tracing to the dated decision records. "Author" marks exclusions made by the author after the **AI** decision (one at report-to-study linkage and two at extraction).

Table D1. Reports excluded after full-text assessment (n = 117), with the reason for exclusion.

Register ID	Year	Title (as recorded in the register)	DOI	Search arm	Reason for exclusion
CAN-0003	2009	Analyses of the poultry value chain and its linkages and interactions with HPAI risk factors in Nigeria	–	Databases	No laboratory-confirmed avian AIV
CAN-0020	2010	Distribution of Aerobic Bacteria in Visceral Organs of Poultry Affected By Highly Pathogenic Avian Influenza (H5N1) in Nigeria	–	Databases	No epidemiological, molecular or spatiotemporal data

Register ID	Year	Title (as recorded in the register)	DOI	Search arm	Reason for exclusion
CAN-0026	2008	Epidemiology of avian influenza in Africa (EPIAAF) survey : Midterm report LoA PR 37212 between FAO and CIRAD, May 2008	–	Databases	No laboratory-confirmed avian AIV
CAN-0045	2009	Mapping the Likelihood of Introduction and Spread of Highly Pathogenic Avian Influenza Virus H5N1 in Africa and Indonesia using Multicriteria Decision Modelling	–	Databases	No laboratory-confirmed avian AIV
CAN-0046	2009	Mapping the Likelihood of Introduction and Spread of Highly Pathogenic Avian Influenza Virus H5N1 in Africa, Ghana, Ethiopia, Kenya and Nigeria using Multicriteria Decision Modelling	–	Databases	No laboratory-confirmed avian AIV
CAN-0049	2011	Participatory animal disease surveillance, Panacea to the bane of animal disease under-reporting in Nigeria	–	Databases	No laboratory-confirmed avian AIV
CAN-0052	2009	Prevalence and risk factors for avian influenza and Newcastle disease viruses in domestic poultry in seven African countries with outbreaks of H5N1 HPAI	–	Databases	No Nigeria-specific data
CAN-0054	2011	Prevalence of Avian Origin H5 and H7 Influenza Virus Antibodies in Dogs in Ibadan and Sagamu, Southwestern Nigeria	–	Databases	No avian-host field data
CAN-0055	2010	Qualitative Risk Assessment on the transmission of HPAI (H5N1) virus from backyard and medium-scale commercial farms to household free-range poultry in Nigeria	–	Databases	No primary data
CAN-0056	2008	Reassortant avian influenza virus (H5N1) in Poultry, Nigeria, 2007 (Emerging Infectious Diseases 14, 4)	–	Databases	Duplicate dataset (primary report included)
CAN-0062	2009	Serological evidence of avian influenza viruses in pigs in south-western Nigeria.	–	Databases	No avian-host field data
CAN-0063	2008	Serological evidence of influenza virus infections in domestic animals and birds in North-Eastern Nigeria	–	Databases	No laboratory-confirmed avian AIV
CAN-0071	2013	Studies of some risk factors for re-introduction and spread of highly-pathogenic avian influenza in two states of Nigeria	–	Databases	No laboratory-confirmed avian AIV
CAN-0076	2009	The role of value chain analysis in animal disease impact studies: methodology and case studies of Rift Valley fever in Kenya and avian influenza in Nigeria	–	Databases	No laboratory-confirmed avian AIV
CAN-0097	2026	Cost of migration increased during an outbreak of highly pathogenic avian influenza virus	10.1002/jav.03616	Databases	No Nigeria-specific data
CAN-0098	2013	Serologic evidence of avian influenza virus infections among Nigerian agricultural workers	10.1002/jmv.23520	Databases	No avian-host field data
CAN-0113	2017	Awareness and mitigation measures on highly pathogenic avian influenza in pastoral poultry flocks of North-central Nigeria: Any challenging gap?	10.1002/vms3.67	Databases	No avian-host field data
CAN-0115	2011	Epidemiology of Highly Pathogenic Avian Influenza Virus Strain Type H5N1	10.1007/978-1-4419-7077-0_10	Databases	No primary data
CAN-0116	2013	Modeling the Spread and Outbreak Dynamics of Avian Influenza (H5N1) Virus and Its Possible Control	10.1007/978-1-4614-9224-5_9	Databases	No primary data
CAN-0126	2011	Identification of amino acids in highly pathogenic avian influenza H5N1 virus hemagglutinin that determine avian influenza species specificity	10.1007/s00705-011-1056-2	Databases	No Nigeria-specific data
CAN-0151	2011	Host-specific genetic variation of highly pathogenic avian influenza viruses (H5N1)	10.1007/s11262-011-0583-y	Databases	No Nigeria-specific data
CAN-0156	2022	Environmental factors and spatiotemporal distribution characteristics of the global outbreaks of the highly pathogenic avian influenza H5N1	10.1007/s11356-022-19016-1	Databases	No Nigeria-specific data
CAN-0187	2025	Effectiveness of non-pharmaceutical control strategies in highly pathogenic avian influenza (H5N1, H5N8) outbreaks	10.1016/j.clinpr.2025.100504	Databases	No primary data
CAN-0232	2023	Global epidemiology of animal influenza infections with explicit virus subtypes until 2016: A spatio-temporal descriptive analysis	10.1016/j.onehlt.2023.100514	Databases	No avian-host field data
CAN-0234	2023	An assessment of the highly pathogenic avian influenza resurgence at human-poultry-environment interface in North-central Nigeria: Sociocultural determinants and One Health implications	10.1016/j.onehlt.2023.100574	Databases	No avian-host field data
CAN-0246	2016	Estimating the sensitivity of passive surveillance for HPAI H5N1 in Bayelsa state, Nigeria	10.1016/j.prevetmed.2016.05.003	Databases	No laboratory-confirmed avian AIV
CAN-0258	2022	First isolation of influenza a subtype H5N8 in ostrich: pathological and genetic characterization	10.1016/j.psj.2022.102156	Databases	No Nigeria-specific data
CAN-0269	2024	Avian Influenza outbreaks: Human infection risks for beach users - One health concern and environmental surveillance implications	10.1016/j.scitotenv.2024.173692	Databases	No primary data
CAN-0340	2010	Serosurveillance study on transmission of H5N1 virus during a 2006 avian influenza epidemic	10.1017/s095026880999166x	Databases	No avian-host field data
CAN-0343	2013	Distribution and dynamics of risk factors associated with highly pathogenic avian influenza H5N1	10.1017/s0950268813000101	Databases	No Nigeria-specific data
CAN-0345	2014	Avian influenza surveillance in Central and West Africa, 2010–2014	10.1017/s0950268814003586	Databases	No Nigeria-specific data

Register ID	Year	Title (as recorded in the register)	DOI	Search arm	Reason for exclusion
CAN-0356	2025	A Geospatial Perspective Toward the Role of Wild Bird Migrations and Global Poultry Trade in the Spread of Highly Pathogenic Avian Influenza H5N1	10.1029/2024gh001296	Databases	No Nigeria-specific data (author)
CAN-0381	2025	Surveillance of avian influenza through bird guano in remote regions of the global south to uncover transmission dynamics	10.1038/s41467-025-59322-z	Databases	No Nigeria-specific data
CAN-0388	2018	Evidence of exposure of domestic pigs to Highly Pathogenic Avian Influenza H5N1 in Nigeria	10.1038/s41598-018-24371-6	Databases	No avian-host field data
CAN-0390	2021	Dynamics of inter-farm transmission of highly pathogenic avian influenza H5N6 integrating vehicle movements and phylogenetic information	10.1038/s41598-021-03284-x	Databases	No Nigeria-specific data
CAN-0396	2026	The role of wild birds in the global highly pathogenic avian influenza H5 panzootic, 2020–2023	10.1038/s44185-025-00114-5	Databases	No primary data
CAN-0399	2016	Southward autumn migration of waterfowl facilitates cross-continental transmission of the highly pathogenic avian influenza H5N1 virus	10.1038/srep30262	Databases	No Nigeria-specific data
CAN-0416	2009	Impacts of avian influenza virus on animal production in developing countries.	10.1079/pavsnr20083080	Databases	No primary data
CAN-0483	2024	Comparative examination of a rapid immunocytochemical test for the detection of highly pathogenic avian influenza virus in domestic birds in field outbreaks	10.1080/03079457.2024.2320699	Databases	No Nigeria-specific data
CAN-0485	2007	Genomic Analysis and Geographic Visualization of the Spread of Avian Influenza (H5N1)	10.1080/10635150701266848	Databases	No Nigeria-specific data
CAN-0492	2021	Antigenic and molecular characterization of low pathogenic avian influenza A(H9N2) viruses in sub-Saharan Africa from 2017 through 2019	10.1080/22221751.2021.1908097	Databases	No Nigeria-specific data
CAN-0496	2024	A multi-species, multi-pathogen avian viral disease outbreak event: Investigating potential for virus transmission at the wild bird – poultry interface	10.1080/22221751.2024.2348521	Databases	No Nigeria-specific data
CAN-0499	2006	Influenza Outbreak among Health Care Workers in an Avian Influenza (H5N1)-Endemic Setting	10.1086/508885	Databases	No Nigeria-specific data
CAN-0536	2016	Antibody responses to avian influenza viruses in wild birds broaden with age	10.1098/rspb.2016.2159	Databases	No Nigeria-specific data
CAN-0543	2014	Seasonality dynamics of avian influenza occurrences in Central and West Africa	10.1101/007740	Databases	No Nigeria-specific data
CAN-0549	2023	Association of poultry vaccination with the interspecies transmission and molecular evolution of H5 subtype avian influenza virus	10.1101/2023.12.20.572711	Databases	No Nigeria-specific data
CAN-0570	2010	Potential spread of highly pathogenic avian influenza H5N1 by wildfowl: dispersal ranges and rates determined from large-scale satellite telemetry	10.1111/j.1365-2664.2010.01845.x	Databases	No Nigeria-specific data
CAN-0599	2007	Control versus no control: Options for avian influenza H5N1 in Nigeria	10.1111/j.1863-2378.2007.01044.x	Databases	No avian-host field data
CAN-0630	2020	Factors influencing chicken farmers' decisions to implement prevention and control measures to reduce avian influenza virus spread under endemic conditions	10.1111/tbed.13757	Databases	No Nigeria-specific data
CAN-0636	2015	Development of Disease-specific, Context-specific Surveillance Models: Avian Influenza (H5N1)-Related Risks and Behaviours in African Countries	10.1111/zph.12200	Databases	No avian-host field data
CAN-0639	2026	Biosecurity Uptake and Perceived Risk of Avian Influenza Among People in Contact With Birds	10.1111/zph.70034	Databases	No Nigeria-specific data
CAN-0640	2025	Association of poultry vaccination with interspecies transmission and molecular evolution of H5 subtype avian influenza virus	10.1126/sciadv.ado9140	Databases	No Nigeria-specific data
CAN-0646	2022	Resurgence of avian influenza virus	10.1126/science.abo1232	Databases	No Nigeria-specific data
CAN-0648	2016	Outbreak of H7N8 Low Pathogenic Avian Influenza in Commercial Turkeys with Spontaneous Mutation to Highly Pathogenic Avian Influenza	10.1128/genomea.00457-16	Databases	No Nigeria-specific data
CAN-0651	2024	Spatiotemporal genotype replacement of H5N8 avian influenza viruses contributed to H5N1 emergence in 2021/2022 panzootic	10.1128/jvi.01401-23	Databases	No Nigeria-specific data
CAN-0707	2012	Situation-based survey of avian influenza viruses in possible "bridge" species of wild and domestic birds in Nigeria.	10.1155/2012/567601	Databases	No laboratory-confirmed avian AIV
CAN-0738	2025	Avian Influenza Screening in Captive Wild Birds and Biosecurity Appraisal of Zoological Gardens in Southwestern Nigeria	10.1155/vmi/3419266	Databases	No laboratory-confirmed avian AIV
CAN-0779	2012	Improving global influenza surveillance: trends of A(H5N1) virus in Africa and Asia	10.1186/1756-0500-5-62	Databases	No avian-host field data
CAN-0851	2014	Adaptive Evolution and Environmental Durability Jointly Structure Phylodynamic Patterns in Avian Influenza Viruses	10.1371/journal.pbio.1001931	Databases	No Nigeria-specific data
CAN-0864	2009	Chances and Limitations of Wild Bird Monitoring for the Avian Influenza Virus H5N1 — Detection of Pathogens Highly Mobile in Time and Space	10.1371/journal.pone.0006639	Databases	No Nigeria-specific data
CAN-0871	2010	Spatio-Temporal Data Comparisons for Global Highly Pathogenic Avian Influenza (HPAI) H5N1 Outbreaks	10.1371/journal.pone.0015314	Databases	No Nigeria-specific data
CAN-0885	2012	The Evolutionary Pattern of Glycosylation Sites in Influenza Virus (H5N1) Hemagglutinin and Neuraminidase	10.1371/journal.pone.0049224	Databases	No Nigeria-specific data

Register ID	Year	Title (as recorded in the register)	DOI	Search arm	Reason for exclusion
CAN-0891	2013	Metapopulation Dynamics Enable Persistence of Influenza A, Including A/H5N1, in Poultry	10.1371/journal.pone.0080091	Databases	No Nigeria-specific data
CAN-0896	2015	Adamantane-Resistant Influenza A Viruses in the World (1902–2013): Frequency and Distribution of M2 Gene Mutations	10.1371/journal.pone.0119115	Databases	No Nigeria-specific data
CAN-0915	2019	Phylogeographic evidence for the inter- and intracontinental dissemination of avian influenza viruses via migration flyways	10.1371/journal.pone.0218506	Databases	No Nigeria-specific data
CAN-0922	2007	Spatial, Temporal, and Species Variation in Prevalence of Influenza A Viruses in Wild Migratory Birds	10.1371/journal.ppat.0030061	Databases	No Nigeria-specific data
CAN-0923	2007	Influenza in Migratory Birds and Evidence of Limited Intercontinental Virus Exchange	10.1371/journal.ppat.0030167	Databases	No Nigeria-specific data
CAN-0925	2008	Evolutionary Dynamics and Emergence of Panzootic H5N1 Influenza Viruses	10.1371/journal.ppat.1000161	Databases	No Nigeria-specific data
CAN-0930	2014	Genetics, Receptor Binding Property, and Transmissibility in Mammals of Naturally Isolated H9N2 Avian Influenza Viruses	10.1371/journal.ppat.1004508	Databases	No Nigeria-specific data
CAN-0935	2017	Public health concerns of highly pathogenic avian influenza H5N1 endemicity in Africa	10.14202/vetworld.2017.1194-1204	Databases	No primary data
CAN-0943	2017	Human avian influenza virus serotypes in a tertiary hospital in Nigeria	10.14303/jmms.2017.018	Databases	Non-AIV
CAN-0946	2014	Evolution and Spread of Avian Influenza H5N1 Viruses	10.14737/journal.aavs/2014/2.4s.33.41	Databases	No Nigeria-specific data
CAN-0948	2016	Retrospective Study of Selected Endemic Viral Diseases of Poultry Diagnosed in Maiduguri North-Eastern Nigeria	10.14737/journal.jahp/2016/4.2.60.64	Databases	No laboratory-confirmed avian AIV
CAN-1033	2016	Ecology of Avian Influenza Virus in Wild Birds in Tropical Africa	10.1637/11149-051115-review	Databases	No Nigeria-specific data
CAN-1107	2007	Computational Identification of Reassortments in Avian Influenza Viruses	10.1637/7625-042706r1.1	Databases	No Nigeria-specific data
CAN-1152	2010	Isolation and Identification of Highly Pathogenic Avian Influenza Virus Subtype H5N1 in Peafowl (Pavo cristatus)	10.1637/8814-040109-resnote.1	Databases	No Nigeria-specific data
CAN-1296	2023	The episodic resurgence of highly pathogenic avian influenza H5 virus	10.21203/rs.3.rs-2434849/v1	Databases	No Nigeria-specific data
CAN-1346	2026	Risk Factors, Perception, and Knowledge of Highly Pathogenic Avian Influenza among Poultry Handlers in Selected Live Bird Markets in Nigeria	10.25259/rvsm_3_2026	Databases	No laboratory-confirmed avian AIV
CAN-1352	2021	INVESTIGATING THE BIOSECURITY MEASURES' APPLICATIONS IN POULTRY FARMS AND ITS RELATIONSHIP WITH THE OCCURENCE OF AVIAN INFLUENZA	10.26873/svr-1451-2021	Databases	No Nigeria-specific data
CAN-1416	2019	Estimation of Pathological and Molecular Findings in Vaccinated and Non-Vaccinated Chickens Challenged with Highly Pathogenic Avian Influenza H5N1 Virus	10.29261/pakvetj/2018.112	Databases	No Nigeria-specific data
CAN-1442	2007	Avian Influenza Viruses in Water Birds, Africa1	10.3201/cid1304.061011	Databases	No Nigeria-specific data
CAN-1449	2008	Duck Migration and Past Influenza A (H5N1) Outbreak Areas	10.3201/cid1407.071477	Databases	No laboratory-confirmed avian AIV
CAN-1457	2009	Ducks as Sentinels for Avian Influenza in Wild Birds	10.3201/cid1510.090439	Databases	No Nigeria-specific data
CAN-1466	2012	Surveillance for Influenza Viruses in Poultry and Swine, West Africa, 2006–2008	10.3201/cid1809.111296	Databases	No Nigeria-specific data
CAN-1502	2023	Avian Influenza Virus A Subtype H5N1	10.33096/umj.v8i1.246	Databases	No Nigeria-specific data
CAN-1534	2023	Serological evidence of virus infection in Eidolon helvum fruit bats: implications for bushmeat consumption in Nigeria	10.3389/fpubh.2023.1283113	Databases	No avian-host field data
CAN-1537	2021	Estimation of the Basic Reproduction Numbers of the Subtypes H5N1, H5N8, and H5N6 During the Highly Pathogenic Avian Influenza Epidemic Spread Between Farms	10.3389/fvets.2021.597630	Databases	No Nigeria-specific data
CAN-1556	2025	Occupational Risk from Avian Influenza Viruses at Different Ecological Interfaces Between 1997 and 2019	10.3390/microorganisms13061391	Databases	No primary data
CAN-1558	2019	The Emergence and Decennary Distribution of Clade 2.3.4.4 HPAI H5Nx	10.3390/microorganisms7060156	Databases	No Nigeria-specific data (author)
CAN-1563	2021	Serologic Evidence of Occupational Exposure to Avian Influenza Viruses at the Wildfowl/Poultry/Human Interface	10.3390/microorganisms9102153	Databases	No Nigeria-specific data
CAN-1567	2022	Seroprevalence of Influenza A Virus in Dromedaries in North-Western Nigeria	10.3390/pathogens11121476	Databases	No avian-host field data
CAN-1610	2025	Detection of Avian Influenza Virus in Pigeons	10.3390/v17040585	Databases	No primary data (author)
CAN-1623	2018	Avian Respiratory Coinfection and Impact on Avian Influenza Pathogenicity in Domestic Poultry: Field and Experimental Findings	10.3390/vetsci5010023	Databases	No primary data
CAN-1627	2020	Performance of Avian Influenza Surveillance System, Ogun State Nigeria, 2015-2019	10.34257/gjmgvol20is1pg25	Databases	No laboratory-confirmed avian AIV
CAN-1651	2020	Serosurveillance for Avian Influenza in Local Chickens in Households and Live Bird Markets in Enugu State, Nigeria	10.37284/eajab.1.1.52	Databases	No laboratory-confirmed avian AIV
CAN-1692	2018	Prevention of Avian Influenza Virus by Ultraviolet Radiation and Prediction of Outbreak by Satellite Parameters	10.4236/jbise.2018.117015	Databases	No Nigeria-specific data

Register ID	Year	Title (as recorded in the register)	DOI	Search arm	Reason for exclusion
CAN-1713	2011	Salmonella Gallinarum Infection in Poultry Affected by Highly Pathogenic Avian Influenza Virus H5N1 in Nigeria	10.4314/nvj.v31i3.68961	Databases	No epidemiological, molecular or spatiotemporal data
CAN-1720	2018	Intravenous pathogenicity of influenza virus A/H5N1/2014 isolated from pig in Ogbomoso, Nigeria	10.4314/ovj.v8i3.16	Databases	Experimental only, without field relevance
CAN-1725	2012	Serosurvey of antibody to highly pathogenic avian influenza (H5N1) in pigs, north central Nigeria	10.4314/sokjvs.v10i2.9	Databases	No avian-host field data
CAN-1732	2020	Biosecurity challenges in the control of avian influenza in Nigeria	10.4314/sokjvs.v18i3.5	Databases	No laboratory-confirmed avian AIV
CAN-1745	2021	Effect of Avian Influenza H9N2 Subtype Virus Infection on Backyard Poultry Production	10.47262/sl/9.1.132021005	Databases	No Nigeria-specific data
CAN-1799	2008	Risk Mapping of Highly Pathogenic Avian Influenza Distribution and Spread	10.5751/es-02532-130215	Databases	No primary data
CAN-1806	2020	Features of Highly Pathogenic Avian Influenza (HPAI) H5N1 in Domestic Poultry	10.5772/intechopen.86098	Databases	No primary data
CAN-1812	2015	Distribution and characteristics of poultry farms in Enugu State, Nigeria after the 2007 outbreak of highly pathogenic avian influenza	10.5897/ijlp.14.0234	Databases	No laboratory-confirmed avian AIV
CAN-1817	2018	EFFECTS OF AVIAN INFLUENZA H5N1 ON POULTRY PRODUCTION IN WUDIL TOWN OF KANO STATE, NIGERIA.	10.59331/jasd.v1i1.42	Databases	No laboratory-confirmed avian AIV
CAN-1836	2025	Global risk mapping of highly pathogenic avian influenza H5N1 and H5Nx in the light of epidemic episodes occurring from 2020 onwards	10.7554/elife.104748	Databases	No avian-host field data
CAN-1837	2016	Global mapping of highly pathogenic avian influenza H5N1 and H5Nx clade 2.3.4.4 viruses with spatial cross-validation	10.7554/elife.19571	Databases	No avian-host field data
CAN-1871	2025	Serological Survey of Equine and Avian Influenza Viruses in Horses in Zaria, Kaduna State	10.4314/nvj.v46i2.1	Databases	No avian-host field data
CAN-1948	2014	A retrospective study of poultry diseases diagnosed in Maiduguri, North-East, Nigeria	–	Other methods	No laboratory-confirmed avian AIV
CAN-2026	2010	Antigenic and genetic characteristics of influenza A (H5N1) and influenza A (H9N2) viruses and candidate vaccine viruses developed for potential use in human ...	–	Other methods	No Nigeria-specific data
CAN-2044	2009	A case-control study to identify risk factors for the occurrence of highly pathogenic avian influenza H5N1 in Nigeria	–	Other methods	No primary data
CAN-2068	2022	Redesign and validation of a real-time RT-PCR to improve surveillance for avian influenza viruses of the H9 subtype	–	Other methods	Experimental only, without field relevance
CAN-2082	2008	Pro-poor HPAI risk reduction strategies in Nigeria-Background Paper	–	Other methods	No primary data
CAN-2094	2006	Anatidae migration in the western Palearctic and spread of highly pathogenic avian influenza H5N1 virus	–	Other methods	No Nigeria-specific data
CAN-2149	2023	Bird flu: Nigeria is on major migratory bird routes, new strains keep appearing	https://doi.org/10.64628/aaj.qddm3ft7c	Other methods	
CAN-2153	2024	Participatory Epidemiology and Sero-Surveillance for Avian Influenza in Local Chicken at Live bird Market and House Holds in Enugu State, Nigeria	–	Other methods	
CAN-2154	2024	Use of Geographic Information System (GIS) to Determine Pattern of Spread of Highly Pathogenic Avian Influenza in Nigeria: A Strategy for Public Health Surveillance	–	Other methods	

Appendix E. Risk-of-bias item tables

Answers: Y, yes; N, no; U, unclear; NA, not applicable. Scores were never summed. Prevalence items: 1 sample frame; 2 sampling method; 3 sample size; 4 subjects and setting described; 5 coverage of identified sample; 6 valid identification method; 7 standard, reliable measurement; 8 statistical analysis; 9 response rate or losses. Case series items: 1 inclusion criteria; 2 standard, reliable measurement; 3 valid identification; 4 consecutive inclusion; 5 complete inclusion; 6 case characteristics; 7 clinical information; 8 outcomes or follow-up; 9 sites and dates; 10 statistical analysis. Analytical cross-sectional items: 1 inclusion criteria; 2 subjects and setting; 3 exposure measurement; 4 objective criteria for the condition; 5 confounders identified; 6 strategies for confounding; 7 outcome measurement; 8 statistical analysis. Molecular items: M1 sampling of

sequenced viruses; M2 assay transparency; M3 sequencing transparency; M4 accessions available; M5 phylogenetic method and support; M6 clade nomenclature; M7 reassortment discipline.

Table E1. JBI Prevalence checklist (55 studies).

Reference	1	2	3	4	5	6	7	8	9	Overall	HR
[111]	Y	N	N	Y	N	U	U	Y	U	Inc	–
[115]	Y	Y	N	Y	Y	Y	U	N	Y	Inc	–
[89]	N	N	N	Y	N	Y	U	N	U	Inc	–
[32]	N	N	N	N	Y	N	U	N	Y	Inc-HR	yes
[61]	N	Y	U	N	Y	Y	U	N	Y	Inc	–
[62]	Y	Y	Y	Y	Y	Y	U	Y	Y	Inc	–
[128]	Y	NA	NA	N	N	Y	U	N	U	Inc	–
[129]	N	Y	U	Y	N	Y	N	N	U	Inc	–
[60]	Y	U	Y	N	N	Y	U	N	U	Inc	–
[13]	Y	N	N	Y	N	Y	Y	N	Y	Inc	–
[83]	Y	NA	NA	N	Y	Y	U	N	U	Inc	–
[33]	Y	NA	NA	U	U	Y	U	U	U	Seek	yes
[122,123]	Y	Y	U	N	N	Y	Y	N	Y	Inc	–
[35]	Y	U	U	U	U	Y	U	N	U	Seek	yes
[132]	Y	NA	NA	N	Y	U	U	N	U	Inc	–
[40]	Y	NA	NA	N	U	N	N	N	U	Inc	yes
[101]	Y	N	N	Y	U	Y	U	Y	U	Inc	–
[49]	Y	U	U	Y	U	Y	U	N	U	Inc	–
[12]	Y	N	N	Y	Y	Y	Y	N	Y	Inc	–
[125]	Y	Y	N	N	Y	U	U	N	Y	Inc	–
[98]	Y	U	N	N	Y	Y	Y	N	Y	Inc	–
[90]	Y	NA	NA	Y	Y	U	U	N	N	Inc	–
[133]	Y	NA	NA	Y	U	U	U	N	N	Inc	–
[84]	Y	NA	NA	N	N	U	U	N	U	Inc	–
[41]	Y	N	U	N	N	N	N	N	N	Inc	yes
[87]	Y	Y	N	Y	Y	Y	U	Y	Y	Inc	–
[103]	N	U	N	N	Y	Y	Y	N	Y	Inc	–
[100]	N	NA	NA	Y	Y	U	U	N	N	Inc	–
[63]	Y	N	N	Y	N	Y	U	N	Y	Inc	–
[112]	Y	N	N	Y	U	Y	U	N	U	Inc	–
[68]	Y	N	N	Y	Y	Y	U	N	Y	Inc	–
[135]	Y	NA	NA	Y	U	U	U	N	N	Inc	–
[52]	Y	N	Y	N	U	Y	U	N	N	Inc	–
[126]	Y	N	N	Y	Y	Y	U	N	Y	Inc	–
[65–67]	Y	U	Y	N	U	Y	U	N	U	Inc	–
[138]	Y	NA	NA	N	U	U	U	N	U	Inc	–
[124]	Y	N	Y	Y	N	Y	U	N	U	Inc	–
[139]	Y	NA	NA	N	U	U	U	N	U	Inc	–

Reference	1	2	3	4	5	6	7	8	9	Overall	HR
[140]	Y	NA	NA	Y	Y	U	N	N	Y	Inc	–
[74,75]	Y	Y	N	N	U	Y	U	N	U	Inc	–
[117]	Y	N	N	N	Y	Y	U	N	Y	Inc	–
[119]	N	Y	N	Y	U	Y	U	N	U	Inc	–
[141]	Y	NA	NA	Y	Y	Y	U	N	U	Inc	–
[99]	Y	N	N	N	U	Y	Y	N	U	Inc	–
[102]	N	N	N	N	U	U	U	N	U	Inc	–
[114]	N	N	N	N	Y	U	U	N	Y	Inc	–
[120]	Y	U	Y	Y	Y	Y	U	N	Y	Inc	–
[113]	Y	N	N	Y	Y	Y	Y	N	Y	Inc	–
[121]	Y	N	Y	N	Y	Y	U	N	Y	Inc	–
[116]	N	N	N	Y	Y	U	U	N	Y	Inc	–
[85]	Y	NA	NA	Y	Y	Y	U	N	U	Inc	–
[70]	N	N	N	Y	Y	Y	Y	N	Y	Inc	–
[127]	Y	Y	Y	N	Y	Y	U	N	Y	Inc	–
[43]	N	N	N	N	U	N	U	Y	U	Inc	yes
[144]	Y	NA	NA	N	U	U	U	N	U	Inc	–
[37]	U	U	U	U	U	U	U	U	U	Seek	yes

Table E2. JBI Case Series checklist (30 studies).

Reference	1	2	3	4	5	6	7	8	9	10	Overall	HR
[44,45]	U	Y	Y	U	N	Y	Y	Y	Y	Y	Inc	–
[46]	N	Y	Y	N	N	N	N	NA	N	N	Inc	–
[38,39]	Y	Y	N	Y	Y	Y	Y	Y	Y	NA	Inc	yes
[16]	U	U	U	U	U	U	U	U	U	U	Seek	yes
[81,82]	U	Y	Y	U	U	Y	N	N	Y	N	Inc	–
[78,79]	Y	Y	Y	N	U	N	N	NA	Y	Y	Inc	–
[58]	Y	U	Y	U	U	Y	Y	N	N	Y	Inc	–
[19]	N	U	Y	U	U	N	N	N	Y	N	Inc	–
[104]	Y	U	Y	N	Y	N	N	N	Y	N	Inc	–
[107]	Y	U	Y	N	Y	N	N	N	Y	Y	Inc	–
[73]	N	U	Y	U	U	N	N	N	N	Y	Inc	–
[105]	Y	U	Y	N	Y	N	N	N	Y	N	Inc	–
[106]	N	U	Y	U	U	N	N	N	Y	N	Inc	–
[53]	Y	U	Y	U	U	N	N	N	N	Y	Inc	–
[76]	Y	U	Y	U	U	Y	Y	N	Y	NA	Inc	–
[77]	Y	U	Y	U	Y	N	N	N	Y	Y	Inc	–
[11]	Y	U	Y	U	Y	Y	Y	Y	N	N	Inc	–
[136]	N	U	Y	NA	U	Y	Y	N	Y	N	Inc	–
[137]	Y	U	Y	U	Y	Y	Y	Y	Y	N	Inc	–
[80]	Y	U	U	U	Y	Y	Y	Y	Y	N	Inc	–

Reference	1	2	3	4	5	6	7	8	9	10	Overall	HR
[54]	Y	U	Y	U	Y	Y	Y	N	Y	N	Inc	–
[69]	N	U	Y	NA	U	Y	Y	Y	Y	NA	Inc	–
[56]	N	U	U	U	U	Y	Y	Y	Y	N	Inc	–
[57]	Y	Y	Y	U	U	Y	Y	Y	Y	N	Inc	–
[18]	Y	U	Y	NA	Y	Y	Y	Y	Y	N	Inc	–
[91]	N	Y	Y	U	N	Y	Y	Y	N	N	Inc	–
[71,72]	Y	U	Y	N	Y	Y	N	Y	Y	N	Inc	–
[92–95]	Y	U	Y	U	Y	Y	Y	Y	N	N	Inc	–
[59]	Y	Y	Y	U	Y	Y	Y	Y	Y	N	Inc	–
[145]	Y	Y	Y	U	Y	Y	Y	Y	N	N	Inc	–
[36]	U	U	U	U	U	U	U	U	U	U	Seek	yes

Table E3. JBI Analytical Cross-sectional checklist (5 studies).

Reference	1	2	3	4	5	6	7	8	Overall	HR
[96,97]	N	Y	N	U	N	N	U	N	Inc	–
[27]	N	N	N	Y	N	Y	U	N	Inc-HR	yes
[131]	Y	Y	U	U	Y	Y	U	N	Inc	–
[142]	N	Y	N	U	N	N	U	N	Inc	–
[42]	Y	Y	U	Y	N	N	N	N	Inc	yes
[30]	U	U	U	U	U	U	U	U	Seek	yes

Table E4. Secondary and modelling studies (no JBI checklist; structured note on data-source limitations).

Reference	Overall	HR	Data-source note (abridged)
[47]	Inc	–	Nigerian evidence is limited to public GenBank HA1 sequences that the authors re-analysed alongside their Burkina Faso viruses; the number, hosts, states and dates of the Nigerian sequences are not stated beyond two coded south-western farms (SO, BA) and "northern Nigeria". Representativeness depends on what had been deposited by late 2006 (largely the ref. 2 Nature dataset), so the Nigerian diversity shown is that of a few early outbreaks. No de-duplication or sequence-quality screening of the downloaded data is described.
[31]	Inc	–	Global agro-ecological niche model in which Nigeria is one national unit; the outcome (seasonal persistence) comes from OIE WAHID/FAO EMPRES-I official reports, which the authors themselves note suffer "major discrepancies in surveillance and reporting", so under-reporting and reporting lag in Nigeria directly bias its persistence index. Country-level predictors (FAOSTAT densities, purchasing power) hide within-Nigeria variation, and the paper's sub-national statements about northern Nigerian wetlands are not supported by its national-level analysis. Nigeria's own index value sits in an unread supplement, so the Nigeria-specific output is only a niche assignment.
[130]	Inc	–	The model is fitted to OIE state-by-month outbreak reports for 2006-2007 (follow-up report #6), so incomplete or delayed official reporting, and the coarse state/month resolution, carry straight into the fitted transmission parameters; the counts themselves are not printed. The Ogun premises model uses a synthetic farm map (19 known infected locations plus 695 randomly placed farms) built from survey proportions, and it assumes village flocks play no part, so premises-level outputs are scenario estimates, not observed Nigerian data. The fitted distance parameters (about 1e-10 km) have unclear units, and the Lagos January introduction assumed in the model is not documented in the report.
[14]	Inc	–	Outbreak counts come from a Federal Ministry of Agriculture database "following surveys in live bird markets", with completeness, case definition and laboratory confirmation not described and per-year or per-state counts given only in figures; they cover 2006-2017 only. The 289 OpenFlu sequences are summarised by subtype without years, hosts, states or de-duplication details, and the 11 Nigerian genomes analysed (2006-2008, 2015-2016, 2021) reflect what earlier studies deposited, with no 2009-2014 or 2017-2020 viruses and no state of origin. Text errors (e.g. "21 sequences isolated in 2021" for one genome) reduce confidence in the narrative description of trees.
[86]	Inc	–	Outbreak counts are OIE-notified, NVRI RRT-PCR-confirmed outbreaks, so they inherit the completeness and lag limits of passive official reporting; the 2006-2008 source is not stated and the monthly counts exist only as figure bar labels. The same 465 outbreaks and 2.77 million birds are tied to two different end dates (May 2016 vs May 2017), and the counts duplicate other NVRI/OIE-based studies, so they must not be added. The economic inputs rely on unstructured data from donors and experts, as the authors acknowledge, and headline benefit-cost ratios differ between Abstract (68x) and Results (24x).

Reference	Overall	HR	Data-source note (abridged)
[34]	Seek	yes	Only the abstract is available: it reports 106 Nigerian complete HA sequences within a 751-sequence comparison but not whether they were generated in-study or drawn from public databases, nor their years, states or hosts. Nigeria enters only as a "non-vaccinating" comparator, so its representativeness depends on which 2006-2008 viruses were sequenced, and no Nigeria-specific evolutionary rate is visible. Possible duplication with other included Nigerian sequence studies (same IZSvE/NVRI authorship) cannot be checked.
[64]	Inc	–	Nigerian evidence comes from public GISAID/GenBank HA sequences plus 7 newly sequenced NVRI samples, so representativeness reflects what earlier studies deposited (clade 2.2 to 2008, 2.3.2.1c from 2015, a single H5N8). The authors note that sequencing covered about 29% of reported outbreaks globally and varied by year, and country-level discrete traits treat Nigeria as one unit. Supplement counts do not match the text (35 vs 37 new H5N1, 5 vs 3 H5N8), and one Nigerian genome labelled new duplicates a published one (CAN-0226), so duplication across studies must be checked.
[48]	Inc	–	Nigerian input is 19 GISAID H9N2 HA sequences from 2019 (16 kept after random one-per-cluster down-sampling), all previously reported by Sulaiman et al. 2021, so they duplicate [65–67] and must not be counted again. They come from a single 2019 surveillance effort, so they say nothing about H9N2 in Nigeria before or after 2019, and state codes in strain names are not decoded. Inferences are regional (Morocco to Western Africa), with Nigeria treated as point locations only.
[134]	Inc	–	The Nigerian data are an Ebonyi State line list of four earlier confirmed outbreaks (2016-2019) plus one new live-bird-market H5N1 confirmation at NVRI Jos, with no assay, denominator or completeness information for either source. The 17 farms were purposively selected across the three senatorial zones, so the biosecurity and stock-sourcing percentages describe those farms only and carry no measure of association. Reported figures are partly irreproducible: the farm proportions total 101%, 14 of 17 is given as 80%, and the projected state poultry population of about 15 million cannot be derived from the two quoted census figures. The forward-looking risk statement is a scenario-tree judgement, not an observed outcome.
[50]	Inc	–	Nigeria contributes 135 of 873 public GenBank clade 2.2 HA sequences, all deposited by earlier studies, so they duplicate primary Nigerian reports and must not be counted again. The authors state that records lacking temporal information came mainly from Egypt, Nigeria and Germany, so Nigerian dating depends on the FAO outbreak series rather than the sequences, and no Nigerian state, host, species or outbreak count is reported. The FAO series carries the usual official-reporting limits of under-ascertainment and uneven reporting, which the authors acknowledge when calling their four routes "possible scenarios". Nigeria is treated as a single spatial unit, so nothing in the study speaks to within-country distribution.
[108]	Inc	–	Nigeria enters as one of 28 localities with 13 public HA sequences, all from 2006 and all deposited by earlier studies (Ducatez and Salzberg series), so they duplicate primary Nigerian reports and must not be counted again. Duplicate sequences within a locality were removed, so 13 is not the number of Nigerian isolates then available, and no Nigerian state, date beyond the year, or host system is recorded. The analysis depends on what GenBank held in 2006, when Nigerian sampling was concentrated in the first epidemic wave, so later years are absent entirely.
[109]	Inc	–	Nigerian evidence consists of eight public GenBank HA sequences from 2006, all deposited by earlier Nigerian studies, so they duplicate those reports and must not be counted again. The paired outbreak data are a global georeferenced compilation of official reports for November 2003 to December 2006, with no Nigerian case count given and the usual under-reporting and lag of official notification. The authors note that most sequences carry location and time only to country and year, so sequences and outbreak clusters cannot be matched exactly, and Nigeria is treated as a single country unit throughout.
[51]	Inc	–	Nigeria contributes 48 H5N1 proteomes downloaded from the Influenza Research Database, all deposited by earlier Nigerian genome studies, so they duplicate those reports and must not be counted again. The records carry only country, year and host species, with no state, pathotype, clade or accession number, and 28 of the 48 (the 2006 strains) are never mentioned in the text. The 20 strains labelled suspected zoonotic are so labelled because a human case occurred in Nigeria that year, not because of anything measured in them, and every Nigerian strain was predicted avian for all 11 proteins. The output is a model prediction, not an observation.
[88]	Inc	–	The Nigerian data are annual counts of H5N1 poultry outbreaks reported to FAO EMPRES-i for 2004-2016, so they reflect official notification and inherit its under-ascertainment, lag and the differing reporting conventions the authors themselves note when justifying the log transformation. The counts duplicate the same outbreaks recorded in WAHIS and NVRI-based studies and must not be added to them. No production system, state, species detail or diagnostic basis accompanies the counts, and zero years cannot be distinguished from non-reporting. All regression coefficients are estimated for ECOWAS as a bloc over 180 country-years, so they are not Nigerian estimates even though Nigeria supplies most of the outbreaks.
[110]	Inc	–	The Nigerian content is four reassortment events compiled from published primary reports through Google Scholar and PubMed keyword searches and citation chasing, with no stated screening or duplicate-checking protocol. The authors note that a primary report reflects first detection and may carry surveillance bias, and that observation bias is likely much stronger for reassortants than conversions, so Nigeria's apparent share of African events tracks sequencing effort. Table S1 assigns host production systems (for example LBM surveillance) that the primary Nigerian reports do not state, and two counted Nigerian events list no segment incongruence. The compilation adds no new Nigerian data and should be used only as a pointer to the primary studies.
[28]	Inc	–	The outbreak data are the FAO EMPRES-i and OIE WAHID official databases, so they capture only outbreaks reported and confirmed through the Nigerian veterinary services, with backyard and remote flocks under-represented and detection concentrated near roads and towns, as the paper itself notes. The two databases give different totals for overlapping periods (150 poultry outbreaks to 2007 vs 294 bird outbreaks to August 2008) and are not reconciled, so counts from this paper must not be added to other sources covering the same epidemic. Reporting lag and the retrospective nature of the December 2007 field interviews also mean the state-level persistence measure reflects reporting intensity as much as infection. The accompanying backyard poultry and duck denominators are National Bureau of Statistics estimates, whose accuracy at state level is not assessed.
[29]	Inc	–	The only data source is OIE official H5N1 outbreak reports for December 2003 to December 2006, so Nigerian coverage reflects what the national veterinary service reported and confirmed during the first epidemic, when diagnosis was new and backyard flocks were under-reported. Outbreak records carry reporting lag and are positioned by administrative centre in at least some countries, with errors of tens of kilometres, which blurs cluster boundaries; the Nigerian records are not described in this respect. Clusters also track surveillance intensity, so the three Nigerian clusters should not be read as the full extent or timing of infection. No Nigerian denominators or counts can be taken from this paper.
[55]	Inc	–	The Nigerian content is a compilation of published literature, agency reports and public GenBank/GISAID sequences that other included studies produced, so it duplicates them and adds no counts, denominators or new isolates. Sequence selection is not reproducible: the Methods give 100 sequences, then 78, then 28, with no accessions and no inclusion rule, so coverage of the available Nigerian sequences is unknown and biased towards what was deposited. Expert opinion was collected by e-mail from respondents selected through professional associations and online networks, which is a convenience sample presented alongside published evidence. Clade labels are taken from the sources but one (Gombe 2008 as clade 2.2.3) does not match its primary report.

Reference	Overall	HR	Data-source note (abridged)
[118]	Inc	–	The Nigerian avian influenza content is a short narrative summary, by the project lead, of the national active and live-bird-market surveillance of 2007-2008 and of official outbreak spread to March 2008, with no methods, assays, sampling dates, states or per-state results, so none of the figures can be checked. The same surveillance is reported as primary data elsewhere in the register (CAN-1357, CAN-0751) with different sample counts, so these figures are a duplicate secondary source and should not be added to the primary estimates. The statements on outbreak extent (97 LGAs in 25 states and the FCT) rest on official notifications of that period, which were incomplete and unevenly confirmed.
[143]	Inc	–	All Nigerian data are official HPAI H5N1 notifications to OIE WAHIS, FAO EMPRES-i and WHO (2006-2016), which depend on national reporting that was incomplete and uneven between waves and states, and which are subject to reporting lag and possible duplication across the three databases; cross-matching with national records is mentioned but not described. The chapter 3 national aggregates use the susceptible birds of affected premises as denominators, so the reported morbidity and mortality are within-outbreak rates, not prevalence, and the Nigerian outbreak count is not stated. The chapter 5 hotspot map rests on expert-weighted raster overlays whose state-level inputs and weights sit in appendices missing from the repository deposit, so the spatial claims cannot be verified.

Table E5. Molecular transparency check (37 studies). A "No" on M4 or M5 caps molecular claims at "suggestive".

Reference	M1	M2	M3	M4	M5	M6	M7	Cap
[44,45]	N	N	N	Y	N	Y	N	suggestive
[46]	Y	Y	Y	N	NA	Y	NA	suggestive
[47]	N	NA	U	N	Y	N	Y	suggestive
[16]	U	U	U	U	U	U	NA	none
[14]	Y	NA	Y	Y	Y	N	Y	none
[78,79]	Y	Y	Y	Y	Y	Y	Y	none
[34]	U	U	U	U	U	U	NA	none
[64]	Y	N	Y	Y	Y	Y	Y	none
[58]	Y	Y	Y	Y	Y	N	NA	none
[48]	Y	NA	U	N	Y	Y	NA	suggestive
[101]	Y	Y	Y	Y	Y	NA	Y	none
[49]	N	Y	Y	Y	N	N	Y	suggestive
[12]	N	Y	Y	Y	Y	Y	Y	none
[19]	N	N	Y	Y	Y	Y	N	none
[104]	Y	N	Y	Y	Y	Y	Y	none
[50]	Y	NA	U	Y	N	Y	N	suggestive
[87]	N	Y	Y	Y	Y	NA	N	none
[108]	Y	NA	U	Y	Y	N	N	none
[107]	Y	Y	Y	Y	Y	Y	Y	none
[109]	Y	NA	U	Y	Y	N	Y	none
[51]	Y	NA	U	N	NA	NA	NA	suggestive
[63]	Y	Y	Y	Y	Y	NA	N	none
[52]	N	N	N	N	N	Y	N	suggestive
[73]	N	N	Y	Y	Y	N	Y	none
[105]	Y	N	Y	Y	Y	Y	Y	none
[106]	N	Y	Y	Y	Y	Y	Y	none
[53]	Y	Y	N	N	Y	Y	Y	suggestive
[76]	Y	N	Y	Y	Y	Y	Y	none
[77]	Y	N	Y	Y	Y	Y	Y	none
[65–67]	N	N	Y	Y	Y	Y	N	none
[11]	Y	Y	Y	U	Y	Y	Y	none

Reference	M1	M2	M3	M4	M5	M6	M7	Cap
[54]	Y	Y	N	N	NA	NA	NA	suggestive
[74,75]	Y	Y	Y	Y	Y	Y	Y	none
[55]	N	NA	NA	N	Y	N	N	suggestive
[56]	N	N	N	N	N	Y	N	suggestive
[57]	Y	Y	N	N	NA	Y	NA	suggestive
[71,72]	Y	Y	Y	Y	Y	N	N	none

Appendix F. Synthesis tables

The tables below are the full SWiM synthesis tables from which the condensed main-text tables were drawn. Each entry cites the contributing studies by reference number; where a study contributes several findings rows, the reference is given once per row group. Laboratory-confirmed means sequence, isolate, RNA or laboratory-confirmed case (method not reported). § marks official reports without a stated assay (not counted); † antibody-positive rows (exposure only); ‡ antigen-only rows. Within cells, studies are ordered low-concern first (high-risk and "suggestive"-capped studies last).

T1. First-detection chronology (laboratory-confirmed evidence)

Laboratory-confirmed = sequence, isolate, RNA or lab-confirmed case (method NR). Official reports (assay not stated) are shown separately (§) and never counted. Earliest date per key; [evidence type at the earliest date]; rows with their date ranges. A window that ends before the earliest dated row is reported as '(window)'.

T1-Subtype

Subtype	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
H5 (N not determined)	includes epidemic-level subtype statements	2006 (earliest month-dated row 2006-12; other rows of 2006 dated to the year or a within-year range) [isolate] [69], [49] (2006-01 to 2006-08); wider-window rows: [46] (2006 to 2007), [46] (2006 to 2007)	sequence: 2006 to 2007 (window); isolate: 2006-12; RNA: 2006	-
H5N1	includes epidemic-level subtype statements	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	sequence: 2006-01; isolate: 2006-01; RNA: 2007-04; lab-confirmed case (method NR): 2006-01	§ 2006-01 to 2006-03 (range) [139] (2006-01 to 2006-03), [139] (2006-01 to 2006-03), [139] (2006-01 to 2006-03), [86] (month NR), [132] (2006-01 to 2006-06), [88] (month NR), [52] (2006-01 to 2006-06), [40] (month NR), [40] (month NR), [40] (month NR), [40] (month NR)
H5N2		2007-02 [sequence] [63]	sequence: 2007-02; isolate: 2011-09 to 2012-02 (window); RNA: 2008-03 to 2008-04 (range)	-

Subtype	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
H5N8		2016-11 [sequence] [64]	sequence: 2016-11; isolate: 2019-02; RNA: 2021 (earliest month-dated row 2021-02; other rows of 2021 dated to the year or a within-year range)	§ 2017-01 [138]
H9N2		2019 [sequence] [65–67] (2019-04 to 2019-12)	sequence: 2019; isolate: 2020 to 2021 (window); RNA: 2020 to 2021 (window)	-
H5N6		2019-06 [isolate] [65–67]	isolate: 2019-06	-
H9 (N not determined)		2024-10 to 2025-07 (window) [RNA] [70] (2024-10 to 2025-07)	RNA: 2024-10 to 2025-07 (window)	-

T1-Clade (WHO/OIE/FAO where the paper uses it)

Clade (WHO/OIE/FAO where the paper uses it)	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
2.2	shown, stated	2006 (earliest month-dated row 2006-03; other rows of 2006 dated to the year or a within-year range) [sequence] [71,72], [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [52] (month NR); wider-window rows: [46] (2006 to 2007), [46] (2006 to 2007), [52] (2006 to 2007), [104] (2006-01 to 2008-07)	sequence: 2006 (earliest month-dated row 2006-03; other rows of 2006 dated to the year or a within-year range); isolate: 2006; RNA: 2006 to 2007 (window)	-
Qinghai lineage (pre-WHO name; not mapped by the paper)	shown	2006-01 [sequence] [58]	sequence: 2006-01; isolate: 2006-01	-
2.2.2	shown	2006-01 [sequence] [53]	sequence: 2006-01	-
2.2.1	shown	2006-01 to 2008-07 (window) [sequence] [104] (2006-01 to 2008-07)	sequence: 2006-01 to 2008-07 (window)	-
2.3.2.1c	shown	2015-01 [sequence] [76], [77], [78,79] (2015-01 to 2015-12), [44,45] (month NR), [16] (month NR)	sequence: 2015-01; RNA: 2015-01	-
2.3.4.4 group B (as labelled)	shown	2016-11 [sequence] [64]	sequence: 2016-11	-
H9 G1 lineage	shown	2019 [sequence] [65–67] (2019-04 to 2019-12)	sequence: 2019	-
2.3.4.4	stated	2019-02 [isolate] [57]	isolate: 2019-02	-
2.3.4.4b	shown, stated	2019-06 [isolate] [65–67]	sequence: 2019-07; isolate: 2019-06; RNA: 2021	-

T1-Sublineage / genotype

Sublineage / genotype	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
EMA-1/EMA-2 (clade 2.2 sublineages)	shown	2006 [sequence] [73] (month NR), [73] (month NR)		-
sublineage I/II-N (clade 2.2, local)	shown	2006-01 to 2008-07 (window) [sequence] [104] (2006-01 to 2008-07)		-
EMA3 / sublineage III (clade 2.2)	shown, stated	2008-05 to 2008-07 (range) [isolate] [74,75] (2008-05 to 2008-07); wider-window rows: [104] (2006-01 to 2008-07)		-

Sublineage / genotype	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
2.3.2.1c WA2	shown	2015 [sequence] [78,79] (2015-01 to 2015-12), [16] (month NR)		-
2.3.2.1c WA1	shown	2015-01 [sequence] [77], [78,79] (2015-01 to 2015-12)		-
H5N1/H9N2 reassortant (H5N2)	shown	2021 [sequence] [11] (month NR), [11] (month NR)		-
2.3.4.4b A/duck/Chelyabinsk/1207-1/2020-like (H5N8)	shown	2021 [sequence] [11] (month NR)		-
2.3.4.4b A/Eurasian_Wigeon/Netherlands/1/2020-like	shown	2022-03 to 2022-04 (range) [sequence] [12] (2022-03 to 2022-04); wider-window rows: [11] (2021 to 2022)		-
2.3.4.4b genotype EA-2020-C	shown	2023 [sequence] [19] (2023-01 to 2023-07)		-
2.3.4.4b genotype EA-2024-DV	shown	2024 [sequence] [19] (2024-02 to 2024-12)		-

T1-Host system

Host system	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
L	live-bird market	2006 [sequence] [71,72] (2006-02 to 2006-06), [52] (month NR); wider-window rows: [41] (2006 to 2009), [52] (2006 to 2007), [92–95] (2006 to 2008), [104] (2006-01 to 2008-07)	sequence: 2006; isolate: 2006; RNA: 2007-10 to 2007-11 (range); lab-confirmed case (method NR): 2007-10 to 2007-11 (range)	§ 2006 [52] (2006-01 to 2006-06)
B	backyard/village poultry	2006 (earliest month-dated row 2006-02; other rows of 2006 dated to the year or a within-year range) [sequence] [80] (2006-02 to 2006-05), [128] (2006-01 to 2006-12), [71,72] (2006-02 to 2006-06), [145] (month NR), [52] (month NR); wider-window rows: [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007), [92–95] (2006 to 2008), [104] (2006-01 to 2008-07), [100] (2006-01 to 2007-12), [92–95] (2006-01 to 2007-03)	sequence: 2006; isolate: 2006 (earliest month-dated row 2006-12; other rows of 2006 dated to the year or a within-year range); RNA: 2007-04; lab-confirmed case (method NR): 2006 (earliest month-dated row 2006-02; other rows of 2006 dated to the year or a within-year range)	§ 2006 [52] (2006-01 to 2006-06)
C	commercial poultry	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [112] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (2006-01 to 2006-08), [49] (month NR), [49] (month NR); wider-window rows: [46] (2006 to 2007), [46] (2006 to 2007), [46] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007), [92–95] (2006 to 2008)	sequence: 2006-01; isolate: 2006-01; RNA: 2006; lab-confirmed case (method NR): 2006-01	§ 2007 [130] (month NR)
Wr	wild, resident	2006-03 [sequence] [71,72]	sequence: 2006-03; isolate: 2022-03 to 2022-04 (range); RNA: 2007-02	-
Cap	captive wild/zoo/ornamental	2006-04 [sequence] [71,72]	sequence: 2006-04; RNA: 2021-08; lab-confirmed case (method NR): 2015 to 2017 (window)	§ 2015 [84] (month NR), [84] (month NR), [84] (month NR), [84] (month NR)

Host system	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
O	other (ostrich farms, imports, quail flock, etc.)	2021 (earliest month-dated row 2021-02; other rows of 2021 dated to the year or a within-year range) [RNA] [81,82], [11] (month NR); wider-window rows: [11] (2021 to 2022), [11] (2021-01 to 2022-12)	sequence: 2021 to 2022 (window); RNA: 2021 (earliest month-dated row 2021-02; other rows of 2021 dated to the year or a within-year range)	-
Wm	wild, migratory	2022-03 to 2022-04 (range) [sequence] [12] (2022-03 to 2022-04), [12] (2022-03 to 2022-04), [12] (2022-03 to 2022-04)	sequence: 2022-03 to 2022-04 (range); isolate: 2022-03 to 2022-04 (range); RNA: 2022-03 to 2022-04 (range)	-

T1-State

State	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
Benue	NC	2006 [sequence] [107] (2006-01 to 2006-06); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Lagos	SW	2006 [sequence] [71,72] (2006-02 to 2006-06); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006; isolate: 2015-01; RNA: 2015-01; lab-confirmed case (method NR): 2006 to 2007 (window)	§ 2015-01 [84]
Taraba	NE	2006 [sequence] [107] (2006-01 to 2006-06); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006-01 to 2007-03)	sequence: 2006; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
FCT	NC	2006 [sequence] [107] (2006-01 to 2006-06); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006; lab-confirmed case (method NR): 2006 to 2007 (window)	§ 2016 [84] (month NR), [84] (month NR)
Bauchi	NE	2006 (earliest month-dated row 2006-02; other rows of 2006 dated to the year or a within-year range) [sequence] [80] (2006-02 to 2006-05), [107] (2006-01 to 2006-06); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006-01 to 2007-03)	sequence: 2006; isolate: 2021-12 to 2022-03 (range); RNA: 2015; lab-confirmed case (method NR): 2006-02 to 2006-05 (range)	§ 2006 [40] (month NR)
Edo	SS	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008), [92–95] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2008); earliest dated: 2015 [44,45] (2015-01 to 2015-12); other windows ending before the first dated row (may be narrower): [92–95] (2006-01 to 2007-03)	sequence: 2016; RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Enugu	SE	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2008); earliest dated: 2015 [44,45] (2015-01 to 2015-12)	sequence: 2016; RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Kwara	NC	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2008)	lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)

State	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
Nasarawa	NC	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008), [92–95] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2008); earliest dated: 2015 [44,45] (2015-01 to 2015-12); other windows ending before the first dated row (may be narrower): [92–95] (2006-01 to 2007-03)	sequence: 2017-01 to 2017-02 (range); isolate: 2021-12 to 2022-03 (range); RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Niger	NC	2006 to 2008 (window) [lab-confirmed case (method NR)] [42] (2006 to 2008), [42] (2006 to 2008)	lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Rivers	SS	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2008); earliest dated: 2015 [78,79] (2015-01 to 2015-12), [44,45] (2015-01 to 2015-12)	sequence: 2015; RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2015 [84] (month NR), [84] (month NR), [135] (month NR)
Yobe	NE	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008)	lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Zamfara	NW	2006 to 2008 (window) [lab-confirmed case (method NR)] [92–95] (2006 to 2008); earliest dated: 2015 [78,79] (2015-01 to 2015-12), [44,45] (2015-01 to 2015-12)	sequence: 2015; RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Kaduna	NW	2006-01 [sequence] [58], [59], [107] (2006-01 to 2006-06), [71,72] (2006-02 to 2006-06); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006-01; isolate: 2006-01; RNA: 2007-04; lab-confirmed case (method NR): 2006-01	§ 2006-01 to 2006-03 (range) [139] (2006-01 to 2006-03), [40] (month NR)
Kano	NW	2006-01 [lab-confirmed case (method NR)] [59]; wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2007-01 to 2007-02 (range); isolate: 2006-12; RNA: 2015; lab-confirmed case (method NR): 2006-01	§ 2006-01 to 2006-03 (range) [139] (2006-01 to 2006-03), [40] (month NR)
Oyo	SW	2006-01 [sequence] [53], [107] (2006-01 to 2006-06), [71,72] (2006-02 to 2006-06); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006-01; isolate: 2011-09 to 2012-02 (window); RNA: 2011-09 to 2012-02 (window); lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2017 [84] (month NR), [84] (month NR)
Plateau	NC	2006-01 [sequence] [59], [107] (2006-01 to 2006-06), [112] (month NR), [71,72] (2006-02 to 2006-06); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2006 (earliest month-dated row 2006-03; other rows of 2006 dated to the year or a within-year range); isolate: 2019-02; RNA: 2015; lab-confirmed case (method NR): 2006-01	§ 2006 [40] (month NR)
Katsina	NW	2006-01 to 2007-03 (window) [lab-confirmed case (method NR)] [92–95] (2006-01 to 2007-03); earliest dated: 2008-06 to 2008-07 (range) [52] (2008-06 to 2008-07)	sequence: 2015; isolate: 2008-06 to 2008-07 (range); RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2006-03 (range) [139] (2006-01 to 2006-03), [40] (month NR)
Ekiti	SW	2007 [sequence] [106] (2007-06 to 2007-11)	sequence: 2007	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Sokoto	NW	2007-01 [sequence] [107] (2007-01 to 2007-02), [105] (2007-01 to 2007-02), [71,72]; wider-window rows: [92–95] (2006 to 2008)	sequence: 2007-01; isolate: 2019-06; RNA: 2008-10 to 2009-03 (window); lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)

State	Zone/label/basis	Earliest evidence	Earliest by evidence tier	Official report (§, not counted)
Adamawa	NE	2007-01 to 2007-02 (range) [lab-confirmed case (method NR)] [140] (2007-01 to 2007-02); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006-01 to 2007-03)	lab-confirmed case (method NR): 2007-01 to 2007-02 (range)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Anambra	SE	2007-01 to 2007-02 (range) [sequence] [107] (2007-01 to 2007-02), [105] (2007-01 to 2007-02); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2007-01 to 2007-02 (range); RNA: 2015; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Borno	NE	2007-01 to 2007-02 (range) [sequence] [107] (2007-01 to 2007-02), [105] (2007-01 to 2007-02); wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2007-01 to 2007-02 (range); RNA: 2021-11 to 2021-12 (range); lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-11 to 2007-01 (range) [36] (2006-11 to 2007-01), [36] (2006-11 to 2007-01)
Jigawa	NW	2007-02 [sequence] [63]; wider-window rows: [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006 to 2008), [92–95] (2006-01 to 2007-03)	sequence: 2007-02; RNA: 2007-02; lab-confirmed case (method NR): 2006 to 2008 (window)	§ 2006-01 to 2006-03 (range) [139] (2006-01 to 2006-03)
Ogun	SW	2007-06 to 2007-08 (range) [sequence] [106] (2007-06 to 2007-08); wider-window rows: [131] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [92–95] (2006 to 2008)	sequence: 2007-06 to 2007-08 (range); RNA: 2015; lab-confirmed case (method NR): 2006 to 2007 (window)	§ 2007 [130] (month NR)
Gombe	NE	2008-05 to 2008-07 (range) [isolate] [74,75] (2008-05 to 2008-07)	sequence: 2008-07; isolate: 2008-05 to 2008-07 (range); RNA: 2015	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Kebbi	NW	2008-06 to 2008-07 (range) [isolate] [52] (2008-06 to 2008-07)	sequence: 2024; isolate: 2008-06 to 2008-07 (range)	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Abia	SE	2015 [RNA] [44,45] (2015-01 to 2015-12)	sequence: 2019; RNA: 2015	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Bayelsa	SS	2015 [sequence] [78,79] (2015-01 to 2015-12)	sequence: 2015	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Delta	SS	2015 [sequence] [78,79] (2015-01 to 2015-12), [44,45] (2015-01 to 2015-12)	sequence: 2015; RNA: 2015	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Imo	SE	2015 [RNA] [44,45] (2015-01 to 2015-12)	sequence: 2019; RNA: 2015	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)
Ebonyi	SE	2022-01 [lab-confirmed case (method NR)] [134]	lab-confirmed case (method NR): 2022-01	§ 2006-01 to 2023-06 (window) [90] (2006-01 to 2023-06)

States with official reports only (§): Akwa Ibom, Kogi, Osun

T1 notes:

background_statements_not_counted: [59] = background statement without source (129 outbreaks / 15 states by mid-2006); [134] = background statement; source NR (4 Ebonyi outbreaks 2016-2019)

official_reports: Official confirmed reports (assay not stated, rule R-f) are never counted as laboratory-confirmed; their earliest date per key is shown separately as 'official_report_earliest(\$)'.

excluded_subset_rows: [81,82] = subset row of the three CAN-1882 rows (published-abstract summary; do not add)

excluded_from_host_keys: [92–95] = ostriches were included only for computation of losses, not necropsied (thesis); cannot confirm host system O or species ostrich

H5N1 2006-01: The 2006-01 date of [58]/#1 (and the January farm cases of [59]) is the outbreak-report date; laboratory confirmation is dated 6-7 February 2006 (Padova; [38,39], [60]) and official notification 8 February 2006. T1 dates are row start dates.

evidence tag: The bracketed tag (e.g. [sequence]) gives the best evidence tier among the listed rows; the list can include method-NR rows.

2.3.4.4 labels: T1-Clade dates 2.3.4.4b from 2019-06 ([65–67]). '2.3.4.4 group B' ([64], 2016-11) is the study's own label; its mapping to 2.3.4.4b is ours, so the two rows are kept separate.

wild host keys: Wm/Wr keys are species-aware where a row states the positive species (MIGRATORY_SPECIES / RESIDENT_WILD_SPECIES). [63] positives are white-faced whistling duck and spur-winged goose (residents), so it supports Wr only.

[44,45]: The 2011-2014 Kano H5N1 entry rests on deaths of 24 Dec 2014; specimens were received in 2015-01.

[53]: Strain A/chicken/Nigeria/228-5/2005 carries 2005 in its name; the consensus row dates the isolates 2006-01 (retrospective clinic isolates obtained in or near January 2006). The earliest H5N1 date is kept at 2006-01.

clade 2.2 sublineages: 2.2.1, 2.2.2 and the EMA-1/2/3, I/II/III/IV and A-H labels are sublineages within clade 2.2; none is an independent clade and none can pre-date its parent clade 2.2.

2.3.4.4 group B: '2.3.4.4 group B' ([64]) corresponds to the clade now named 2.3.4.4b; the study label is preserved.

[71,72]: Species conflict: the isolate is A/chicken/Nigeria/VRD184/2006 in thesis Table 2.1 but A/vulture/Ng/VRD184/06 in the alignment and is listed as a hooded vulture. The Wr (wild resident) earliest date 2006-03 and the 'vulture' tag rest on this row.

undated antibody rows: H9 and H7 antibody rows without a date ([35], [103], [114], [121]) cannot be placed against the earliest dated H9/H7 antibody evidence (2013, [68]).

T1b. Authors' 'first' claims, with our earliest evidence alongside

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[111]	No previous information on AIV, IBV and NDV antibodies in these four wild bird species in Kano State (knowledge-gap statement, not a formal 'first detection')	not a detection claim (not compared)	-	
[89]	First report of A antibody in Pycnonotus barbatus, Ardea cinerea, Streptopelia senegalensis and Ciconia ciconia in Nigeria	not a detection claim (not compared)	-	
[89]	First reported extensive survey for AIV distribution in wild birds in Kaduna State	not a detection claim (not compared)	-	
[44,45]	Re-introduction of HPAI H5N1 into Nigeria for a second time in 9 years (first official report since July 2008)	clade: 2.3.2.1c	2015-01 [sequence] [76], [77], [78,79] (2015-01 to 2015-12), [44,45] (month NR), [16] (month NR)	
[44,45]	A new clade of H5N1 introduced into Nigeria in 2015	clade: 2.3.2.1c	2015-01 [sequence] [76], [77], [78,79] (2015-01 to 2015-12), [44,45] (month NR), [16] (month NR)	

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[47]	First African HPAI H5N1 outbreak reported on 7 February 2006 from a farm in Kaduna State	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.
[47]	First molecular characterisation of HPAI H5N1 detected in African wild birds (hooded vultures, Burkina Faso; not Nigeria)	not a detection claim (not compared)	-	
[32]	Nigeria was the first African country affected by H5N1, initially reported at a commercial farm in Kaduna State in January 2006	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.
[38,39]	Nigeria's outbreak was the first H5N1 Asian-strain outbreak in Africa; first clinically diagnosed at ABU VTH Zaria; NVRI confirmed type A on 6 February 2006 and Padova confirmed HPAI H5N1 (PQGERRRKKRG	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Earlier evidence in our data: 2006-01 ([58], [59]) precedes the claimant's own evidence 2007-04 ([38,39]).
[38,39]	April 2007 waterfowl outbreak was outside the cold harmattan months of earlier Nigerian outbreaks, suggesting HPAI may be becoming endemic in Zaria	not a detection claim (not compared)	-	
[61]	First confirmed H5N1 HPAI outbreak in Africa, January 2006, commercial farm in Kaduna State	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[62]	Evidence of AI (H5) antibodies in Kogi State, an area earlier reported free of the disease during the 2006-2008 HPAI outbreaks	not a detection claim (not compared)	-	
[62]	HPAI on the African continent first confirmed in Nigeria on 8 February 2006	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.
[128]	First HPAI outbreak in Africa reported in Nigeria in January 2006 (commercial flocks in Kaduna, Plateau and Kano); Nigeria's first outbreak on 16 January 2006, confirmed at NVRI and validated by OIE r	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	
[128]	First and only human H5N1 case in Nigeria confirmed January 2007, Lagos State	not a detection claim (not compared)	-	
[129]	AI first reported in Nigeria in February 2006; one human case in a woman from Lagos, February 2007	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.
[60]	Nigeria's first HPAI outbreak, February 2006, commercial farm in northern Nigeria (Sambawa farms, Jaji, Kaduna, reported January 2006); confirmed H5N1 (PQGERRRKKRGLFG) by Padova on 6 February 2006, no	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	
[96,97]	HPAI H5N1 occurred for the first time in Gombe State in 2015 (index case Kwadon)	state: Gombe	2008-05 to 2008-07 (range) [isolate] [74,75] (2008-05 to 2008-07)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2008-05 to 2008-07 (range).

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[96,97]	Association of low altitude and low rainfall with A1 seroprevalence not previously documented in Nigeria	not a detection claim (not compared)	-	
[31]	No previous study has investigated the impact of agro-ecological conditions on HPAI H5N1 epidemiology at the global scale (not a Nigeria first)	not a detection claim (not compared)	-	
[83]	Re-emergence first detected at a live bird market (Sabon Gari, Kano, 24 December 2014), after Nigeria was considered free of HPAI H5N1 since July 2008	clade: 2.3.2.1c	2015-01 [sequence] [76], [77], [78,79] (2015-01 to 2015-12), [44,45] (month NR), [16] (month NR)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2015-01.
[83]	Social network and value chain analysis combined with epidemiological surveillance described as a novel approach for Nigeria	not a detection claim (not compared)	-	
[78,79]	Emergence of intra-clade (WA1 x WA2) reassortant H5N1 genotypes in Nigeria, one of which (first detected in Katsina State) became predominant by 2016	sublineage: 2.3.2.1c WA2	2015 [sequence] [78,79] (2015-01 to 2015-12), [16] (month NR)	
[131]	Nigeria was the first African country to report HPAI (Feb 2006, commercial farm in Kaduna) - background, cited, not shown by this study; authors also say epidemiologic aspects of HPAI H5N1 in African	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	
[58]	First incursion of Asian H5N1 HPAI virus in Africa (Kaduna index outbreak, mid-January 2006) and first H5N1 isolate from the African continent	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	
[19]	First report of the new clade 2.3.4.4b H5N1 reassortant genotype EA-2024-DV in Nigeria and West Africa; not previously reported in Europe or West Africa	sublineage: 2.3.4.4b genotype EA-2024-DV	2024 [sequence] [19] (2024-02 to 2024-12)	
[19]	Kano area acted as a major source of viral diffusion within Nigeria, with onward spread towards Togo	not a detection claim (not compared)	-	
[104]	A major new sublineage in Nigeria, I/II-Nigeria, phylogenetically distinguishable from all previously identified sublineages of clade 2.2, which evolved locally in Nigeria	sublineage: sublineage I/II-N (clade 2.2, local)	2006-01 to 2008-07 (window) [sequence] [104] (2006-01 to 2008-07)	

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[104]	Two novel reassortment events, R2 and R4, in addition to the three described previously	not a detection claim (not compared)	-	
[104]	Nigeria is the only African country in which clade 2.2.1 and all three sublineages (I, II and III) of clade 2.2 have been detected	not a detection claim (not compared)	-	
[98]	First serological survey for avian influenza in turkeys in Nigeria is implied: the authors state there has been little or no report of surveillance in turkeys, which form almost 2% of the poultry popu	not a detection claim (not compared)	-	
[84]	The hotspots of avian influenza in Nigeria in 2015-2017 clustered in six states across five geopolitical zones, with the index case in Eti-Osa LGA, Lagos	not a detection claim (not compared)	-	
[84]	The H5N8 isolated from backyard poultry in Kano in week 52 of 2016 was "the first time HPAI H5N8 would be reported in Nigeria since 2006"	subtype: H5N8	2016-11 [sequence] [64]	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2016-11.
[41]	Live-bird markets are possible reservoirs of H5N1 in Nigeria, shown by detection in markets in two states that had recorded no outbreak	not a detection claim (not compared)	-	
[50]	Nigeria is a key region of genetic mutation on the first transmission route, where genetic distance from A/goose/Guangdong/1996 is greatest, which the authors call a more challenging or precarious sit	not a detection claim (not compared)	-	
[50]	Nigeria acted as the African entry point and onward source for West Africa and Sudan on route 1	not a detection claim (not compared)	-	
[87]	Detection of circulating LPAI H5N2 in domestic waterfowl four years after AI was considered eradicated from Nigeria (not framed as a formal 'first')	not a detection claim (not compared)	-	
[103]	Serological evidence of LPAI H3N8, H5N2 and H9N7 infection in Japanese quails and of H9N7 in a laughing dove in southwest Nigeria, species for which the authors say there is little previous Nigerian r	not a detection claim (not compared)	-	
[100]	First description of the spatial distribution and clustering of the free-of-charge diagnostic submission programme during the Nigerian HPAI H5N1 outbreaks	not a detection claim (not compared)	-	
[108]	Nigeria was a secondary epicentre of H5N1 that seeded tertiary outbreaks; the study is presented as the first quantification of H5N1 diversity across geographic space (global, not Nigeria-specific)	not a detection claim (not compared)	-	
[107]	Since the first Nigerian outbreak in early 2006, the Qinghai lineage (clade 2.2) was the only clade isolated in Africa up to early 2008	not a detection claim (not compared)	-	

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[107]	The Nigerian 2007 reassortant (I/II-2:6) spread beyond Nigeria: an identical genotype was found in the Benin outbreak of December 2007	not a detection claim (not compared)	-	
[63]	Rare evidence of non-lethal natural infection of healthy wild birds by an AIV with an HPAI viral genotype (not framed as a first for Nigeria)	not a detection claim (not compared)	-	
[112]	First serological survey for A antibodies in previously affected Plateau State farms since control of HPAI in 2008	not a detection claim (not compared)	-	
[135]	First documented report on the trend and distribution of HPAI in Rivers State for 2015-2022	not a detection claim (not compared)	-	
[52]	EMA3 sublineage of clade 2.2 detected in Gombe and Kebbi in June-July 2008 is novel to the African continent	sublineage: EMA3 / sublineage III (clade 2.2)	2008-05 to 2008-07 (range) [isolate] [74,75] (2008-05 to 2008-07); wider-window rows: [104] (2006-01 to 2008-07)	Earlier evidence in our data: 2008-05 to 2008-07 (range) ([74,75] (2008-05 to 2008-07)) precedes the claimant's own evidence 2008-06 to 2008-07 (range) ([52] (2008-06 to 2008-07)).
[73]	Among the first complete genomes of H5N1 viruses sequenced outside Asia; first demonstration of a reassortant between two EMA sublineages, in a Nigerian isolate	sublineage: EMA-1/EMA-2 (clade 2.2 sublineages)	2006 [sequence] [73] (month NR), [73] (month NR)	
[73]	The Nigerian isolates fall into two clades (EMA-1 and EMA-2), implying at least two introductions, against the three introductions reported by Ducatez et al. 2006	not a detection claim (not compared)	-	
[105]	Only report of a reassortant generated by clade 2.2 H5N1 viruses spreading extensively in poultry; new genotype emerged in <7 months	not a detection claim (not compared)	-	
[106]	The sublineages first imported into Nigeria in 2006 have been gradually replaced by reassortants; three distinct reassortants among 8 strains, including a reassortant between two reassortants	not a detection claim (not compared)	-	
[106]	All Nigerian reassortants carry sublineage C-derived NS genes with two amino acids (NS1 V194, NS2 R34) never seen in sublineage A, suggesting a fitness or adaptation advantage	not a detection claim (not compared)	-	
[53]	These are early Nigerian H5N1 isolates from surveillance that pre-dated the first official January 2006 report	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[53]	Only two clades (2.2.2 and 2.2.3) are present among Nigerian isolates, implying two introductions rather than the three reported previously	not a detection claim (not compared)	-	
[76]	A novel clade, 2.3.2.1c, has reached Nigeria, marking the reemergence of H5N1 after seven years	clade: 2.3.2.1c	2015-01 [sequence] [76], [77], [78,79] (2015-01 to 2015-12), [44,45] (month NR), [16] (month NR)	
[77]	Two co-circulating genetic groups (WA1 and WA2) within clade 2.3.2.1c in West Africa, implying at least two independent introductions	not a detection claim (not compared)	-	
[65–67]	First detection of HPAI H5N6 on the African continent	subtype: H5N6	2019-06 [isolate] [65–67]	
[65–67]	First report of AIV H9N2 (G1 lineage) in Nigeria	subtype: H9N2	2019 [sequence] [65–67] (2019-04 to 2019-12)	
[11]	Putative H5N1/H9N2 reassortant (H5N2) in a mixed-species commercial farm; HA/NA donor combination 'appears to be unique to Nigeria'; first H5N2 in commercial chickens in Nigeria (previous H5N2 detecti	subtype_host: H5N2/C	2021 [sequence] [11] (month NR), [11] (month NR)	
[74,75]	First detection in Africa of an H5N1 genotype of sublineage III (clade 2.2), a novel introduction into Nigeria	sublineage: EMA3 / sublineage III (clade 2.2)	2008-05 to 2008-07 (range) [isolate] [74,75] (2008-05 to 2008-07); wider-window rows: [104] (2006-01 to 2008-07)	
[119]	First serological evidence of influenza A subtype H7 in local birds and waterfowls sold at live-bird markets in Nigeria	antibody: H7	2013 [NA] [68] (2013-03 to 2013-10)	Earlier evidence in our data: 2013 ([68] (2013-03 to 2013-10)) precedes the claimant's own evidence 2018-05 to 2018-07 (range) ([119] (2018-05 to 2018-07)).
[99]	First report of A1 antibodies (exposure) in free-flying wild birds in Kogi State	not a detection claim (not compared)	-	
[120]	First serological indication of influenza A/H9 in Plateau State and Nigeria (Abstract); first detection of H9 from local chickens in Jos Plateau (Discussion)	antibody: H9	2013 [NA] [68] (2013-03 to 2013-10)	Earlier evidence in our data: 2013 ([68] (2013-03 to 2013-10)) precedes the claimant's own evidence 2020 ([120] (2020-03 to 2020-07)).
[18]	First report of an H9N2 outbreak in commercial poultry in southern Nigeria	subtype_host: H9N2/C	2020 to 2021 (window) [isolate] [91] (2020 to 2021), [91] (2020 to 2021); earliest dated: 2024-02 [18]	Earlier evidence in our data: 2020 to 2021 (window) ([91] (2020 to 2021), [91] (2020 to 2021)) precedes the claimant's own evidence 2024-02 ([18]). Scope: [18]'s claim concerns southern Nigeria (Oyo); the [91] rows are from Kaduna/Kano (northern), so they do not contradict a southern 'first'.
[91]	Implied first: H9N2-associated outbreaks in commercial poultry in Nigeria had not been detected until now	subtype_host: H9N2/C	2020 to 2021 (window) [isolate] [91] (2020 to 2021), [91] (2020 to 2021); earliest dated: 2024-02 [18]	
[118]	First nation-wide active disease surveillance in poultry in Nigeria	not a detection claim (not compared)	-	

Reference	Claim (as claimed by the authors)	Compared with	Our earliest evidence	Conflict note
[143]	HPAI H5N1 was first reported in Nigeria and in Africa in 2006 (background, cited)	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Claimant has no dated laboratory-confirmed row for this key; our earliest is 2006-01.
[71,72]	HPAI H5N1 was reported for the first time on the African continent in Nigeria in February 2006	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	Earlier evidence in our data: 2006-01 ([58], [59]) precedes the claimant's own evidence 2006 (earliest month-dated row 2006-03; other rows of 2006 dated to the year or a within-year range) ([71,72] (2006-02 to 2006-06)).
[71,72]	The dataset represents the widest spatio-temporal and species diversity of Nigerian H5N1 HA sequences publicly available (only 15 Nigerian isolates in public databases before this study)	not a detection claim (not compared)	-	
[92–95]	First documentation of state-by-state and region-by-region HPAI losses in commercial and backyard poultry in Nigeria	not a detection claim (not compared)	-	
[59]	First confirmed outbreaks of avian influenza in domestic birds in Nigeria: commercial chickens in Kaduna, Jos and Kano, January 2006	subtype: H5N1	2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [60] (month NR), [107] (2006-01 to 2006-06), [112] (month NR), [73] (month NR), [73] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (month NR), [49] (month NR), [52] (month NR); wider-window rows: [131] (2006 to 2007), [33] (2006 to 2008), [41] (2006 to 2009), [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007)	

T2. State x epoch matrix (laboratory-confirmed evidence)

Cell: subtypes; host codes; best evidence; rows (with flags such as spanning / inferred state / high-risk). § = official report, assay not stated (not counted); † = antibody-positive rows (exposure only); ‡ = antigen-only rows. Rows whose date range spans more than two epochs or exceeds 24 months across a boundary go to 'multi-epoch'.

State (zone)	2006-2010	2011-2014	2015-2020	2021-2026	multi-epoch	date NR
Benue (NC)	H5N1; C,B; sequence; [107], [92–95], [42] (high-risk study), [42] (high-risk study)				§ [90]	
FCT (NC)	H5N1; C,B; sequence; [131], [107], [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1; C,L; sequence; [78,79] ; § [84]	H5N1; host NR; sequence; [19]	§ [90]	
Kogi (NC)	† [62]	† [124]			§ [90]	† [99]
Kwara (NC)	H5N1; C,B; lab-confirmed case (method NR); [92–95], [42] (high-risk study), [42] (high-risk study)		† [13]		§ [90]	
Nasarawa (NC)	H5N1; C,B; lab-confirmed case (method NR); [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1; C,B,L; sequence; [64], [44,45]	H5N1; C; isolate; [54]	§ [90]	
Niger (NC)	H5N1; C,B; lab-confirmed case (method NR); [42] (high-risk study), [42] (high-risk study)				§ [90]	
Plateau (NC)	H5N1; C,B,L,Wr,Cap; sequence; [131], [107], [112], [105], [71,72], [92–95], [59], [42] (high-risk study), [42] (high-risk study) ; § [40]	† [112]	H5N1/H5N8; C,B,L; sequence; [78,79], [64], [44,45], [57] ; § [84], [141], [85], [144] ; † [119], [120]	H5N1/H5N8; C,B; sequence; [19], [137], [54] ; § [136], [141], [85]	§ [90]	type A / subtype NR; C; isolate; [32] (high-risk study)
Adamawa (NE)	H5N1; C,B; lab-confirmed case (method NR); [140], [92–95], [42] (high-risk study), [42] (high-risk study)				§ [90]	
Bauchi (NE)	H5N1; C,B; sequence; [131], [107], [105], [80], [92–95], [42] (high-risk study), [42] (high-risk study) ; § [40]	† [96,97]	H5N1; C,B,L; sequence; [78,79], [64], [44,45] ; § [84] ; † [102]	H5N1/H5N2/H5N8; C,O; sequence; [81,82], [19], [11], [54]	§ [90] ; † [125]	H5 (N not determined); C,B,L,Wr,Cap; RNA; [96,97] ; † [96,97]
Borno (NE)	H5N1; C,B; sequence; [107], [105], [92–95], [42] (high-risk study), [42] (high-risk study) ; § [36]			type A / subtype NR; B; RNA; [126]	§ [90]	type A / subtype NR; B,L; RNA; [127] ; † [37]
Gombe (NE)	H5N1; B,L; sequence; [74,75], [52]	† [96,97]	H5N1; C,B,L; RNA; [44,45] ; † [102]		§ [90] ; † [125]	H5 (N not determined); C,B,L,Wr,Cap; RNA; [96,97] ; † [96,97]
Taraba (NE)	H5N1; C,B; sequence; [107], [92–95], [42] (high-risk study), [42] (high-risk study)				§ [90]	
Yobe (NE)	H5N1; B; lab-confirmed case (method NR); [92–95]				§ [90]	
Jigawa (NW)	H5 (N not determined)/H5N1/H5N2; C,B,Wr; sequence; [63], [92–95], [42] (high-risk study), [42] (high-risk study) ; § [139] ; † [115]		H5N1; C,B,L; sequence; [78,79], [44,45]		§ [90]	H5N1; B; isolate; [61] ; † [61]

State (zone)	2006-2010	2011-2014	2015-2020	2021-2026	multi-epoch	date NR
Kaduna (NW)	H5N1; C,B,L; sequence; [131], [58], [107], [105], [71,72], [92–95], [59], [38,39] (inferred state; high-risk study), [42] (high-risk study), [42] (high-risk study) ; § [139], [40] ; ‡ [38,39]	† [89]	H5 (N not determined)/H5N1/H9N2; C,B,L,W; sequence; [78,79], [64], [65–67], [116], [44,45] ; † [102], [116], [91]	H5 (N not determined)/H5N1/H9 (N not determined); C; sequence; [19], [136], [70], [54] ; † [136], [91] ; ‡ [70]	§ [90]	† [122,123], [114], [121]
Kano (NW)	H5 (N not determined)/H5N1; C,B; sequence; [131], [107], [105], [69], [92–95], [59], [52], [42] (high-risk study), [42] (high-risk study) ; § [139], [142], [40] ; ‡ [69]	H5N1; L; isolate; [44,45] ; § [83] ; ‡ [111]	H5N1/H5N8; C,B,L; sequence; [78,79], [64], [76], [44,45] ; § [83], [84], [138] ; † [91]	H5N1; host NR; sequence; [19] ; § [136] ; † [91]	§ [90]	
Katsina (NW)	H5N1; C,B; isolate; [92–95], [52], [42] (high-risk study), [42] (high-risk study) ; § [139], [40]		H5N1; C,B,L; sequence; [78,79], [44,45]	H5N1; host NR; sequence; [19]	§ [90]	
Kebbi (NW)	H5N1; B,L; isolate; [52]			H5N1; host NR; sequence; [19]	§ [90]	
Sokoto (NW)	H5N1; C,L; sequence; [129], [107], [105], [71,72], [92–95]		H5N1/H5N6/H9N2; C,B,L; sequence; [78,79], [65–67], [44,45]		§ [90]	
Zamfara (NW)	H5N1; B; lab-confirmed case (method NR); [92–95]		H5N1; C,B,L; sequence; [78,79], [44,45]	H5N1; host NR; sequence; [19]	§ [90]	
Abia (SE)			H5N1/H9N2; C,B,L; sequence; [65–67], [44,45]		§ [90]	
Anambra (SE)	H5N1; C,B; sequence; [107], [105], [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1; C,B,L; RNA; [44,45]		§ [90]	
Ebonyi (SE)				H5N1; L; lab-confirmed case (method NR); [134]	§ [90]	
Enugu (SE)	H5N1; C,B; lab-confirmed case (method NR); [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1/H9N2; C,B,L; sequence; [78,79], [133], [65–67], [44,45]		§ [90]	
Imo (SE)			H5N1/H9N2; C,B,L; sequence; [65–67], [44,45]		§ [90]	
Akwa Ibom (SS)					§ [90]	
Bayelsa (SS)			H5N1; C,L; sequence; [78,79]		§ [90]	
Delta (SS)			H5N1; C,B,L; sequence; [78,79], [44,45]	H5N1; Cap; RNA; [81,82]	§ [90]	
Edo (SS)	H5N1; C,B; lab-confirmed case (method NR); [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1; C,B,L; sequence; [78,79], [44,45]		§ [90]	
Rivers (SS)	H5N1; C,B; lab-confirmed case (method NR); [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1/H9N2; C,B,L; sequence; [78,79], [65–67], [44,45] ; § [84], [135]	§ [135]	§ [90]	
Ekiti (SW)	H5N1; C; sequence; [106]			H5N1; host NR; sequence; [19]	§ [90]	

State (zone)	2006-2010	2011-2014	2015-2020	2021-2026	multi-epoch	date NR
Lagos (SW)	H5N1; C,B,L; sequence; [131], [107], [105], [106], [71,72], [92–95], [42] (high-risk study), [42] (high-risk study)		H5N1/H9N2; C,B,L; sequence; [78,79], [76], [77], [65–67], [44,45] (window start overridden by exact-month rows) ; § [83], [84] ; † [117]	H5N1; C,B,L; sequence; [81,82], [19], [56]	§ [90]	
Ogun (SW)	H5N1; C,B; sequence; [131], [106], [92–95], [42] (high-risk study), [42] (high-risk study) ; § [130]		H5N1/H5N8/H9N2; C,B,L; sequence; [78,79], [65–67], [44,45]		§ [90]	† [35]
Osun (SW)					§ [90]	† [98], [103]
Oyo (SW)	H5N1; C,B,L; sequence; [107], [106], [71,72], [92–95], [53], [42] (high-risk study), [42] (high-risk study)	H5 (N not determined)/H5N2; L; sequence; [87] ; † [68]	H5N1/H9N2; C,L; sequence; [78,79], [65–67] ; § [84]	H9N2; C; RNA; [18] ; † [113]	§ [90]	† [98], [103], [35]

States with confirmed evidence per column: 2006-2010: 27 (26 states + FCT; sequence/isolate: 17 states + FCT); 2011-2014: 2 (2 states; sequence/isolate: 2 states); 2015-2020: 22 (21 states + FCT; sequence/isolate: 19 states + FCT); 2021-2026: 15 (14 states + FCT; sequence/isolate: 10 states + FCT); multi-epoch: 0 (0 states; sequence/isolate: 0 states); date NR: 5 (5 states; sequence/isolate: 2 states)

States with official reports only (§) per column: 2006-2010: -; 2011-2014: -; 2015-2020: -; 2021-2026: Rivers; multi-epoch: Abia, Adamawa, Akwa Ibom, Anambra, Bauchi, Bayelsa, Benue, Borno, Delta, Ebonyi, Edo, Ekiti, Enugu, FCT, Gombe, Imo, Jigawa, Kaduna, Kano, Katsina, Kebbi, Kogi, Kwara, Lagos, Nasarawa, Niger, Ogun, Osun, Oyo, Plateau, Rivers, Sokoto, Taraba, Yobe, Zamfara; date NR: -

Background statements, not counted: [59] (background statement without source (129 outbreaks / 15 states by mid-2006)); [134] (background statement; source NR (4 Ebonyi outbreaks 2016-2019))

T2 notes:

[129]: Sokoto 2008-09 type-A RNA: the 2 chicken positives were negative on Luxembourg re-test; the pigeon positive is from the Luxembourg test.

[126]: Borno 2021: type-A (M-gene) RNA in apparently healthy ducks, not subtyped.

[32]: Plateau (date NR): high-risk study; 'isolates' identified by HA plus a rapid kit, subtype not shown.

[135]: Rivers 2015-2022 rows are official reports (assay not stated): § mark only.

[44,45]: Kano 2011-2014 H5N1 entry rests on deaths of 24 Dec 2014 (specimens received 2015-01).

placement rule: A row spanning an epoch boundary is not placed in the earlier epoch when exact-month confirmed rows for the same state inside its window date it later (B4).

Rows with positives that cannot be assigned to one state: [49] (Lagos or Ogun or Oyo; E1); [49] (Lagos or Ogun or Oyo; E1); [49] (Lagos or Ogun or Oyo; E1); [12] (Jigawa or Yobe; E4); [12] (Jigawa or Yobe; E4); [12] (Jigawa or Yobe; E4); [91] (Kaduna or Kano; E3,E4); [91] (Kaduna or Kano; E3,E4)

T3. Host range

Host system	Status	Confirmed detection (sequence/isolate/RNA/lab case)	Official report (§, not counted)	Antibody positive (exposure only)
C (commercial poultry)	confirmed detection	H5 (N not determined)/H5N1/H5N2/H5N8/H9 (N not determined)/H9N2; earliest 2006-01 [sequence] [58], [59], [53], [128] (2006-01 to 2006-12), [112] (month NR), [71,72] (2006-02 to 2006-06), [145] (month NR), [49] (2006-01 to 2006-08), [49] (month NR), [49] (month NR); wider-window rows: [46] (2006 to 2007), [46] (2006 to 2007), [46] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007), [92–95] (2006 to 2008); earliest by subtype: H5 (N not determined): 2006; H5N1: 2006-01; H5N2: 2021; H5N8: 2019-02; H9 (N not determined): 2024-10 to 2025-07 (window); H9N2: 2020 to 2021 (window); type A / NR: 2006 to 2007 (window); species stated positive (single-host rows): broiler, chicken, layer; rows: [128], [96,97], [83], [78,79], [58], [104], [133], [100], [112], [106], [11], [136], [137], [80], [18], [91], [70], [71,72], [92–95], [59], [145], [44,45], [46], [49], [52], [53], [54], [56], [57], [32], [42]	§ [130], [83], [90], [84], [100], [135], [141], [142], [85], [144]	H3N8/H5 (N not determined)/H5N2/H7 (N not determined)/H9 (N not determined)/H9N7/type A; species (single-host rows): broiler, chicken, layer, quail, turkey; rows: [96,97], [125], [98], [103], [112], [68], [136], [117], [102], [91], [37]
B (backyard/village poultry)	confirmed detection	H5 (N not determined)/H5N1/H5N8; earliest 2006 (earliest month-dated row 2006-02; other rows of 2006 dated to the year or a within-year range) [sequence] [80] (2006-02 to 2006-05), [128] (2006-01 to 2006-12), [71,72] (2006-02 to 2006-06), [145] (month NR), [52] (month NR); wider-window rows: [52] (2006 to 2007), [42] (2006 to 2008), [42] (2006 to 2007), [42] (2006 to 2007), [92–95] (2006 to 2008), [104] (2006-01 to 2008-07), [100] (2006-01 to 2007-12), [92–95] (2006-01 to 2007-03); earliest by subtype: H5 (N not determined): 2006-12; H5N1: 2006 (earliest month-dated row 2006-02; other rows of 2006 dated to the year or a within-year range); H5N8: 2019-02; type A / NR: 2007-01 to 2007-02 (range); species stated positive (single-host rows): chicken, duck, goose, guinea fowl, pigeon, turkey; rows: [61], [128], [96,97], [83], [81,82], [104], [100], [76], [126], [11], [80], [140], [69], [127], [71,72], [92–95], [145], [44,45], [52], [54], [56], [57], [38,39], [42]	§ [83], [90], [100], [85], [144], [52]	H3N8/H5 (N not determined)/H5N2/type A; species (single-host rows): chicken, duck, turkey; rows: [61], [62], [96,97], [122,123], [125], [98], [124], [113], [37]

Host system	Status	Confirmed detection (sequence/isolate/RNA/lab case)	Official report (§, not counted)	Antibody positive (exposure only)
L (live-bird market)	confirmed detection	H5 (N not determined)/H5N1/H5N2/H5N6/H5N8/H9N2; earliest 2006 [sequence] [71,72] (2006-02 to 2006-06), [52] (month NR); wider-window rows: [41] (2006 to 2009), [52] (2006 to 2007), [92–95] (2006 to 2008), [104] (2006-01 to 2008-07); earliest by subtype: H5 (N not determined): 2011-09 to 2012-02 (window); H5N1: 2006; H5N2: 2012; H5N6: 2019-06; H5N8: 2019-07; H9N2: 2019; type A / NR: 2007-10 to 2007-11 (range); species stated positive (single-host rows): chicken, duck, guinea fowl, layer, pigeon; rows: [129], [60], [96,97], [83], [78,79], [104], [133], [134], [87], [76], [65–67], [74,75], [116], [127], [71,72], [92–95], [44,45], [52], [56], [41]	§ [83], [84], [138], [52]	H5 (N not determined)/H7 (N not determined)/H9 (N not determined)/NR/type A; species (single-host rows): chicken, duck, goose, guinea fowl, turkey; rows: [115], [89], [62], [13], [125], [124], [119], [102], [114], [120], [121], [116], [52], [35]
Wm (wild, migratory)	confirmed detection (stated positive species)	H5N1; earliest 2022-03 to 2022-04 (range) [sequence] [12] (2022-03 to 2022-04), [12] (2022-03 to 2022-04), [12] (2022-03 to 2022-04); earliest by subtype: H5N1: 2022-03 to 2022-04 (range); species stated positive (single-host rows): ruff, white stork; rows: [12]	-	type A; species (single-host rows): -; rows: [99]
Wr (wild, resident)	confirmed detection	H5 (N not determined)/H5N1/H5N2; earliest 2006-03 [sequence] [71,72]; earliest by subtype: H5 (N not determined): 2007-02; H5N1: 2006-03; H5N2: 2007-02; type A / NR: 2007-02; species stated positive (single-host rows): duck, goose, jacana, nightjar, spur-winged, vulture, whistling duck; rows: [96,97], [101], [12], [63], [116], [71,72]	-	H5 (N not determined)/H9 (N not determined)/type A; species (single-host rows): dove, egret, parrot, pigeon; rows: [96,97], [99], [102], [114], [116]
W? (wild, migratory status unresolved)	official report only (assay not stated; §)	none	§ [90]	H5 (N not determined)/type A; species (single-host rows): dove, egret, pigeon; rows: [89], [96,97]
Cap (captive wild/zoo/ornamental)	confirmed detection	H5 (N not determined)/H5N1; earliest 2006-04 [sequence] [71,72]; earliest by subtype: H5 (N not determined): None; H5N1: 2006-04; type A / NR: 2015 to 2017 (window); species stated positive (single-host rows): duck; rows: [96,97], [81,82], [133], [71,72]	§ [84]	H5 (N not determined)/type A; species (single-host rows): crane, duck, goose, ostrich, parrot, peacock, peafowl, stork; rows: [96,97], [13], [102]
O (other (ostrich farms, imports, quail flock, etc.))	confirmed detection	H5N1/H5N8; earliest 2021 (earliest month-dated row 2021-02; other rows of 2021 dated to the year or a within-year range) [RNA] [81,82], [11] (month NR); wider-window rows: [11] (2021 to 2022), [11] (2021-01 to 2022-12); earliest by subtype: H5N1: 2021 to 2022 (window); H5N8: 2021 (earliest month-dated row 2021-02; other rows of 2021 dated to the year or a within-year range); species stated positive (single-host rows): pheasant; rows: [81,82], [11]	-	H5 (N not determined)/H5N2/H9N7/type A; species (single-host rows): dove, quail, white stork; rows: [89], [103], [43], [37]

Rows excluded from host keys: [92–95] (ostriches were included only for computation of losses, not necropsied (thesis); cannot confirm host system O or species ostrich)

T3b. Species keywords

Species are keyword tags from the consensus species/host text. 'Species stated positive' = the row's only species, or a clause naming the species with a positive count, 'positive' or 'isolate from'. The vulture tag rests on [71,72], whose strain is named chicken in thesis Table 2.1.

[83]: 'layer' tag comes from supplier text ('point-of-lay pullets, spent layers'); LBM row, status unchanged

[128]: 'chicken' tag comes from host text ('poultry farms and local chicken communities'); status unchanged

[92–95]: ostriches were included only for computation of losses, not necropsied (thesis); cannot confirm host system O or species ostrich

[58]: 'isolate from chicken': only chicken is stated positive; geese and ostriches were sampled

Species tag	Status	Rows where species stated positive	Rows naming species (positivity not established)	Antibody rows
black stork	antibody only (exposure)	-	-	[99]
broiler	confirmed detection (species stated positive)	[92–95]	[137], [18], [70], [59], [44,45], [54], [57]	[117], [37]
chicken	confirmed detection (species stated positive)	[61], [128], [129], [64], [58], [133], [134], [107], [73], [106], [76], [77], [65–67], [71,72], [44,45], [46], [49], [53], [54]	[96,97], [78,79], [19], [107], [73], [105], [11], [136], [137], [80], [140], [69], [18], [92–95], [44,45], [54], [56], [57], [42]	[115], [61], [62], [96,97], [13], [122,123], [125], [112], [68], [136], [124], [117], [119], [102], [120], [121], [35], [37]
crane	named in a row with a confirmed detection (species positivity not established)	-	[96,97], [80], [56]	[96,97], [102]
dove	antibody only (exposure)	-	-	[96,97], [103], [102], [114], [116]
duck	confirmed detection (species stated positive)	[81,82], [64], [12], [87], [63], [126], [65–67], [74,75], [71,72], [92–95]	[129], [78,79], [12], [19], [105], [11], [80], [140], [92–95], [56], [38,39], [42]	[115], [96,97], [13], [124], [119], [102], [113], [37]
egret	antibody only (exposure)	-	-	[96,97], [102], [114], [116]
goose	confirmed detection (species stated positive)	[81,82], [12], [63], [92–95]	[78,79], [58], [101], [12], [80], [140], [69], [92–95], [59], [54], [56], [38,39], [42]	[96,97], [119], [102], [116], [37]
guinea fowl	confirmed detection (species stated positive)	[64], [65–67], [71,72], [92–95]	[129], [78,79], [107], [73], [11], [80], [140], [42]	[115], [96,97], [119], [102]
jacana	confirmed detection (species stated positive)	[12]	[12]	-
layer	confirmed detection (species stated positive)	[83], [91], [71,72], [59], [145], [49]	[19], [11], [136], [137], [18], [70], [92–95], [59], [44,45], [54]	[112], [68], [136], [117], [102], [91], [37]
mallard	antibody only (exposure)	-	-	[116]
muscovy	named in a row with a confirmed detection (species positivity not established)	-	[56]	[102]
nightjar	confirmed detection (species stated positive)	[12]	[12]	-
ostrich	named in a row with a confirmed detection (species positivity not established)	-	[58], [107], [73], [80], [92–95], [59], [42]	[96,97]
parrot	confirmed detection (species stated positive)	[116]	-	[96,97], [102], [116]
peacock	named in a row with a confirmed detection (species positivity not established)	-	[56]	[96,97]
peafowl	named in a row with a confirmed detection (species positivity not established)	-	[54]	[13]
pheasant	confirmed detection (species stated positive)	[81,82]	[42]	-
pigeon	confirmed detection (species stated positive)	[129], [127], [71,72]	[80], [140], [42]	[96,97], [102], [114], [116]

Species tag	Status	Rows where species stated positive	Rows naming species (positivity not established)	Antibody rows
quail	antibody only (exposure)	-	-	[103], [43]
ruff	confirmed detection (species stated positive)	[12]	[12]	-
spur-winged	confirmed detection (species stated positive)	[12], [63]	[101], [12]	-
stork	antibody only (exposure)	-	-	[13]
turkey	confirmed detection (species stated positive)	[71,72], [92-95]	[129], [78,79], [80], [69], [59], [42]	[96,97], [98], [124], [119]
vulture	confirmed detection (species stated positive)	[71,72]	-	-
weaver	antibody only (exposure)	-	-	[114]
whistling duck	confirmed detection (species stated positive)	[12], [63]	[12]	-
white stork	confirmed detection (species stated positive)	[12]	[12]	[89]

T4. Clade transitions

Verdicts follow the language rules: a reassortant needs segment incongruence; an introduction needs a phylogeographic link; capped studies 'suggest'.

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2006-2008 wave	H5 (N not determined)	2.2	clade stated without sequence data shown ('suggests')	[46]	stated	claimed from cleavage-site motif only	HA2 fragment, 154 nt (positions 1531-1684); HA cleavage site (cloned)	NA	M3 Yes / M4 No / M5 N/A	suggestive	none claimed	none claimed
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[14]	reused		all 8 (HA, NA, MP, NP, NS, PA, PB1, PB2)	1000 bootstrap replicates. CAN-0155 Methods: "The robustness	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[64]	reused	four West African lineages	Complete genomes generated; HA gene analysed	Bayes factor (>5), posterior probability, 95% HPD	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogeographic link (meets the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[108]	reused	Qinghai-like	HA (1716 positions) and NA (1419 positions)	Bootstrap, 1000 replicates; Monte Carlo upper-tail tests wit	M3 Unclear / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[107]	reused	I/II-2:6 reassortant (cited); sublineages I, II, IV	Complete coding sequences of all eight segments (HA and NA for 67 strains; full	Bootstrap 1000 replicates; Bayesian posterior probabilities	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[73]	own	EMA-1, EMA-2 (EMA mapped to clade 2.2 in consensus clade_system); EMA-1/EMA-2 4:4 reassortant (1047-62); EMA mapped to clade 2.2 in consensus	Complete genomes, all 8 segments	Bootstrap, 1000 replicates	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[105]	own	EMA1/EMA2-2:6 reassortant (10/12); EMA2 (2/12)	All 8 segments (whole genome)	Bootstrap, 1,000 replicates (Figure legends)	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	none claimed
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[106]	own	AC HA/NS reassortant (EMA1/EMA2); AC NS reassortant (EMA1/EMA2); sublineage A (EMA2); authors map A/C to EMA2/EMA1 (clade 2.2)	All 8 segments, complete for most viruses	Bootstrap, 1000 replicates by maximum likelihood	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	none claimed
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[110]	reused	EMA1/EMA2 2:6 reassortant; EMA1/EMA2 4:4 reassortant	NA	NA	M3 NA / M4 NA / M5 NA	NA	inherited genotype (shown elsewhere)	none claimed
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[74,75]	own	sublineage III	Full genome (8 segments) of A/duck/Nigeria/3724-2/2008; HA of A/duck/Nigeria/372	Posterior probabilities (Bayesian trees)	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[71,72]	own	study HA sublineages A, D, E, F, G, H; sublineage H	HA only (nt 92-1633 of 1,730)	1,000 bootstrap replicates; >70 taken as strong support	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[50]	reused		HA (NA sequences also collected; analysis is of HA)	Bootstrap values greater than 60 marked on branches	M3 Unclear / M4 Yes / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[52]	stated	EMA3	NA	NA	M3 No / M4 No / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2	clade assignment shown (own sequences with phylogenetic support; study not capped)	[55]	reused		HA (public sequences)	Bootstrap, 1000 replicates	M3 N/A / M4 No / M5 Yes	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2.2	clade assignment from own sequences but support/cap limits apply ('suggests')	[53]	own		Coding regions of all 8 segments of 3 viruses	Bootstrap, 1000 replicates	M3 No / M4 No / M5 Yes	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	2.2; 2.2.1	clade assignment shown (own sequences with phylogenetic support; study not capped)	[104]	own	sublineages I, II, III, I/II-N; reassortants R2, R4	All eight segments, complete coding sequences	Bootstrap 1000 NJ replicates, Bayesian posterior probability	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	Qinghai lineage (pre-WHO name; not mapped by the paper)	clade assignment shown (own sequences with phylogenetic support; study not capped)	[58]	own		All 8 segments (polygenic sequencing; Table 1 primers)	Bootstrap 1,000	M3 Yes / M4 Yes / M5 Yes	none	none claimed	none claimed
2006-2008 wave	H5N1	no WHO clade (EMA / West African sublineages)	re-analysis of other studies' sequences only (not independent evidence)	[109]	reused	EMA / West African sublineages	HA, full or partial length	aLRT (SH-like) and 1000 bootstrap replicates; values of 50%	M3 Unclear / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	no WHO clade (reassortant of reassortants)	re-analysis of other studies' sequences only (not independent evidence)	[110]	reused	reassortant of reassortants	NA	NA	M3 NA / M4 NA / M5 NA	NA	inherited genotype (shown elsewhere)	none claimed
2006-2008 wave	H5N1	no WHO clade (study clusters A/B/C (I/II/III))	re-analysis of other studies' sequences only (not independent evidence)	[47]	reused	study clusters A/B/C (I/II/III)	HA (HA1 analysed)	100 bootstraps, 3 jumbles	M3 Unclear / M4 No / M5 Yes	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	no WHO clade (study sublineages A and B (WHO clade not named))	clade assignment from own sequences but support/cap limits apply ('suggests')	[49]	own	study sublineages A and B (WHO clade not named)	All 8 segments (full genomes); partial genomes for further birds	NR	M3 Yes / M4 Yes / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N1	no WHO clade (study sublineages A/C (comparators))	re-analysis of other studies' sequences only (not independent evidence)	[49]	reused	study sublineages A/C (comparators)	All 8 segments (full genomes); partial genomes for further birds	NR	M3 Yes / M4 Yes / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2006-2008 wave	H5N2	no WHO clade (LPAI H5N2 Eurasian gene pool; NS allele B (African))	clade assignment shown (own sequences with phylogenetic support; study not capped)	[101]	own	LPAI H5N2 Eurasian gene pool; NS allele B (African)	All 8 segments	Posterior probability (>0.75 shown); NJ bootstrap 1,000	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	phylogenetic clustering only (does not meet the introduction rule)
2006-2008 wave	H5N2	no WHO clade (LPAI-like Eurasian H5 lineage with HP genotype (H5N2))	clade assignment shown (own sequences with phylogenetic support; study not capped)	[63]	own	LPAI-like Eurasian H5 lineage with HP genotype (H5N2)	HA (segment 4) and NA (segment 6) complete ORF	Bootstrap, 1,000 replicates	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2009-2014 inter-wave	H5N2	no WHO clade (LPAI H5 wild-duck-associated cluster)	clade assignment shown (own sequences with phylogenetic support; study not capped)	[87]	own	LPAI H5 wild-duck-associated cluster	HA partial (cleavage-site region); NA subtyped by N1-N9 primer panel	Bootstrap, 1,000 replicates	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[78,79]	own	WA1 and WA2; reassortants R0-R8; WA2 subgroup (reassortant genotype predominant)	Complete genomes (all 8 segments)	Bootstrap 1,000 replicates	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[64]	own/reused		Complete genomes generated; HA gene analysed	Bayes factor (>5), posterior probability, 95% HPD	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogeographic link (meets the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[76]	own	HA sequenced; PB2 from Asian H9N2 (inherited genotype)	Complete genome, all 8 segments (one virus); HA for both	Bootstrap, 100 replicates; values above 60 shown	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[77]	own	WA1	Complete coding sequences for 39 viruses; partial for 4	Non-parametric bootstrap, 100 replicates; values above 60 sh	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[110]	reused	H9N2-PB2 reassortant	NA	NA	M3 NA / M4 NA / M5 NA	NA	inherited genotype (shown elsewhere)	none claimed
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[44,45]	own		Complete genome of index virus (CAN-1902); HA phylogeny cited from Monne et al.	NR	M3 No / M4 Yes / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[55]	reused		HA (public sequences)	Bootstrap, 1000 replicates	M3 N/A / M4 No / M5 Yes	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1c	clade assignment shown (own sequences with phylogenetic support; study not capped)	[16]	own	WA2; clusters I/II	HA (complete gene)	NR	M3 Unclear / M4 Unclear / M5 Unclear	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N1	2.3.2.1f (as labelled; BV-BRC)	re-analysis of other studies' sequences only (not independent evidence)	[14]	reused		all 8 (HA, NA, MP, NP, NS, PA, PB1, PB2)	1000 bootstrap replicates. CAN-0155 Methods: "The robustness	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2014/15-2017 wave	H5N8	2.3.4.4 group B (as labelled)	clade assignment shown (own sequences with phylogenetic support; study not capped)	[64]	own	WA-Introduction 1	Complete genomes generated; HA gene analysed	Bayes factor (>5), posterior probability, 95% HPD	M3 Yes / M4 Yes / M5 Yes	none	none claimed	none claimed
2018-2020	H5N6	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[65–67]	own	European 2017-2018 H5N6 cluster	Whole genome (8 segments)	Ultrafast bootstrap 1000 (CAN-1585); ultrafast bootstrap >80	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)
2018-2020	H5N8	2.3.4.4	clade stated without sequence data shown ('suggests')	[57]	stated	sequence data not shown	HA (cleavage site at least; clade assignment reported)	NA	M3 No / M4 No / M5 N/A	suggestive	none claimed	hypothesised only
2018-2020	H5N8	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[65–67]	own	South Africa 2018 H5N8 genotype	Whole genome (8 segments)	Ultrafast bootstrap 1000 (CAN-1585); ultrafast bootstrap >80	M3 Yes / M4 Yes / M5 Yes	none	inherited genotype (shown elsewhere)	phylogenetic clustering only (does not meet the introduction rule)
2018-2020	H9N2	H9 G1 lineage	clade assignment shown (own sequences with phylogenetic support; study not capped)	[65–67]	own	two West African clusters	Whole genome (8 segments)	Ultrafast bootstrap 1000 (CAN-1585); ultrafast bootstrap >80	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2018-2020	H9N2	H9 G1 lineage	clade assignment shown (own sequences with phylogenetic support; study not capped)	[48]	reused		HA used for phylogeography	Bootstrap 1,000 (ML); posterior probability >0.75 (MCC)	M3 Unclear / M4 No / M5 Yes	suggestive	none claimed	none claimed

Period	Subtype	Clade	Group clade verdict	Reference	Basis (own/reused/stat ed)	Sublineage/ genotype	Segments	Support	RoB M3/M4/M5	Cap	Reassortment (study x subtype)	Introduction (study x subtype)
2021-2022 wave	H5N1	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[14]	reused		all 8 (HA, NA, MP, NP, NS, PA, PB1, PB2)	1000 bootstrap replicates. CAN-0155 Methods: "The robustness	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2021-2022 wave	H5N1	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[12]	own	groups A/B; A/Eurasian_Wigeon/Netherlands/1/2020-like	Whole genome (8 segments)	Ultrafast bootstrap, 1000 replicates (values >80 shown)	M3 Yes / M4 Yes / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2021-2022 wave	H5N1	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[11]	own	A/ Eurasian_Wigeon /Netherlands/ 1/2020-like; HA Group 1/1-A	Whole genome (8 segments); partial genomes for 7	Ultrafast bootstrap, 1000 replicates (values >80% shown)	M3 Yes / M4 Unclear / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2021-2022 wave	H5N1	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[56]	stated	sequences not shown	NR	NA	M3 No / M4 No / M5 No	suggestive	none claimed	hypothesised only
2021-2022 wave	H5N2	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[11]	own	H5N1/H9N2 reassortant; H5N1/H9N2 reassortant (NA, M from H9N2 G1)	Whole genome (8 segments); partial genomes for 7	Ultrafast bootstrap, 1000 replicates (values >80% shown)	M3 Yes / M4 Unclear / M5 Yes	none	demonstrated (segment incongruence shown)	none claimed
2021-2022 wave	H5N8	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[11]	own	A/duck/ Chelyabinsk/ 1207-1/2020-like; HA Group 2	Whole genome (8 segments); partial genomes for 7	Ultrafast bootstrap, 1000 replicates (values >80% shown)	M3 Yes / M4 Unclear / M5 Yes	none	none claimed	phylogenetic clustering only (does not meet the introduction rule)
2023-2025 wave	H5N1	2.3.4.4b	clade assignment shown (own sequences with phylogenetic support; study not capped)	[19]	own/stated	EA-2024-DV in 19/28 outbreaks; genotype EA-2020-C; genotype EA-2024-DV; genotypes EA-2020-C and EA-2024-DV	Complete genomes (8 segments)	Bootstrap values above 80 shown on the HA tree	M3 Yes / M4 Yes / M5 Yes	none	demonstrated (segment incongruence shown)	phylogeographic link (meets the introduction rule)
NR	H5N1	2.2	re-analysis of other studies' sequences only (not independent evidence)	[50]	reused		HA (NA sequences also collected; analysis is of HA)	Bootstrap values greater than 60 marked on branches	M3 Unclear / M4 Yes / M5 No	suggestive	none claimed	phylogenetic clustering only (does not meet the introduction rule)

Notes: 2.2.1, 2.2.2 and the EMA-1/2/3, I/II/III/IV and A-H labels are sublineages within clade 2.2; none is an independent clade and none can pre-date its parent clade 2.2. '2.3.4.4 group B' ([64]) corresponds to the clade now named 2.3.4.4b; the study label is preserved.

T4b. Clades and subtypes per period (own-data groups)

Period	Subtypes	Clades	Rows
2006-2008 wave	H5 (N not determined), H5N1, H5N2	2.2, 2.2.2, 2.2; 2.2.1, Qinghai lineage (pre-WHO name; not mapped by the paper), no WHO clade (LPAI H5N2 Eurasian gene pool; NS allele B (African)), no WHO clade (LPAI-like Eurasian H5 lineage with HP genotype (H5N2)), no WHO clade (study sublineages A and B (WHO clade not named))	[14], [64], [58], [101], [104], [108], [107], [63], [73], [105], [106], [110], [74,75], [71,72], [46], [49], [50], [52], [53], [55]
2009-2014 inter-wave	H5N2	no WHO clade (LPAI H5 wild-duck-associated cluster)	[87]
2014/15-2017 wave	H5N1, H5N8	2.3.2.1c, 2.3.4.4 group B (as labelled)	[78,79], [64], [76], [77], [110], [44,45], [55], [16]
2018-2020	H5N6, H5N8, H9N2	2.3.4.4, 2.3.4.4b, H9 G1 lineage	[65–67], [48], [57]
2021-2022 wave	H5N1, H5N2, H5N8	2.3.4.4b	[14], [12], [11], [56]
2023-2025 wave	H5N1	2.3.4.4b	[19]

T5. Accession and methods audit (one row per molecular study)

Reference	Platform	Segments	n sequences	Phylogeny support	Accessions	M3	M4	M5	Cap	High risk
[14]	NA (public sequences)	all 8 (HA, NA, MP, NP, NS, PA, PB1, PB2)	61 complete genomes (11 Nigerian)	1000 bootstrap replicates. CAN-0155 Methods: "The robustness	list (Table S2: accessions for all 8 segments of the 11 Nigerian and 50 comparator genomes; the 289 OpenFlu sequences ar	Yes	Yes	Yes	none	no
[78,79]	Illumina MiSeq 300-bp paired-end (Nextera XT); consensus by	Complete genomes (all 8 segments)	100 genomes; GenBank MF112254-MF113041	Bootstrap 1,000 replicates	range: GenBank MF112254-MF113041 (100 genomes)	Yes	Yes	Yes	none	no
[64]	Illumina MiSeq 2x250 bp (Nextera XT); consensus via BWA/LoFr	Complete genomes generated; HA gene analysed	40 new African genomes (7 Nigerian); >1,200 HA seq	Bayes factor (>5), posterior probability, 95% HPD	list: GISAID EPI1398275-EPI1398280 (6 Nigerian H5N1 HA generated here); reused Nigerian sequences listed by GISAID ID in	Yes	Yes	Yes	none	no
[58]	Sanger (BigDye Terminator v3.1, ABI 3100 Avant)	All 8 segments (polygenic sequencing; Table 1 primers)	1 complete genome (A/ck/Nigeria/641/06)	Bootstrap 1,000	list: DQ406728, DQ52929[0]-DQ529297 (as printed 'DQ52929'), DQ780008	Yes	Yes	Yes	none	no
[101]	Sanger (overlapping PCR fragments; SeqScape 2.5)	All 8 segments	3 complete genomes (24 sequences, FR771823-FR77184	Posterior probability (>0.75 shown); NJ bootstrap 1,000	range: FR771823-FR771846	Yes	Yes	Yes	none	no

Reference	Platform	Segments	n sequences	Phylogeny support	Accessions	M3	M4	M5	Cap	High risk
[12]	Illumina MiSeq 2x300 bp (IZSvE, Italy); BWA/GATK/LoFreq cons	Whole genome (8 segments)	17 whole genomes	Ultrafast bootstrap, 1000 replicates (values >80 shown)	range: GISAID EPI_ISL_18462524- 18462541 (Tables S1 and S2)	Yes	Yes	Yes	none	no
[19]	Illumina MiSeq	Complete genomes (8 segments)	34 complete genomes (35 viruses listed in Table S1)	Bootstrap values above 80 shown on the HA tree	list: GISAID EPI_ISL_20316058 to EPI_ISL_20316116 for the 35 Nigerian viruses (Table S1); comparison set EPI_SET_260113s	Yes	Yes	Yes	none	no
[104]	Sanger: BigDye Terminator v3.1, ABI Prism 3130xl; SeqScape v	All eight segments, complete coding sequences	106 whole genomes	Bootstrap 1000 NJ replicates, Bayesian posterior probabiliti	list: GenBank CY047976- CY048647, CY016276- CY016283, CY017179- CY017186, CY016284- CY016291, CY016907- CY016954, EU148356-EU	Yes	Yes	Yes	none	no
[87]	Sanger, BigDye Terminator v3.1 (at IZSvE Padua)	HA partial (cleavage- site region); NA subtyped by N1-N9 prim	2 isolates sequenced (HA)	Bootstrap, 1,000 replicates	KJ917546 (partial HA)	Yes	Yes	Yes	none	no
[108]	NA (sequences downloaded)	HA (1716 positions) and NA (1419 positions)	481 HA and 429 NA globally; 13 Nigerian HA (7 with	Bootstrap, 1000 replicates; Monte Carlo upper-tail tests wit	list: GenBank accessions per sequence in Table S1 (Nigeria: AM262541, AM262543, AM262546, AM262547, AM262553, AM262572,	Unclear	Yes	Yes	none	no
[107]	Sanger: BigDye Terminator v3.1, ABI PRISM 3130xl	Complete coding sequences of all eight segments (HA and NA f	494 sequences in all; 21 Nigerian strains, 8 segme	Bootstrap 1000 replicates; Bayesian posterior probabilities	list: GenBank per strain in Table S2 (Nigeria: CY016276- CY016283, CY017179- CY017186, CY016284- CY016291, CY016907-CY01695	Yes	Yes	Yes	none	no
[109]	NA (sequences downloaded)	HA, full or partial length	185 HA globally; 8 Nigerian (Table S1)	aLRT (SH-like) and 1000 bootstrap replicates; values of 50%	list: GenBank per sequence in Table S1 (Nigeria: AM503005, EF631175, EF631180, EF631183, EF631184, EF631186, EF631187, E	Unclear	Yes	Yes	none	no

Reference	Platform	Segments	n sequences	Phylogeny support	Accessions	M3	M4	M5	Cap	High risk
[63]	Direct Sanger-type sequencing of overlapping RT-PCR amplicon	HA (segment 4) and NA (segment 6) complete ORF	7 (GenBank EU544242-EU544248) from 4 birds	Bootstrap, 1,000 replicates	EU544242-EU544248 (HA and NA complete ORF)	Yes	Yes	Yes	none	no
[73]	High-throughput Sanger sequencing at TIGR, with Elvira/Minim	Complete genomes, all 8 segments	36 genomes, of which 9 Nigerian (71 segment access	Bootstrap, 1000 replicates	list: GenBank per segment in the Appendix Table (Nigeria: CY016276-CY016291, CY016907-CY016954, CY017179-CY017186; the r	Yes	Yes	Yes	none	no
[105]	Sanger, ABI 3100 Avant Genetic Analyzer	All 8 segments (whole genome)	12 genomes (96 segment sequences, EU148356-EU14845	Bootstrap, 1,000 replicates (Figure legends)	EU148356-EU148451 (8 segments x 12 strains)	Yes	Yes	Yes	none	no
[106]	Sanger (method as in Ducatez et al. 2006)	All 8 segments, complete for most viruses	8 Nigerian strains (accessions FM160635-FM160642,	Bootstrap, 1000 replicates by maximum likelihood	range: GenBank FM160635-FM160642 and FM164800-FM164855	Yes	Yes	Yes	none	no
[76]	Illumina MiSeq, from RNA eluted from FTA cards	Complete genome, all 8 segments (one virus); HA for both	1 complete genome (8 segments)	Bootstrap, 100 replicates; values above 60 shown	list: GISAID EPI556504 and EPI567299-EPI567305 (one complete genome)	Yes	Yes	Yes	none	no
[77]	Illumina MiSeq	Complete coding sequences for 39 viruses; partial for 4	43 West African viruses; 1 Nigerian complete genom	Non-parametric bootstrap, 100 replicates; values above 60 sh	list: GenBank KU971593-KU971600 for the Nigerian virus (full West African list in the Table and the Technical Appendix)	Yes	Yes	Yes	none	no
[65–67]	Illumina MiSeq (all three reports; CAN-1484 in Appendix)	Whole genome (8 segments)	19 H9N2 genomes (MZ150582-MZ150733); 1 H5N6 genome	Ultrafast bootstrap 1000 (CAN-1585); ultrafast bootstrap >80	range: MZ150582-MZ150733 (CAN-1585); range: MN889503-MN889510 (CAN-0491); list: GISAID EPI_ISL_405278 (CAN-1484)	Yes	Yes	Yes	none	no
[11]	Illumina MiSeq, 2x250 bp, Nextera XT (IZSVe); ultradeep sequ	Whole genome (8 segments); partial genomes for 7	97 viruses: 90 whole genomes (78 H5N1, 11 H5N8, 1	Ultrafast bootstrap, 1000 replicates (values >80% shown)	exist, listed only in an unavailable supplement (GISAID EpiFlu IDs in Table S1); exist, listed only in an unavailable su	Yes	Unclear	Yes	none	no

Reference	Platform	Segments	n sequences	Phylogeny support	Accessions	M3	M4	M5	Cap	High risk
[74,75]	NR (sequencing at IZSve Padova)	Full genome (8 segments) of A/duck/Nigeria/3724-2/2008; HA o	9 segment sequences from 2 viruses (8 + 1)	Posterior probabilities (Bayesian trees)	list: GISAID EPI161701-EPI161708 (8 segments of A/duck/Nigeria/3724-2/2008); HA of 3724-10/2008 accession not given	Yes	Yes	Yes	none	no
[71,72]	Sanger, BigDye Terminator v3.1 on ABI 3130 Genetic Analyzer	HA only (nt 92-1633 of 1,730)	32 isolates sequenced (thesis); 35 'characterised'	1,000 bootstrap replicates; >70 taken as strong support	range: EF631164-EF631187 (24 HA sequences; identical sequences excluded)	Yes	Yes	Yes	none	no
[44,45]	NR (performed at FAO/OIE reference laboratory)	Complete genome of index virus (CAN-1902); HA phylogeny cite	NR (one named virus A/chicken/Nigeria/15 VIR339-2/2)	NR	GISAID EPI556504; EPI567299-EPI567305 (CAN-1902)	No	Yes	No	suggestive	no
[46]	Direct Sanger sequencing (Starick 2008); GCG v11.1 assembly	HA2 fragment, 154 nt (positions 1531-1684); HA cleavage site	4 HA2 fragments (A/07, B/07, D/06, I/06); cleavage	NA	none	Yes	No	N/A	suggestive	no
[47]	NR	HA (HA1 analysed)	Nigerian: NR (from GenBank); Burkina Faso: 8 HA1	100 bootstraps, 3 jumbles	none	Unclear	No	Yes	suggestive	no
[48]	NA for Nigeria (Malian genomes: Illumina MiSeq)	HA used for phylogeography	19 Nigerian HA sequences from GISAID (16 in final	Bootstrap 1,000 (ML); posterior probability >0.75 (MCC)	Nigerian sequences cited by GISAID isolate name only (Supplementary Table S2); no Nigerian accessions generated	Unclear	No	Yes	suggestive	no
[49]	NR (RT-PCR and sequencing 'as described previously', Ducatez	All 8 segments (full genomes); partial genomes for further b	Full genomes of 10 Nigerian strains (plus 2 Burkin	NR	range: AM262524-AM262578 and AM502998-AM503074	Yes	Yes	No	suggestive	no
[50]	NA (sequences downloaded from GenBank)	HA (NA sequences also collected; analysis is of HA)	873 clade 2.2 HA sequences globally; 135 Nigerian	Bootstrap values greater than 60 marked on branches	list: GenBank accession numbers of all 873 sequences in Additional file 1 (not broken down by country)	Unclear	Yes	No	suggestive	no
[51]	NA (sequences downloaded)	Complete proteome: 11 proteins (HA, M1, M2, NA, NP, NS1, NS2	12,624 strains globally; 48 Nigerian (S1 Dataset)	NA	none (strain names only in S1 Dataset; no accession numbers)	Unclear	No	N/A	suggestive	no
[52]	NA	NA	NA	NA	none listed (52 HA sequences said to be deposited in GenBank/EMBL, per reference 13)	No	No	No	suggestive	no
[53]	NR (Sanger; platform not stated)	Coding regions of all 8 segments of 3 viruses	3 Nigerian genomes	Bootstrap, 1000 replicates	none (no accession numbers given in the article or Technical Appendix)	No	No	Yes	suggestive	no

Reference	Platform	Segments	n sequences	Phylogeny support	Accessions	M3	M4	M5	Cap	High risk
[54]	NR	HA cleavage site only	NR (stated for confirmation of HPAI; number not gi	NA	none (cleavage-site sequence not deposited or shown)	No	No	N/A	suggestive	no
[55]	NA (no new sequencing)	HA (public sequences)	28 in final tree (10 Nigerian)	Bootstrap, 1000 replicates	none new (re-analysed public GenBank/GISAID sequences; accession numbers not listed, strain names in Fig. 1); none new (	N/A	No	Yes	suggestive	no
[56]	NR (IZSVe Padova)	NR	NR	NA	none (sequences from IZSVe not listed)	No	No	No	suggestive	no
[57]	NR	HA (cleavage site at least; clade assignment reported)	NR	NA	none	No	No	N/A	suggestive	no
[16]	NR	HA (complete gene)	14	NR	NR in abstract (full text not accessed)	Unclear	Unclear	Unclear	none	yes
[34]	NR	HA (complete open reading frame)	106 Nigerian HA sequences (751 in total)	NR	NR in abstract (full text not accessed)	Unclear	Unclear	Unclear	none	yes

S5. Exposure and surveillance in apparently healthy birds (never pooled)

Proportion n/N with exact Clopper-Pearson 95% CI computed here. 'Indep.' = no: subset/second-assay row, not counted as an independent estimate.

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
B	H5 antibody (HI)	[122,123]	Kaduna	NR (before 2012-05)	14 / 480	0.029	0.016-0.049	yes	
B	H5 antibody (HI)	[124]	Kogi	2012-09 to 2013-02	23 / 547	0.042	0.027-0.062	yes	
B	NP/type-A antibody (ELISA/AGID)	[62]	Kogi	2009-10 to 2010-01	90 / 450	0.200	0.164-0.240	yes	
B	NP/type-A antibody (ELISA/AGID)	[124]	Kogi	2012-09 to 2013-02	92 / 309	0.298	0.247-0.352	yes	
B	NP/type-A antibody (ELISA/AGID)	[113]	Oyo	2021-09 to 2022-08	85 / 200	0.425	0.356-0.497	yes	
B	RNA (RT-PCR)	[101]	NR (Dagona wildlife sanctuary, Hadejia-N	2008-03 to 2008-04	0 / 373	0.000	0.000-0.010	yes	
B	RNA (RT-PCR)	[101]	Plateau	2007-12 to 2008-03	0 / 374	0.000	0.000-0.010	yes	
B	RNA (RT-PCR)	[126]	Borno	2021-11 to 2021-12	17 / 28	0.607	0.406-0.785	yes	
B	RNA (RT-PCR)	[127]	Borno	NR	2 / 41	0.049	0.006-0.165	yes	
B+L	RNA (RT-PCR)	[99]	Kogi	NR	0 / 710 oropharyngeal swabs (tested in pools of 10)	0.000	0.000-0.005 (pools 0/71: 0.000-0.051)	yes	pooled swabs (710 swabs tested in 71 pools of 10); per-swab CI assumes independent swabs; pool-level CI shown alongside

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
C	H3 antibody (HI)	[103]	Osun	NR	14 / 101	0.139	0.078-0.222	no: consensus marks this row as a subset of another row of the same study (do not add)	
C	H5 antibody (ELISA)	[102]	Kaduna	2015 to 2016	6 / 6	1.000	0.541-1.000	yes	
C	H5 antibody (HI)	[103]	Osun	NR	3 / 101	0.030	0.006-0.084	no: consensus marks this row as a subset of another row of the same study (do not add)	
C	H5 antibody (HI)	[112]	Plateau	2011-08 to 2011-09	NR / 377	NR	NR	no: same specimens as [112] tested with a second assay	
C	H5 antibody (HI)	[68]	Oyo	2013-03 to 2013-10	11 / 461	0.024	0.012-0.042	no: same specimens as [68] tested with a second assay	subtype test run on screen-positive sera only; N is all sera screened
C	H5 antibody (HI)	[117]	Lagos	2015 to 2016	3 / 161	0.019	0.004-0.053	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
C	H7 antibody (HI)	[68]	Oyo	2013-03 to 2013-10	1 / 461	0.002	0.0001-0.012	no: same specimens as [68] tested with a second assay	subtype test run on screen-positive sera only; N is all sera screened
C	H9 antibody (HI)	[103]	Osun	NR	19 / 101	0.188	0.117-0.278	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
C	H9 antibody (HI)	[68]	Oyo	2013-03 to 2013-10	39 / 461	0.085	0.061-0.114	no: same specimens as [68] tested with a second assay	subtype test run on screen-positive sera only; N is all sera screened
C	NP/type-A antibody (ELISA/AGID)	[98]	Oyo	NR	19 / 315	0.060	0.037-0.093	yes	
C	NP/type-A antibody (ELISA/AGID)	[98]	Osun	NR	4 / 95	0.042	0.012-0.104	yes	
C	NP/type-A antibody (ELISA/AGID)	[98]	Ondo	NR	0 / 110	0.000	0.000-0.033	yes	
C	NP/type-A antibody (ELISA/AGID)	[103]	Osun	NR	19 / 101	0.188	0.117-0.278	yes	
C	NP/type-A antibody (ELISA/AGID)	[112]	Plateau	2011-08 to 2011-09	119 / 377	0.316	0.269-0.365	yes	
C	NP/type-A antibody (ELISA/AGID)	[68]	Oyo	2013-03 to 2013-10	49 / 461	0.106	0.080-0.138	yes	
C	NP/type-A antibody (ELISA/AGID)	[117]	Lagos	2015 to 2016	8 / 161	0.050	0.022-0.096	yes	
C	NP/type-A antibody (ELISA/AGID)	[117]	Ogun	2015 to 2016	0 / 120	0.000	0.000-0.030	yes	
C	NP/type-A antibody (ELISA/AGID)	[117]	Plateau	2015 to 2016	0 / 140	0.000	0.000-0.026	yes	
C	NP/type-A antibody (ELISA/AGID)	[117]	Bauchi	2015 to 2016	0 / 80	0.000	0.000-0.045	yes	

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
C	RNA (RT-PCR)	[68]	Oyo	2013-03 to 2013-10	0 / 461	0.000	0.000-0.008	no: same specimens as [68] tested with a second assay	
C+B	H3 antibody (HI)	[98]	Osun, Oyo	NR	22 (18 H3N8 only + 4 H3N8 and H5N2) / 520	0.042	0.027-0.063	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
C+B	H5 antibody (ELISA)	[96,97]	Gombe	2013	6 / 450	0.013	0.005-0.029	no: same specimens as [96,97] tested with a second assay	
C+B	H5 antibody (HI)	[96,97]	Gombe	2013	22 / 450	0.049	0.031-0.073	yes	
C+B	H5 antibody (HI)	[98]	Oyo	NR	4 / 520	0.008	0.002-0.020	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
C+B+L	H5 antibody (HI)	[125]	Bauchi	2009; 2011-2012 (split NR)	6 / 200	0.030	0.011-0.064	yes	
C+B+L	H5 antibody (HI)	[125]	Gombe	2009; 2011-2012 (split NR)	2 / 200	0.010	0.001-0.036	no: consensus marks this row as a subset of another row of the same study (do not add)	
C+B+L	antigen (ELISA/rapid test)	[125]	Bauchi, Gombe	2009 to 2012	0 / 400	0.000	0.000-0.009	yes	
C+B+L+Wr+Cap	RNA (RT-PCR)	[96,97]	Bauchi, Gombe	NR	NR (3% stated) / 1000	NR	NR	yes	
C+B+O	H5 antibody (ELISA)	[37]	Borno	NR	NR (not stated; 3.5% of 284 = 9.94, so n / 284	NR	NR	yes	numerator not stated; 3.5% of 284 = 9.94 is not an integer, so no exact numerator is derivable from the abstract
C+B+Wr	H5 antibody (ELISA)	[96,97]	Bauchi	2013	14 / 500	0.028	0.015-0.046	no: same specimens as [96,97] tested with a second assay	
C+B+Wr	H5 antibody (HI)	[96,97]	Bauchi	2013	16 / 500	0.032	0.018-0.051	yes	
Cap	H5 antibody (ELISA)	[96,97]	Bauchi, Gombe	NR (questionnaires dated	NR (11% per the Abstract; no count or ta / NR (37 captive birds listed in Table 1; serum count not given)	NR	NR	no: consensus marks this row as a subset of another row of the same study (do not add)	
Cap	NP/type-A antibody (ELISA/AGID)	[13]	Kwara	2016-03 to 2018-02	5 / 48	0.104	0.035-0.227	yes	
L	H5 antibody (HI)	[115]	Jigawa	2008-08 to 2009-04	48 / 396	0.121	0.091-0.158	yes	
L	H5 antibody (HI)	[124]	Kogi	2012-09 to 2013-02	10 / 232	0.043	0.021-0.078	yes	
L	H5 antibody (HI)	[74,75]	Abia, Akwa Ibom, Bayelsa, Cross River, E	2008-05 to 2008-06	0 / NR (number of sera not given)	NR	NR	no: same specimens as [74,75] tested with a second assay	
L	H5 antibody (HI)	[119]	Plateau	2018-05 to 2018-07	0 / 292	0.000	0.000-0.013	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
L	H5 antibody (HI)	[120]	Plateau	2020-03 to 2020-07	0 / 276	0.000	0.000-0.013	no: consensus marks this row as a subset of another row of the same study (do not add)	
L	H5 antibody (HI)	[121]	Kaduna	NR	1 / 300	0.003	0.0001-0.018	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H5 antibody (HI)	[35]	Ogun, Oyo	NR	NR / 491	NR	NR	no: same specimens as [35] tested with a second assay	
L	H7 antibody (HI)	[119]	Plateau	2018-05 to 2018-07	5 / 292	0.017	0.006-0.040	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H7 antibody (HI)	[120]	Plateau	2020-03 to 2020-07	4 / 276	0.014	0.004-0.037	no: consensus marks this row as a subset of another row of the same study (do not add)	
L	H7 antibody (HI)	[121]	Kaduna	NR	1 / 300	0.003	0.0001-0.018	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H7 antibody (HI)	[35]	Ogun, Oyo	NR	0 / 491	0.000	0.000-0.007	no: same specimens as [35] tested with a second assay	
L	H9 antibody (HI)	[119]	Plateau	2018-05 to 2018-07	0 / 292	0.000	0.000-0.013	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H9 antibody (HI)	[120]	Plateau	2020-03 to 2020-07	72 / 276	0.261	0.210-0.317	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H9 antibody (HI)	[121]	Kaduna	NR	1 / 300	0.003	0.0001-0.018	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L	H9 antibody (HI)	[35]	Ogun, Oyo	NR	NR / 491	NR	NR	no: same specimens as [35] tested with a second assay	
L	NP/type-A antibody (ELISA/AGID)	[89]	Kaduna	2012-03 to 2012-06	6 / 35	0.171	0.066-0.337	yes	
L	NP/type-A antibody (ELISA/AGID)	[62]	Kogi	2009-10 to 2010-01	78 / 300	0.260	0.211-0.314	yes	
L	NP/type-A antibody (ELISA/AGID)	[13]	Kwara	2016-03 to 2018-02	87 / 184	0.473	0.399-0.548	yes	
L	NP/type-A antibody (ELISA/AGID)	[13]	Kwara	2016-03 to 2018-02	0 / 168	0.000	0.000-0.022	yes	
L	NP/type-A antibody (ELISA/AGID)	[103]	Oyo	NR	0 / 30	0.000	0.000-0.116	yes	
L	NP/type-A antibody (ELISA/AGID)	[124]	Kogi	2012-09 to 2013-02	47 / 143	0.329	0.253-0.412	yes	

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
L	NP/type-A antibody (ELISA/AGID)	[74,75]	Abia, Akwa Ibom, Bayelsa, Cross River, E	2008-05 to 2008-06	0 / NR (number of sera not given)	NR	NR	yes	
L	NP/type-A antibody (ELISA/AGID)	[119]	Plateau	2018-05 to 2018-07	15 / 292	0.051	0.029-0.083	yes	
L	NP/type-A antibody (ELISA/AGID)	[120]	Plateau	2020-03 to 2020-07	84 / 276	0.304	0.251-0.362	yes	
L	NP/type-A antibody (ELISA/AGID)	[121]	Kaduna	NR	5 / 300	0.017	0.005-0.038	yes	
L	NP/type-A antibody (ELISA/AGID)	[35]	Ogun, Oyo	NR	51 / 491	0.104	0.078-0.134	yes	
L	RNA (RT-PCR)	[129]	Sokoto	2008-10 to 2009-03	3 / 221	0.014	0.003-0.039	yes	
L	RNA (RT-PCR)	[87]	Oyo	2011-09 to 2012-02 (posit	20 / 155	0.129	0.081-0.192	yes	
L	RNA (RT-PCR)	[127]	Borno	NR	7 / 30	0.233	0.099-0.423	yes	
L	RNA (RT-PCR)	[52]	National (25 infected States and the FCT	2007-10 to 2007-11	3 (tracheal swabs) / 8,753 swabs (tracheal 4,385 + cloacal 4,368)	0.0003	0.0001-0.001	yes	
L	antibody, AGID H5 + HI H5/H7/H9 panel (reactivity NR)	[52]	National (25 infected States and the FCT	2007-10 to 2007-11	6 / 4,228 sera	0.001	0.0005-0.003	yes	
L	virus isolation	[87]	Oyo	2011-09 to 2012-02 (posit	3 isolates / NR (isolation attempted on cloacal swab specimens; number inoculated not stated)	NR	NR	yes	
L	virus isolation	[65-67]	Sokoto	2019-06	1 isolate (A/duck/Nigeria/SK28T_19VIR842 / 3131 samples	0.0003	0.000-0.002	yes	
L	virus isolation	[74,75]	Gombe	2008-05 to 2008-07	2 / 43	0.047	0.006-0.158	yes	
L	virus isolation	[74,75]	NR (11 states: Abia, Akwa Ibom, Bayelsa,	2008-05 to 2008-06	0 (implied; only the two duck isolates a / 1,899	0.000	0.000-0.002	yes	
L	virus isolation	[35]	Ogun, Oyo	NR	0 / 400	0.000	0.000-0.009	yes	
L+Cap	H5 antibody (ELISA)	[102]	Bauchi, Gombe, Kaduna	2015 to 2016	8 / 72	0.111	0.049-0.207	yes	
L+Cap	H5 antibody (HI)	[13]	Kwara	2016-03 to 2018-02	0 / 400	0.000	0.000-0.009	no: follow-up H5 HI on the 92 ELISA-positive sera of rows [13]-#2 (same specimens; screening-then-follow-up rule)	subtype test run on screen-positive sera only; N is all sera screened
L+W?+O	H5 antibody (HI)	[89]	Kaduna	2012-03 to 2012-06	0 / 157 screened by NP-ELISA; H5/H7 HI run on the 14-15 ELISA-positive sera only	0.000	0.000-0.023	no: follow-up H5/H7 HI on the ELISA-positive sera of rows [89]-#2 (same specimens; screening-then-follow-up rule)	
L+W?+O	RNA (RT-PCR)	[89]	Kaduna	2012-03 to 2012-06	0 / 300	0.000	0.000-0.012	yes	
L+Wf	H5 antibody (ELISA)	[116]	Kaduna	2018-04 to 2018-12	9 / 136	0.066	0.031-0.122	yes	

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
L+Wr	H5 antibody (HI)	[114]	Kaduna	NR	1 / 146	0.007	0.0002-0.038	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L+Wr	H9 antibody (HI)	[114]	Kaduna	NR	2 / 146	0.014	0.002-0.049	no: consensus marks this row as a subset of another row of the same study (do not add)	subtype test run on screen-positive sera only; N is all sera screened
L+Wr	NP/type-A antibody (ELISA/AGID)	[114]	Kaduna	NR	5 / 146	0.034	0.011-0.078	yes	
L+Wr	RNA (RT-PCR)	[116]	Kaduna	2018-04 to 2018-12	1 pool / 136 birds in 26 pools	0.038	0.001-0.196	yes	pool-level: 1 of 26 pools (4-5 birds per pool); not bird-level
NR	H7 antibody (HI)	[13]	Kwara	2016-03 to 2018-02	12 / 400	0.030	0.016-0.052	no: follow-up H7 HI on the 92 ELISA-positive sera of rows [13]-#2 (same specimens; screening-then-follow-up rule)	subtype test run on screen-positive sera only; N is all sera screened
O	H5 antibody (HI)	[43]	NR	NR	3 / 203	0.015	0.003-0.043	yes	
O	H5 antibody (HI)	[43]	NR	NR	0 / 79	0.000	0.000-0.046	yes	
O	H9 antibody (HI)	[103]	Oyo	NR	1 / 1	1.000	0.025-1.000	no: consensus marks this row as a subset of another row of the same study (do not add)	
O	NP/type-A antibody (ELISA/AGID)	[89]	Kaduna	2012-03 to 2012-06	4 / 20	0.200	0.057-0.437	yes	
O	NP/type-A antibody (ELISA/AGID)	[103]	Oyo	NR	1 / 1	1.000	0.025-1.000	yes	
O	RNA (RT-PCR)	[52]	National (36 States and the FCT)	2007-10 to 2008-07	0 / 7,861 swabs (tracheal 4,253 + cloacal 3,608)	0.000	0.000-0.0005	yes	
O	antibody, AGID H5 + HI H5/H7/H9 panel (reactivity NR)	[52]	National (36 States and the FCT)	2007-10 to 2008-07	0 / 3,100 sera	0.000	0.000-0.001	yes	
W?	H5 antibody (ELISA)	[96,97]	Bauchi, Gombe	NR (questionnaires dated)	NR (15% per the Abstract; no count or ta / NR (32 free-flying birds listed in Table 1; serum count not given)	NR	NR	no: consensus marks this row as a subset of another row of the same study (do not add)	
W?	NP/type-A antibody (ELISA/AGID)	[89]	Kaduna	2012-03 to 2012-06	4 / 90	0.044	0.012-0.110	yes	
Wm+Wr	H5 antibody (HI)	[99]	Kogi	NR	0 / 44	0.000	0.000-0.080	no: same specimens as [99] tested with a second assay	
Wm+Wr	NP/type-A antibody (ELISA/AGID)	[99]	Kogi	NR	2 / 44	0.045	0.006-0.155	yes	
Wm+Wr	RNA (RT-PCR)	[101]	Plateau	2007-12 to 2008-03	0 / 154	0.000	0.000-0.024	yes	
Wm+Wr	RNA (RT-PCR)	[12]	Jigawa, Yobe	2022-03 to 2022-04	25 / 226	0.111	0.073-0.159	yes	
Wm+Wr	RNA (RT-PCR)	[63]	Jigawa	2007-02	9 birds / 228 birds	0.039	0.018-0.074	yes	

Host	Assay class	Row	State	Date	n / N (as extracted)	Prop.	95% CI (exact)	Indep.	Note
Wm+Wr	RNA (RT-PCR)	[99]	Kogi	NR	0 / 100 cloacal swabs (tested in pools of 10)	0.000	0.000-0.036 (pools 0/10: 0.000-0.308)	yes	pooled swabs (100 swabs tested in 10 pools of 10); per-swab CI assumes independent swabs; pool-level CI shown alongside
Wm+Wr	virus isolation	[12]	Jigawa, Yobe	2022-03 to 2022-04	25 / 226	0.111	0.073-0.159	no: consensus marks this row as a subset of another row of the same study (do not add)	
Wr	H5 antibody (ELISA)	[102]	Bauchi, Gombe, Kaduna	2015 to 2016	19 / 127	0.150	0.092-0.224	yes	
Wr	NP/type-A antibody (ELISA/AGID)	[111]	Kano	2014-11 to 2014-12	0 / 195	0.000	0.000-0.019	yes	
Wr	RNA (RT-PCR)	[101]	NR (Dagona wildlife sanctuary, Hadejia-N	2008-03 to 2008-04	3 / 651 (wild-bird samples at Dagona, derived as 1,024 minus 373 domestic; not printed)	0.005	0.001-0.013	yes	derived denominator: 651 (wild-bird samples at Dagona, derived as 1,024 minus 373 domestic; not printed)
Wr	RNA (RT-PCR)	[63]	Jigawa	2007-02	7 birds / 228 birds	0.031	0.012-0.062	no: same specimens as [63] tested with a second assay	
Wr	antigen (ELISA/rapid test)	[111]	Kano	2014-11 to 2014-12	4 / 204	0.020	0.005-0.049	yes	

S5 ranges per host system x assay class (independent rows with counts)

Host	Assay class	Rows	Independent with counts	Range
B	H5 antibody (HI)	2	2	0.029 (14/480 ([122,123])) to 0.042 (23/547 ([124]))
B	NP/type-A antibody (ELISA/AGID)	3	3	0.200 (90/450 ([62])) to 0.425 (85/200 ([113]))
B	RNA (RT-PCR)	4	4	0.000 (0/373 ([101])) to 0.607 (17/28 ([126]))
B+L	RNA (RT-PCR)	1	1	0.000 (0/710 ([99])) to 0.000 (0/710 ([99]))
C	H3 antibody (HI)	1	0	NR
C	H5 antibody (ELISA)	1	1	1.000 (6/6 ([102])) to 1.000 (6/6 ([102])) [small N: all independent rows N<10]
C	H5 antibody (HI)	4	0	NR
C	H7 antibody (HI)	1	0	NR
C	H9 antibody (HI)	2	0	NR
C	NP/type-A antibody (ELISA/AGID)	10	10	0.000 (0/110 ([98])) to 0.316 (119/377 ([112]))
C	RNA (RT-PCR)	1	0	NR
C+B	H3 antibody (HI)	1	0	NR
C+B	H5 antibody (ELISA)	1	0	NR
C+B	H5 antibody (HI)	2	1	0.049 (22/450 ([96,97])) to 0.049 (22/450 ([96,97]))
C+B+L	H5 antibody (HI)	2	1	0.030 (6/200 ([125])) to 0.030 (6/200 ([125]))
C+B+L	antigen (ELISA/rapid test)	1	1	0.000 (0/400 ([125])) to 0.000 (0/400 ([125]))
C+B+L+Wr+Cap	RNA (RT-PCR)	1	0	NR

Host	Assay class	Rows	Independent with counts	Range
C+B+O	H5 antibody (ELISA)	1	0	NR
C+B+Wr	H5 antibody (ELISA)	1	0	NR
C+B+Wr	H5 antibody (HI)	1	1	0.032 (16/500 ([96,97])) to 0.032 (16/500 ([96,97]))
Cap	H5 antibody (ELISA)	1	0	NR
Cap	NP/type-A antibody (ELISA/AGID)	1	1	0.104 (5/48 ([13])) to 0.104 (5/48 ([13]))
L	H5 antibody (HI)	7	2	0.043 (10/232 ([124])) to 0.121 (48/396 ([115]))
L	H7 antibody (HI)	4	0	NR
L	H9 antibody (HI)	4	0	NR
L	NP/type-A antibody (ELISA/AGID)	11	10	0.000 (0/168 ([13])) to 0.473 (87/184 ([13]))
L	RNA (RT-PCR)	4	4	0.0003 (3/8753 ([52])) to 0.233 (7/30 ([127]))
L	antibody, AGID H5 + HI H5/H7/H9 panel (reactivity NR)	1	1	0.001 (6/4228 ([52])) to 0.001 (6/4228 ([52]))
L	virus isolation	5	4	0.000 (0/400 ([35])) to 0.047 (2/43 ([74,75]))
L+Cap	H5 antibody (ELISA)	1	1	0.111 (8/72 ([102])) to 0.111 (8/72 ([102]))
L+Cap	H5 antibody (HI)	1	0	NR
L+W?+O	H5 antibody (HI)	1	0	NR
L+W?+O	RNA (RT-PCR)	1	1	0.000 (0/300 ([89])) to 0.000 (0/300 ([89]))
L+Wr	H5 antibody (ELISA)	1	1	0.066 (9/136 ([116])) to 0.066 (9/136 ([116]))
L+Wr	H5 antibody (HI)	1	0	NR
L+Wr	H9 antibody (HI)	1	0	NR
L+Wr	NP/type-A antibody (ELISA/AGID)	1	1	0.034 (5/146 ([114])) to 0.034 (5/146 ([114]))
L+Wr	RNA (RT-PCR)	1	1	0.038 (1/26 ([116])) to 0.038 (1/26 ([116]))
NR	H7 antibody (HI)	1	0	NR
O	H5 antibody (HI)	2	2	0.000 (0/79 ([43])) to 0.015 (3/203 ([43]))
O	H9 antibody (HI)	1	0	NR
O	NP/type-A antibody (ELISA/AGID)	2	2	0.200 (4/20 ([89])) to 1.000 (1/1 ([103])) [small N: includes a row with N<10]
O	RNA (RT-PCR)	1	1	0.000 (0/7861 ([52])) to 0.000 (0/7861 ([52]))
O	antibody, AGID H5 + HI H5/H7/H9 panel (reactivity NR)	1	1	0.000 (0/3100 ([52])) to 0.000 (0/3100 ([52]))
W?	H5 antibody (ELISA)	1	0	NR
W?	NP/type-A antibody (ELISA/AGID)	1	1	0.044 (4/90 ([89])) to 0.044 (4/90 ([89]))
Wm+Wr	H5 antibody (HI)	1	0	NR
Wm+Wr	NP/type-A antibody (ELISA/AGID)	1	1	0.045 (2/44 ([99])) to 0.045 (2/44 ([99]))
Wm+Wr	RNA (RT-PCR)	4	4	0.000 (0/154 ([101])) to 0.111 (25/226 ([12]))
Wm+Wr	virus isolation	1	0	NR
Wr	H5 antibody (ELISA)	1	1	0.150 (19/127 ([102])) to 0.150 (19/127 ([102]))
Wr	NP/type-A antibody (ELISA/AGID)	1	1	0.000 (0/195 ([111])) to 0.000 (0/195 ([111]))
Wr	RNA (RT-PCR)	2	1	0.005 (3/651 ([101])) to 0.005 (3/651 ([101]))
Wr	antigen (ELISA/rapid test)	1	1	0.020 (4/204 ([111])) to 0.020 (4/204 ([111]))

Occurrence (count once: designated primary dataset per period)

2006-2007 values are positive diagnostic samples ([52]); 2008 onwards are cases/outbreaks. Units differ; never summed.

Period	Unit	Designated primary rows	Value (as extracted)	Also reported in (do not add)	Note
2006	positive diagnostic samples ([52] Table 1; not outbreaks)	[52]	145 positive diagnostic samples (H5N1)	[60], [86], [40], [41], [88], [128], [145], [132], [59]	[52] primary (S01). Other series give 140 ([60]), 145 OIE ([86], [40]), 129 ([41]), 84 EMPRES ([88]), 134 premises ([128]), 149 ([145]).
2007	positive diagnostic samples ([52] Table 1; not outbreaks)	[52]	154 positive diagnostic samples (H5N1)	[60], [86], [41], [88], [128]	Other series: 160 ([60]), 155 OIE ([86]), 168 ([41]), 63 EMPRES ([88]).
2008	lab-confirmed cases	[60]	2 lab-confirmed cases (H5N1)	[41], [86], [88], [52], [74,75]	[52] covers 2006-2007 only; 2008 taken from [60] (same NVRI series). [41] also 2 (August); OIE 2; EMPRES 4. [52]/084 report June-July 2008 isolates (counts NR or 2 ducks).
2009-2010	lab-confirmed cases	[60]	0 confirmed field outbreaks or surveillance detections	[88]	
2011-2013	NA	-	no primary national dataset; EMPRES compilation reports 0 (context only)	[88]	LPAI H5N2 detected in Ibadan LBM ducks 2011-2012 ([87]) is a surveillance detection, not an outbreak.
2014	outbreaks	[83]	index outbreak, Sabon Gari LBM Kano, 24 December 2014 (count of 2014 outbreaks not reported)	[44,45], [88]	EMPRES compilation lists 1 outbreak in 2014 ([88]).
2015	confirmed cases (premises: farms, LBMs, zoo)	[84]	513 confirmed cases (21 states)	[44,45], [86], [88], [76], [138], [135], [85], [144], [143]	Other series: 542 ([44,45], NVRI), 265 OIE ([86]), 254 EMPRES ([88]).
2016	confirmed cases (premises: farms, LBMs, zoo)	[84]	248 confirmed cases (27 states)	[86], [88], [138], [135], [85], [144]	Other series: 186 OIE ([86]), 219 EMPRES ([88]).
2017	confirmed cases (premises: farms, LBMs, zoo)	[84]	50 confirmed cases (26 states)	[86], [138], [85], [144], [141]	OIE 14 for January-May ([86]).
2018-2020	NA	-	no national primary dataset (NR)	[85], [141], [135], [57]	State-level only: Plateau 2 farms in 2019 ([85]), 0 in 2018 and 2020; Rivers 0 in 2017-2020 ([135]). 2019 LBM surveillance detections ([65–67]) are not outbreak counts.
2021-2022	outbreaks	[11]	454 H5N1 + 12 H5N8 + 1 H5N2 outbreaks (national records; not split by year)	[137], [54], [56], [141], [85], [135], [136]	Subtypes are separate findings of one dataset; the three rows are not overlapping counts of the same outbreaks.
2023	NA	-	no national count (NR); 15 outbreak viruses sequenced ([19]) is not an outbreak count	[19]	
2024 (Jan-Apr)	outbreaks	[19]	28 poultry outbreaks reported January-April 2024	[19]	Partial year.
2025-2026	NA	-	no national count (NR)	[19]	

E1 2006-2010: 2006: 145; 2007: 154 ([52]); 2008: 2 ([60]); 2009-2010: 0 ([60]). Not summed across studies.

E2 2011-2014: No national outbreak dataset for 2011-2013 (EMPRES 0, context); 2014: index outbreak 24 Dec 2014 ([83]).

E3 2015-2020: 2015: 513; 2016: 248; 2017: 50 ([84]); 2018-2020: national NR.

E4 2021-2026: 2021-2022: 467 H5Nx outbreaks ([11]); 2024 Jan-Apr: 28 ([19]); other years NR.

States affected as reported by national datasets: 2006-2007: 25 States and the FCT (names in figure only) ([52]); 2006-2008: 27 states + FCT ([33]); 2015: 21 states ([84]); 2016: 27 states ([84]); 2017: 26 states ([84]); 2021-2022: 31 of 37 administrative regions (all H5Nx; Abstract and 3.1) - the Introduction of the same report says 30 ([11])

Overlap (count-once) map

Each data stream reused across reports is counted once, under one primary study; the other studies are cited as also reporting it and their data are not added. Certainty of each link: (c) confirmed, (p) probable, (?) possible. S11 lists global or national compilations used for context only.

Stream	Data	Primary (counted here)	Also reported in (not added)
S01	NVRI national laboratory-confirmed HPAI H5N1 series, 2006-2007 (and 2008 tail)	[52]	[33] (p), [92–95] (p), [42] (c), [41] (p), [100] (p), [38,39] (p), [60] (p), [130] (p), [128] (p), [118] (c), [71,72] (p), [80] (p), [139] (p), [140] (p), [69] (p), [59] (p), [145] (p), [132] (p), [40] (p), [131] (p), [142] (?), [86] (p), [112] (c), [61] (?), [46] (p), [58] (p), [28] (p), [88] (p), [143] (p)
S02a	2006 Nigerian H5N1 complete genomes, TIGR/Salzberg CY series (CY016276-CY016954, CY017179-CY017186)	[73]	[104] (c), [107] (c), [108] (c), [51] (c), [110] (c), [14] (p), [55] (p), [49] (c)
S02b	2006 south-western farm H5N1 genomes and HA (AM262524-AM262578, AM502998-AM503074)	[49]	[47] (p), [108] (c), [109] (c), [50] (p)
S02c	Kaduna index outbreak, January 2006 (isolate A/c/Nigeria/641/06)	[58]	[59] (p), [73] (p), [49] (c), [104] (p), [139] (p)
S02d	2006 Ibadan clinic H5N1 genomes (228-5, 228-6, 228-10)	[53]	[109] (c), [51] (c)
S02e	2006-2007 NVRI VRD HA sequences (EF631164-EF631187)	[71,72]	[104] (p), [109] (c), [64] (p), [52] (p)
S03a	Early-2007 H5N1 genomes (EU148356-EU148451; 1071-series, EMA1/EMA2-2:6 reassortant)	[105]	[104] (c), [107] (c), [110] (c), [106] (p)
S03b	2006-2008 NVRI genomes new in [104] (CY047976-CY048647)	[104]	[64] (c), [50] (p), [55] (p), [14] (c), [51] (c), [34] (?)
S03c	Mid-2007 south-western genomes (FM160635-FM160642, FM164800-FM164855)	[106]	[110] (c)
S03d	2008 Gombe duck H5N1, sublineage III / EMA3 (EPI161701-EPI161708)	[74,75]	[104] (p), [55] (c), [52] (p)
S04a	HP-genotype H5N2 in wild birds, Hadejia-Nguru (Jigawa), February 2007 (EU544242-EU544248)	[63]	[55] (c), [87] (c)
S04b	LPAI H5N2 in spur-winged geese, Dagona, 2008 (FR771823-FR771846)	[101]	[55] (c)
S05a	2015 index virus, Kano backyard (A/chicken/Nigeria/15VIR339-2/2015; EPI556504, EPI567299-305)	[76]	[44,45] (c), [110] (c)
S05b	2015 Lagos LBM index virus (A/chicken/Nigeria/15VIR339-1/2015; KU971593-KU971600)	[77]	[76] (p), [55] (c), [44,45] (p)
S06a	2015-2016 NVRI genomes (16VIR5840 series; MF112254-MF113041)	[78,79]	[64] (c), [14] (c), [16] (?)
S06b	New 2016-2017 genomes generated in [64] (17RS737-43 H5N8 Kano Nov 2016; 17RS737-44 to -48 H5N1 2017)	[64]	[84] (?), [138] (?)
S07	National outbreak records, December 2014-2017 (NADIS/FDVS, WAHIS)	[84]	[44,45] (p), [86] (p), [90] (p), [133] (c), [88] (p), [135] (c), [138] (c), [143] (p), [83] (p), [134] (p), [76] (p), [117] (p), [144] (p), [85] (p), [141] (p), [14] (p)
S08	National HPAI H5Nx outbreak records and genomes, 2021-2022	[11]	[137] (p), [54] (p), [56] (p), [141] (p), [81,82] (?), [136] (?), [85] (p), [14] (?), [90] (p), [135] (?)
S09	Plateau State Avian Influenza Desk Office records (2015-2022)	[85]	[141] (c), [144] (c), [57] (p), [137] (p), [54] (p)
S10	2019 national LBM active surveillance genomes (H9N2 MZ150582-733; H5N6 MN889503-10; H5N8 EPI_ISL_405278)	[65–67]	[48] (c)
S11	WAHIS / EMPRES-i / OIE global or national compilations (context only; no primary)	context only	[31] (c), [90] (c), [109] (c), [88] (c), [28] (c), [29] (c), [143] (c), [130] (c), [86] (c), [40] (c), [50] (c), [14] (c)
S12	Jos H5N8 farm isolates, February 2019	[57]	[141] (c), [85] (p)
S13	Bauchi/Gombe wild-bird sera (2013 vs 2015-2016)	[96,97]	[102] (?)

Stream	Data	Primary (counted here)	Also reported in (not added)
S14	Kogi State poultry and wild-bird surveys (Ameji group)	[124]	[99] (?), [62] (?)

[143] (Fasanmi PhD thesis) also reports on CAN-0174 (= [27], chapter 2) and CAN-0266 (= [86], chapter 6); those chapters were not re-extracted under [143].
[92–95] (Akanbi PhD thesis, Study 1) also reports on CAN-1907 (= [42], Study 2) and CAN-0031 (= [46], Study 3); each is extracted once, under its own study.
[71,72] (Fasina MSc) GIS chapter is CAN-0339 (= [132]) and is counted under [132] only; [71,72] thesis Table 2.3 bird total matches [59].
[81,82]: CAN-0191 (journal, abstract read) and CAN-1882 (preprint, full text) are the same three events; rows from CAN-0191 are subset rows (do not add).
[44,45]: CAN-1902 (seminar, to Sept 2015) is an earlier cut of CAN-0025 (to Dec 2015); use the CAN-0025 count.
Not-retrieved companions that may be the same surveys: CAN-0204 ([89]), CAN-0209 ([96,97]), CAN-1939 ([116]), CAN-1291 ([11] preprint).

Coverage of included studies

All 113 included studies are cited: 109 contribute to at least one synthesis table (T1–T5, S5 or the occurrence series). The following studies contribute no synthesisable Nigerian data and are listed here for completeness.

Reference	Reason
[27]	Only Nigerian outcome (61/155 infected LBMs and the biosecurity risk factors) is pooled with Egyptian markets; no Nigeria-specific count or date (consensus row [27]).
[28]	No new Nigerian data: EMPRES-i/WAHID compilations of national records (S11, context only); its single wild-vulture case has no species, place or date beyond 2006-2007.
[29]	No Nigerian counts: reports only that three space-time clusters occurred in Nigeria in January-February 2006, from OIE records (S11, context only).
[31]	No new Nigerian outbreak or virus data: national-level agro-ecological niche classification from WAHID/EMPRES-i (S11); cited only in the overlap map.
[30]	Abstract only (W3-029): laboratory methods stated without any result; no Nigeria-specific count, date or test result extractable. The only quantities are Egypt-specific or Africa-wide model outputs. Thematic overlap with CAN-1914/CAN-0174 recorded; resolve if a full text is obtained.

Sensitivity analyses: details

Each table builder was re-run on the filtered findings rows and compared with the main analysis. (i) removes all rows of the 14 high-risk studies; (ii) removes rows whose report is grey literature (theses, preprints, the lecture, the seminar paper, the conference paper and repository records), keeping reports of uncertain status; (iii) as (ii), with the four reports of uncertain status also treated as grey; (iv) post hoc, counts states only where the best evidence in the cell is a sequence or an isolate.

Run	Studies removed	Changes from the main analysis	Conclusion changed?
(i) High-risk studies removed	[32], [38,39], [16], [27], [33], [34], [35], [40], [41], [42], [43], [36], [30], [37]	T1: Keys lost: state: Niger. Earliest dates changed: none. T2 (states with laboratory evidence): 2006-2010: 27 → 26 (lost Niger); date NR: 5 → 4 (lost Plateau). T3: no change. T4: groups lost: none; verdicts changed: none. T5 studies removed: {[16]}, {[34]}. S5: groups lost: C+B+O H5 antibody (ELISA), O H5 antibody (HI); ranges changed: none.	No
(ii) Peer-reviewed reports only	[128], [118], [70], [143], [44,45] (partly), [81,82] (partly), [71,72] (partly), [92–95] (partly)	T1: Keys lost: subtype: H9 (N not determined). Earliest dates changed: clade: 2.2 2006 → 2006; host_system: Wr 2006-03 → 2007-02; host_system: Cap 2006-04 → 2015 to 2017 (window); host_system: O 2021 → 2021; state: Lagos 2006 → 2007-01 to 2007-02 (range); state: Sokoto 2007-01 → 2007-01 to 2007-02 (range). T2 (states with laboratory evidence): 2021-2026: 15 → 14 (lost Delta). T3: Cap: confirmed detection → confirmed detection on mixed-host rows only; O: confirmed detection → confirmed detection on mixed-host rows only; species: pheasant: named in a row with a confirmed detection (species positivity not established); vulture: absent. T4: groups lost: none; verdicts changed: none. T5 studies removed: none. S5: groups lost: none; ranges changed: none. The Delta (2021-2026), Cap, host-O and pheasant changes in this run depend on excluding {[81,82]} from detection tables (review item B5). The published version of {[81,82]} (CAN-0191, journal, abstract read only) states the same events; data unchanged.	No

Run	Studies removed	Changes from the main analysis	Conclusion changed?
(iii) As (ii), uncertain reports treated as grey	[128], [85], [118], [70], [143], [59], [145]	T1: Keys lost: subtype: H9 (N not determined). Earliest dates changed: clade: 2.2 2006 → 2006; host_system: Wr 2006-03 → 2007-02; host_system: Cap 2006-04 → 2015 to 2017 (window); host_system: O 2021 → 2021; state: Kano 2006-01 → 2006-12; state: Lagos 2006 → 2007-01 to 2007-02 (range); state: Plateau 2006-01 → 2006; state: Sokoto 2007-01 → 2007-01 to 2007-02 (range). T2 (states with laboratory evidence): 2021-2026: 15 → 14 (lost Delta). T3: Cap: confirmed detection → confirmed detection on mixed-host rows only; O: confirmed detection → confirmed detection on mixed-host rows only; species: pheasant: named in a row with a confirmed detection (species positivity not established); vulture: absent. T4: groups lost: none; verdicts changed: none. T5 studies removed: none. S5: groups lost: none; ranges changed: none. The Delta (2021-2026), Cap, host-O and pheasant changes in this run depend on excluding {[81,82]} from detection tables (review item B5). The published version of {[81,82]} (CAN-0191, journal, abstract read only) states the same events; data unchanged.	No
(iv) Post hoc: sequence or isolate evidence only	none (cells restricted)	T2 states: 2006-2010: 26 states + FCT → 17 states + FCT (lost Adamawa, Edo, Enugu, Kwara, Nasarawa, Niger, Rivers, Yobe, Zamfara); 2015-2020: 21 states + FCT → 19 states + FCT (lost Anambra, Gombe); 2021-2026: 14 states + FCT → 10 states + FCT (lost Borno, Delta, Ebonyi, Oyo); date NR: 5 states → 2 states (lost Bauchi, Borno, Gombe). Zone effect: 2006–2010 loses the south-south (Edo, Rivers) and keeps only Anambra in the south-east; 2021–2026 loses the south-south and south-east (Delta, Ebonyi).	No